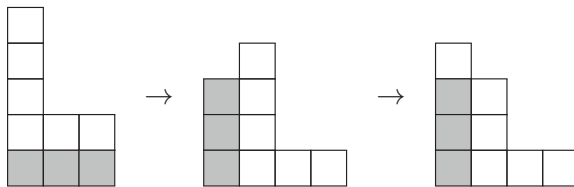


PROCESSES ON INTEGER PARTITIONS AND THEIR LIMIT SHAPES

Markus Jonsson



Mälardalen University Press Dissertations
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School of Education, Culture and Communication

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Abstract

This thesis deals with processes on integer partitions and their limit shapes, with focus on deterministic and stochastic variants on one such process called *Bulgarian solitaire*. The main scientific contributions are the following.

Paper I: Bulgarian solitaire is a dynamical system on integer partitions of n which converges to a unique fixed point if $n=1+2+\dots+k$ is a triangular number. There are few results about the structure of the game tree, but when k tends to infinity the game tree itself converges to a structure that we are able to analyze. Its level sizes turns out to be a bisection of the Fibonacci numbers. The leaves in this tree structure are enumerated using Fibonacci numbers as well. We also demonstrate to which extent these results apply to the case when k is finite.

Paper II: Bulgarian solitaire is played on n cards divided into several piles; a move consists of picking one card from each pile to form a new pile. In a recent generalization, σ -Bulgarian solitaire, the number of cards you pick from a pile is some function σ of the pile size, such that you pick $\sigma(h) < h$ cards from a pile of size h . Here we consider a special class of such functions. Let us call σ well-behaved if $\sigma(1) = 1$ and if both $\sigma(h)$ and $h - \sigma(h)$ are non-decreasing functions of h . Well-behaved σ -Bulgarian solitaire has a geometric interpretation in terms of layers at certain levels being picked in each move. It also satisfies that if a stable configuration of n cards exists it is unique. Moreover, if piles are sorted in order of decreasing size then a configuration is convex if and only if it is a stable configuration of some well-behaved σ -Bulgarian solitaire. If sorted configurations are represented by Young diagrams and scaled down to have unit height and unit area, the stable configurations corresponding to an infinite sequence of well-behaved functions $(\sigma_1, \sigma_2, \dots)$ may tend to a limit shape Φ . We show that every convex Φ with certain properties can arise as the limit shape of some sequence of well-behaved σ_n . For the special case when $\sigma_n(h) = \text{ceil}(q_n h)$ for $0 < q_n \leq 1$ (where ceil is the ceiling function rounding upward to the nearest integer), these limit shapes are triangular (in case $q_n^2 n \rightarrow 0$), or exponential (in case $q_n^2 n \rightarrow \infty$), or interpolating between these shapes (in case $q_n^2 n \rightarrow C > 0$).

Paper III: We introduce p_n -random q_n -proportion Bulgarian solitaire ($0 < p_n, q_n \leq 1$), played on n cards distributed in piles. In each pile, a number of cards equal to the proportion q_n of the pile size rounded upward to the nearest integer are candidates to be picked. Each candidate card is picked with probability p_n , independently of other candidate cards. This generalizes Popov's random Bulgarian solitaire, in which there is a single candidate card in each pile. Popov showed that a triangular limit shape is obtained for a fixed p as n tends to infinity. Here we let both p_n and q_n vary with n . We show that under the conditions $q_n^2 p_n n / \log n \rightarrow \infty$ and $p_n q_n \rightarrow 0$ as $n \rightarrow \infty$, the p_n -random q_n -proportion Bulgarian solitaire has an exponential limit shape.

Paper IV: We consider two types of discrete-time Markov chains where the state space is a graded poset and the transitions are taken along the covering relations in the poset. The first type of Markov chain goes only in one direction, either up or down in the poset (an *up chain* or *down chain*). The second type toggles between two adjacent rank levels (an *up-and-down chain*). We introduce two compatibility concepts between the up-directed transition probabilities (an *up rule*) and the down-directed (a *down rule*), and we relate these to compatibility between up-and-down chains. This framework is used to prove a conjecture about a limit shape for a process on Young's lattice. Finally, we settle the questions whether the reverse of an up chain is a down chain for some down rule and whether there exists an up or down chain at all if the rank function is not bounded.

Sammanfattning

Heltalspartitioner är matematiska objekt som är enkla att förstå. Det rör sig helt enkelt om olika sätt att dela upp ett heltal som en summa av heltalsdelar. T.ex. är $2 + 2$ och $2 + 1 + 1$ två olika partitioner av 4. Ett sätt att representera heltalspartitioner är i s.k. *Youngdiagram*, där delarna (sorterade i fallande storleksordning) motsvaras av kolumner av rutor. Till exempel är  Youngdiagrammet för $2 + 1 + 1$.

Bulgarisk patients är en process som kan representeras med Youngdiagram. Den spelas med n identiska kort som delas upp i ett antal högar—startkonfigurationen. Ett drag består i att ta ett kort från varje hög och bilda en ny hög av de tagna korten. Detta drag upprepas om och om igen. Varje konfiguration av kort kan ses som en heltalspartition av n , vars delar är högstorlekarna. I termer av Youngdiagram består draget i att ta bort raden längs ned och stoppa in den som en ny kolumn. Om antalet kort är ett triangeltal har denna process en intressant egenskap: Oberoende av hur startkonfigurationen ser ut kommer processen till slut att “fastna” i en stabil konfiguration.

Artikel I i denna avhandling studerar den specifika frågan: Hur många konfigurationer finns det med ett givet antal drag d till den stabila konfigurationen? Det visar sig att för tillräckligt små d ges svaret av ett Fibonaccital.

Det finns många andra processer vars tillstånd låter sig representeras av Youngdiagram. För denna typ av processer kan man studera processens *gränsform*. Detta gör man genom att låta antalet rutor i diagrammet öka. Om man hela tiden skalar ned storleken på rutorna så att diagrammets totala area hålls konstant när antalet rutor ökar kan man jämföra diagrammens form. Om denna form konvergerar mot en viss kurva när antalet rutor ökar, kallar man denna kurva processens gränsform.

Artikel II studerar en generalisering av bulgarisk patients där man tillåts ta fler än ett kort ur varje hög. För specialfallet där antal kort som tas ur en hög är en given andel q_n (som beror på n) av högens storlek bevisas gränsformen, vilken beror på den asymptotiska egenskapen hos nq_n^2 när n går mot oändligheten.

Artikel III studerar en stokastisk generalisering av patientsen i artikel II. Här är andelen q_n av korten i varje hög *kandidatkort*, och varje sådant kort tas med sannolikhet p_n , oberoende av andra kandidatkort. Gränsformen i fallet då $nq_n^2p_n$ går mot oändligheten bevisas.

Artikel IV bygger ett ramverk för en generell familj av processer. Där introduceras ett kompatibilitetskoncept som används för att bevisa en existerande förmodan om en gränsform.

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The following papers are included in this thesis.

I. Level sizes of the Bulgarian solitaire game tree

Henrik Eriksson, Markus Jonsson

The Fibonacci Quarterly (in press)

II. Limit shapes of stable configurations of a generalized Bulgarian solitaire

Kimmo Eriksson, Markus Jonsson, Jonas Sjöstrand

Submitted manuscript

III. An exponential limit shape of random q -proportion Bulgarian solitaire

Kimmo Eriksson, Markus Jonsson, Jonas Sjöstrand

Submitted manuscript

IV. Markov chains on graded posets: Compatibility of up-directed and down-directed transition probabilities

Kimmo Eriksson, Markus Jonsson, Jonas Sjöstrand

Order (in press)

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Structure of the thesis

This thesis consists of two parts. Readers who are mostly interested in technical details should go directly to the original research papers in part II of the thesis. Others would probably gain more from reading the more lightweight descriptions in part I. There I present the results, provide the background information required to understand them, and put them in their historical context.

Where necessary, there is also a “Motivation” subsection that explains the thoughts, observations and conjectures that inspired the paper.

Part I

Background and results

1 Preliminaries

Let us begin with an introduction to the objects and concepts used in this thesis: integer partitions and probability distributions on them, Young diagrams and their limit shapes and processes on Young diagrams.

1.1 Integer partitions

Integer partitions are among the mathematical objects that are easiest to understand. It is all about different ways of writing a positive integer as a sum of (weakly) smaller positive integers. For example,

$$3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1 \quad \text{and} \quad 4$$

are different partitions of the integer 4. In partition theory, $2 + 1 + 1$, $1 + 2 + 1$ and $1 + 1 + 2$ are considered the same partition, and one usually writes the parts in (weakly) decreasing order. The listed partitions are therefore *all* partitions of 4. Thus there are five partitions of four, and we write $p(4) = 5$ where $p(n)$ is the *partition function*. The number of partitions of 100 is $p(100) = 190, 569, 292$.

This seemingly innocent concept has raised a number of interesting problems and led to a lot of fascinating mathematical theory, which has since then found applications to statistical mechanics, computer science and other branches of mathematics. For example, it is easy to ask and understand the question *How many partitions $p(n)$ are there of a given positive integer n ?* It is not as easy to understand that the answer is

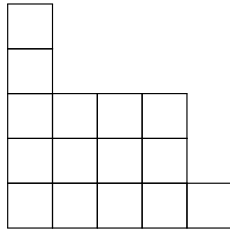
$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \left[\frac{d}{dx} \frac{\sinh\left(\frac{\pi}{k} \sqrt{\frac{2}{3}} \left(x - \frac{1}{24}\right)\right)}{\sqrt{x - \frac{1}{24}}} \right]_{x=n} \quad (1)$$

where each $A_k(n)$ is an explicitly given finite sum of $24k$ th roots of unity. The equation (1) is the so called *Hardy-Ramanujan-Rademacher expansion* of $p(n)$, which was proved in 1937 by Hans Rademacher, building on earlier work by G. H. Hardy and S. Ramanujan. A full proof can be found in [2].

Not surprisingly, the foundations of the theory of integer partitions were laid by Leonhard Euler, who also proved a number of beautiful and significant partition theorems. A good introduction to this fascinating subject is the book *Integer Partitions* [3] by George Andrews and Kimmo Eriksson.

1.2 Young diagrams and their shapes

The partition $5 + 4 + 4 + 1 + 1$ of 15 can be visualized as a geometric shape like this:



The parts 5, 4, 4, 1, 1 of the partition are sorted in weakly decreasing order and we draw rows of equally sized squares where the first row from the bottom has 5 squares, the second row 4 squares, and so on, with a left aligned margin. The result is called a *Young diagram*, after the British mathematician Alfred Young.

Young diagrams can be oriented in different ways. The above way of drawing Young diagrams is sometimes called *French notation*. In paper II and paper III they are instead drawn in such a way that the parts of the partition are represented by columns rather than rows, so that the Young diagram of the partition $5 + 4 + 4 + 1 + 1$ looks like this:



If we denote the partition $5 + 4 + 4 + 1 + 1$ by λ , one usually writes $\lambda = (5, 4, 4, 1, 1)$. There is obviously a one-to-one correspondence between partitions of n and Young diagrams with n squares. Therefore, if λ is a given integer partition, it is common that λ also denotes the corresponding Young diagram. We will be mostly concerned with the *shape* of a Young diagram. To this end, let the squares have unit length and place the diagram in a coordinate system as in Figure 1(a). We denote by $\partial\lambda$ the function that describes the boundary of the Young diagram λ as in Figure 1(b).

All four papers in this thesis deals with Young diagrams in one way or another.

1.3 Distributions on partitions and their limit shapes

There are $p(n)$ integer partitions of a given integer n . Let us denote the set of partitions of n by $\mathcal{P}(n)$. If we assign nonnegative values to each of these partitions such that the sum of all these values is 1, we have a probability distribution on $\mathcal{P}(n)$.

For a given integer n , let us sample a partition λ at random according to the probabilities in some distribution $\nu^{(n)}$ on $\mathcal{P}(n)$, and draw its Young

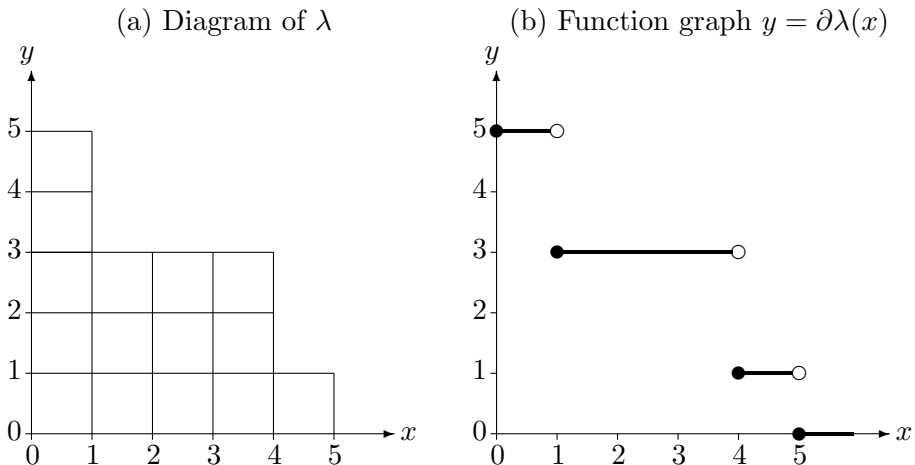


Figure 1: The Young diagram (in French notation) and its boundary function of the partition $\lambda = (5, 4, 4, 1, 1)$.

diagram boundary $\partial\lambda$. Doing this for larger and larger n , we get larger and larger Young diagrams. If we scale down the row lengths of each diagram by a factor $1/a_n$ and the column heights by a factor a_n/n so that the total area of each diagram is 1, then we can compare their shapes. Let us denote by $\partial^{a_n}\lambda$ the boundary function of the diagram rescaled this way. If the boundaries of these rescaled diagrams converge in probability to a certain deterministic curve ϕ as $n \rightarrow \infty$, this curve is the *limit shape* of the sequence of distributions $(\nu^{(n)})_n$, in other words, if

$$\lim_{n \rightarrow \infty} \nu^{(n)} \{ \lambda \in \mathcal{P}(n) : |\partial^{a_n}\lambda(x) - \phi(x)| < \varepsilon \} = 1 \quad (2)$$

for all $x > 0$ and all $\varepsilon > 0$.

Let us first consider the *uniform* distribution. The result of a sampling of partitions from this distribution with the scaling factor $a_n = \sqrt{n}$ can be seen in Figure 2. As we can see, the diagram boundaries seem to approach a limit shape. One of the famous results in this field is that Young diagrams sampled and scaled this way do indeed have a limit shape, namely

$$e^{-(\pi/\sqrt{6})x} + e^{-(\pi/\sqrt{6})y} = 1.$$

Note that this is a completely unbiased sampling: in the uniform distribution on $\mathcal{P}(n)$, each partition has the exact same probability $1/p(n)$ to be picked. The fact that this distribution has a limit shape at all is therefore a rather astonishing result: Almost all partitions of a large integer “look” the same!

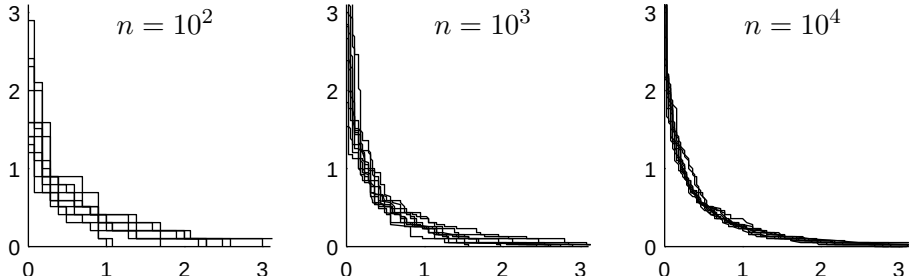


Figure 2: Young diagrams sampled uniformly at random from $\mathcal{P}(n)$, then downscaled with scaling factor $a_n = \sqrt{n}$, for $n = 10^2$, 10^3 , and 10^4 , with ten samples in each graph.

The study of this phenomenon of limit shape formation from the uniform probability distribution on integer partitions has a long history, starting with the paper by Temperley [28] in 1952 with heuristic arguments. In 1977 it was made more formally in [27] although the result was not presented in a modern way. This was done by Vershik and Kerov [32] in 1985. Since then, a large number of distributions have been studied in terms of limit shapes. A famous example is the limit shape of partitions chosen according to the Plancherel distribution [18, 31] in connection to the study of longest increasing subsequences of permutations drawn uniformly at random. A survey of such limit shape results was made by Vershik in [33], who also made a significant contribution [30] in 1996, stressing the close relationships between these problems and statistical physics of ideal gas.

For some probability distributions on partitions there is no limit shape, i.e., no single shape is approached in probability. Vershik calls such cases “non-ergodic”. These non-ergodic cases have been studied in some detail by Yakubovich [34].

1.4 Processes on Young diagrams

In the previous section, we considered the sampling of Young diagrams $\lambda_1, \lambda_2, \dots$ where each $\lambda_k \in \mathcal{P}(k)$ is drawn according to some probability distribution $\nu^{(k)}$ on $\mathcal{P}(k)$, $k = 1, 2, \dots$. This yields a sequence $(\lambda_1, \lambda_2, \dots)$ of growing Young diagrams, where each diagram λ_{k+1} has one more square than the previous λ_k .

An alternative way of obtaining such a sequence of randomly growing Young diagrams is to start with a single square diagram $\lambda_1 = (1)$ and sequentially add squares, one in each discrete time-step. The possible positions at which to add a square to a Young diagram while still maintaining

a Young diagram are the *inner corners*, see Figure 3. Let the inner corner of $\lambda_k \in \mathcal{P}(k)$ at which to add a square to obtain $\lambda_{k+1} \in \mathcal{P}(k+1)$ (for $k = 1, 2, \dots$) be chosen at random. Then the resulting discrete-time random process $(\lambda_1, \lambda_2, \dots)$ is called a *birth process* (or *growth process*) on Young diagrams. Each step in such a process is called a *birth step*.

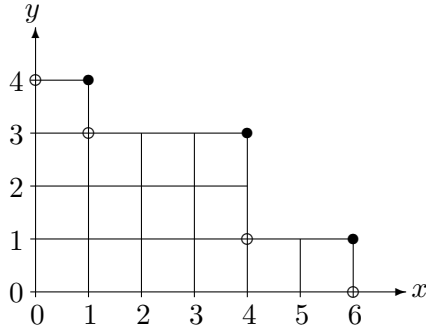


Figure 3: Inner corners (unfilled circles) and outer corners (filled circles) in the Young diagram of the partition $\lambda = (6, 4, 4, 1)$.

Such birth processes have been studied in the past. When the inner corner is picked uniformly at random, the resulting birth process is a description of a discrete-time version of a one-dimensional asymmetric particle system whose limit behaviour was studied by Rost [21]. Also, Green, Nijenhuis and Wilf [12] suggested an efficient algorithm for generating a Plancherel distributed partition by means of such a birth process.

A birth process naturally induces a probability distribution on each $\mathcal{P}(1), \mathcal{P}(2), \dots$, and one can thereby study its limit shape as explained in Section 1.3. This is done by Eriksson and Sjöstrand in [9] where both Rost's model and Nobel prize winning economist Herbert Simon's model of urban growth is formulated as a birth processes on Young diagrams. For example, Rost's model corresponds to a birth process wherein all inner corners are equally probable. The limit shape for this process under the scaling $a_n = \sqrt{n}$ is

$$\sqrt{x} + \sqrt{y} = 6^{1/4}.$$

See Figure 4 for the result of a simulation of this process. Eriksson and Sjöstrand also introduce *death steps* analogous to birth steps where the square at a randomly chosen *outer corner* is removed (see Figure 3.) They study *birth-and-death processes* consisting of alternating births and deaths of squares so that at the end of each birth-and-death period the Young diagram has a fixed size n . A very important model in mathematical population

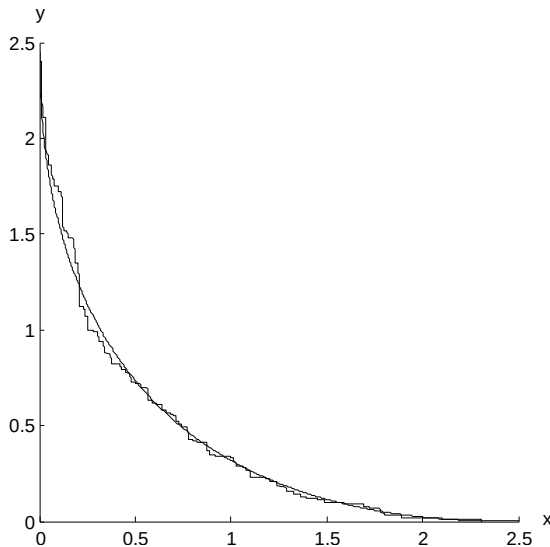


Figure 4: The result of a simulation of 10,000 steps of Rost's model under the scaling $a_n = \sqrt{n}$. The smooth curve is the limit shape $\sqrt{x} + \sqrt{y} = 6^{1/4}$.

genetics is the so-called *Moran model with infinitely many alleles* [11] which they also formulate as a birth-and-death process on Young diagrams.

2 About paper I

2.1 Bulgarian solitaire

Section 1.4 considered processes where each step deals with the birth or death of a *single* square. These *single-square processes* can be extended to *multi-square processes* where each step comprises the birth of multiple squares (a *multi-square birth process*), possibly followed by the death of multiple squares (a *multi-square birth-and-death process*). For instance, Jockusch, Propp and Shor [17] studied the birth process where, in each step, *each* inner corner is filled with a fixed probability p (independently of the other inner corners), and found the limit shape to be a quarter-ellipse (a quarter circle when $p = 1/2$). Similarly, the so called *Bulgarian solitaire* can be interpreted as a deterministic birth-and-death process with multi-square steps.

The game of Bulgarian solitaire is played with a deck of n identical cards divided arbitrarily into several piles. A move consists of picking a card from

each pile and letting these cards form a new pile. This move is repeated over and over again. For information about the earlier history of the game (including its name¹) and a summary of subsequent research, see reviews by Hopkins [14] and Drensky [8].

Let n be the number of cards and let us represent a configuration of cards by an integer partition of n (so that part sizes in a partition correspond to pile sizes in a card configuration). This way, Bulgarian solitaire is a simple dynamical system on $\mathcal{P}(n)$. Using this representation, it can be formulated in terms of Young diagrams (drawn such that column heights represent partition parts) as follows: In each step, remove the longest row (corresponding to picking one card from each pile) and reinsert the same number of squares as a new column (letting the picked cards form a new pile). The new column should be inserted such that the result is a valid Young diagram. This can be accomplished by inserting the new column to the left and then left-shifting all cards. See Figure 5 (and on the cover of this thesis) for an example.

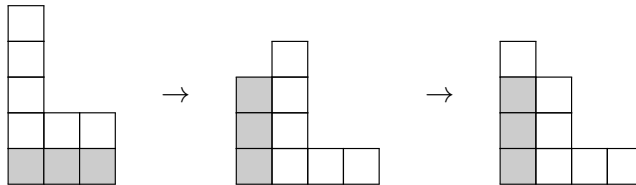


Figure 5: A move in Bulgarian solitaire from $\lambda = (5, 2, 2) \in \mathcal{P}(9)$: The bottom layer is picked to form a new pile with three cards and the cards are then left-shifted.

A remarkable fact about Bulgarian solitaire, which indeed sparked the initial interest in it, is that when played with a triangular number of cards $n = 1 + 2 + \dots + k$ for some positive integer k , the solitaire will eventually reach the stable configuration $(k, k - 1, \dots, 1)$, starting from any initial configuration. This has been known since the 1980s [7, 29]. Brandt [7] conjectured that the maximal number of moves necessary to reach that configuration is $k^2 - k$. This was later proved by Igusa [16] and Etienne [10], apparently independently at about the same time. (Although Etienne’s proof was published much later, his paper is noted as having been received in 1984.) In 1998, Griggs and Ho [13] found new simpler proofs of these facts.

¹Firstly, because the player only follows rules mechanically, and do not choose between different possible courses of action, Bulgarian solitaire is not really a “solitaire” in the sense most people mean the word. Secondly, as it turns out, nor is it particularly Bulgarian!

Let B be the operation on $\mathcal{P}(n)$ corresponding to a move in Bulgarian solitaire. For example, the move depicted in Figure 5 can be represented as $B((5, 2, 2)) = (4, 3, 1, 1)$. The *game graph* (or *state diagram*) of Bulgarian solitaire is a directed graph whose nodes are $\mathcal{P}(n)$ and where there is an edge from λ to λ' if $B(\lambda) = \lambda'$. For example, the game graph for $n = 6$ is shown in Figure 6.

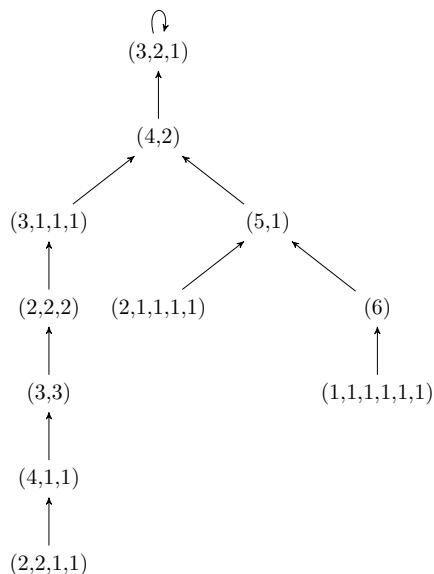


Figure 6: Bulgarian solitaire game graph for $n = \tau_3 = 6$.

For any positive integer k , let $\tau_k = k(k+1)/2$ be the k th triangular number. Let us also denote the staircase partition $(k, k-1, \dots, 2, 1) \in \mathcal{P}(\tau_k)$ by Δ_k . Many results on Bulgarian solitaire can be formulated in terms of the game graph. For example, the mentioned results of Brandt, Igusa and Etienne for $n = \tau_k$ means that the game graph is a tree (which we will call *game tree*) of maximal height $k^2 - k$ and with a loop at the root node Δ_k , which we can see in the game tree for the triangular number $n = 6$ ($k = 3$) in Figure 6.

For any number of cards, it is easy to realize that Bulgarian solitaire must eventually return to an already visited configuration, since $\mathcal{P}(n)$ is finite. Thus, the game graph must have a cycle. As we have seen, for triangular $n = \tau_k$, there is only one cycle and it has length one (the loop at Δ_k), but for general n several cycles of different lengths may occur. The partitions in these cycles are “almost” staircase shaped in the sense that

they can all be constructed by starting with some staircase shape Δ_k and adding at most one card to each pile, and possibly adding one more pile of size 1 [1, 4, 10, 13]. After downscaling, the deviations from the staircase partition tends to zero, so we may say that Bulgarian solitaire has a limit shape that is a straight line.

The smallest game graph with multiple cycles occurs at $n = 8$, where there are two cycles. This game graph therefore has two connected components. It is shown in Figure 7, where we see that the two cycles have lengths four and two. The number of cycles and their lengths are counted

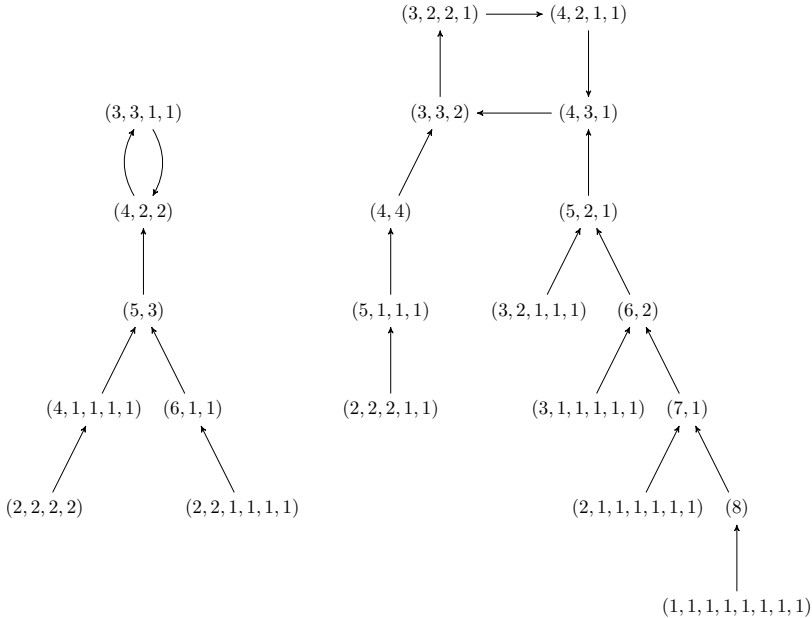


Figure 7: Bulgarian solitaire game graph for $n = 8$.

in [7]. Hopkins [15] enumerates the partitions with no preimage under B , called *Garden of Eden Partitions*. These are the partitions with indegree zero in the game graph (i.e. the leaves in the game tree for triangular n). For example, in Figure 6 we see that the Garden of Eden partitions for $n = 6$ are $(2, 2, 1, 1)$, $(2, 1, 1, 1, 1)$ and $(1, 1, 1, 1, 1, 1)$, and in Figure 7 for $n = 8$ we find seven Garden of Eden partitions.

Let $D(n)$ be the maximal number of moves required to reach a cycle (the longest preperiod length) in Bulgarian solitaire with n cards. We saw earlier that $D(\tau_k) = k^2 - k$. Igusa [16] and Etienne [10] proved that $D(n) \leq k^2 - k$ whenever $n \leq 1+2+\dots+k$ with equality for $n = 1+2+\dots+k$. Griggs and Ho

[13] improved this bound to $D(n) \leq k^2 - 2k - 1$ whenever $n \leq 1 + 2 + \dots + k$ with $k \geq 4$, and present a lower bound which they conjecture to be the actual value. For example, from Figure 7 we see that $D(8) = 5$ because of the path $(1, 1, 1, 1, 1, 1, 1, 1) \rightarrow (8) \rightarrow (7, 1) \rightarrow (6, 2) \rightarrow (5, 2, 1) \rightarrow (4, 3, 1)$.

Paper I considers the case when $n = \tau_k$ is a triangular number, i.e., when the game graph is a tree, rooted at the stable configuration Δ_k . It addresses the question *What is the level size of the game tree at a given level?* In other words; *How many partitions are there with a given number of moves from the stable configuration?* In Figure 6, we see that the level sizes are 1, 1, 2, 3, 2, 1, and 1.

2.2 Motivation

For $m \geq 0$ let B^m be m consecutive repetitions of B . Also, for any k and any $\lambda \in \mathcal{P}(\tau_k)$, let $D_k(\lambda)$ be the number of moves necessary to reach Δ_k starting from λ , i.e., for any $k \in \mathbb{P}$, define the function $D_k : \mathcal{P}(\tau_k) \rightarrow \mathbb{N}$ such that

$$D_k(\lambda) = \min\{d \in \mathbb{N} : B^d(\lambda) = \Delta_k\}.$$

We use the minimum in the definition of D_k since Δ_k is a fixed point of B .

Suppose the number of cards is $n = \tau_k$ and consider the configuration $\lambda := (k - 1, k - 1, k - 2, k - 3, \dots, 3, 2, 1, 1)$. In other words λ is the configuration obtained from Δ_k by removing one card from the biggest pile and forming one more pile of size 1 with that card. The configuration λ has the maximal number of moves to the staircase partition Δ_k — the number of moves is exactly $k^2 - k$ (see for example [10]).

Consider instead the configuration $\tilde{\lambda} := (k + 1, k - 1, k - 2, k - 3, \dots, 3, 2)$ obtained from Δ_k by removing the second biggest pile (that of size $k - 1$) and distributing these $k - 1$ cards to the remaining piles, one in each pile. Starting from $\tilde{\lambda}$ only one single move is required to reach Δ_k .

Moreover, $\tilde{\lambda}$ is unique being one move away from Δ_k ; the inverse image $D_k^{-1}(1) = \{\lambda \in \mathcal{P}(\tau_k) : D_k(\lambda) = 1\}$ is the singleton $\{\tilde{\lambda}\}$ for any k . The partition λ on the other hand is not unique being $k^2 - k$ moves away from Δ_k ; for example, the value of $|D_k^{-1}(k^2 - k)|$ is 3, 16, 65, 293, and 1267 for $k = 4, 5, 6, 7$ and 8, respectively.

The above observation motivated the investigation of $|D_k^{-1}(d)|$ for all d between 1 and $k^2 - k$. In other words, given any positive integer $d \leq k^2 - k$, how many partitions are there in $\mathcal{P}(\tau_k)$ with d moves to Δ_k ? Or, in terms of the Bulgarian solitaire game tree, what are its level sizes?

2.3 Result

The values of $|D_k^{-1}(d)|$ in Table 1 were found programmatically. They sug-

$k \setminus d$	1	2	3	4	5	6	7	8	9	10
1										
2	1	1								
3	1	2	3	2	1	1				
4	1	3	5	5	3	4	4	4	3	3
5	1	3	7	12	12	7	8	8	7	6
6	1	3	8	17	26	26	18	20	24	20
7	1	3	8	20	41	59	55	40	48	58
8	1	3	8	21	50	96	132	121	92	112
9	1	3	8	21	54	124	225	293	265	213
10	1	3	8	21	55	138	303	523	652	581
11	1	3	8	21	55	143	350	735	1207	1448
12	1	3	8	21	55	144	370	879	1768	2773
13	1	3	8	21	55	144	376	952	2190	4220
14	1	3	8	21	55	144	377	979	2433	5413
15	1	3	8	21	55	144	377	986	2540	6177
16	1	3	8	21	55	144	377	987	2575	6561

Table 1: Values of $|D_k^{-1}(d)|$

gest that, as k grows,

$$(|D_k^{-1}(1)|, |D_k^{-1}(2)|, |D_k^{-1}(3)|, \dots)$$

converges to a sequence starting with $(1, 3, 8, 21, 55, 144, 377)$, which equals the bisection $(F_2, F_4, F_6, F_8, F_{10}, F_{12}, F_{14})$ of the famous Fibonacci sequence $(F_0 = 0, F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3, \dots)$. A more careful investigation of Table 1 suggests that

$$|D_k^{-1}(d)| = \begin{cases} F_{2d} & \text{for } 1 \leq d \leq \lfloor k/2 \rfloor, \\ F_{2d} - 1 & \text{for } d = \lfloor k/2 \rfloor + 1, \text{ odd } k, \\ F_{2d} - 1 - k/2 & \text{for } d = \lfloor k/2 \rfloor + 1, \text{ even } k. \end{cases} \quad (3)$$

This is indeed so, which is one of the results in paper I. This is proved by introducing a *quasi-infinite* Bulgarian solitaire, obtained by fixing d and letting $k \rightarrow \infty$, and identifying recursive properties of the quasi-infinite game tree.

Thus, when $n = k(k+1)/2$, for $1 \leq d \leq \lfloor k/2 \rfloor$ the number of partitions with d moves to the staircase partition is the Fibonacci number F_{2d} .

2.4 Future research

Note that (3) determines the value of $|D_k^{-1}(d)|$ only for $1 \leq d \leq \lceil k/2 \rceil + 1$. A natural question for future research is therefore the enumeration of $|D_k^{-1}(d)|$ for *all* possible d , i.e., for $1 \leq d \leq k^2 - k$.

As mentioned in Section 2.1, when the number of cards n is not triangular, the game graph has one cycle per connected component and the game eventually reaches a cycle of partitions. Note that the fixed point partition for triangular n also constitutes a cycle (of length one). The question addressed in paper I can therefore be extended to the case when n is non-triangular. In other words, in Bulgarian solitaire with *any* number of cards, how many partitions are there with a given number of steps to a partition in a cycle?

3 About paper II

3.1 σ -Bulgarian solitaire

Many variants of Bulgarian solitaire have been suggested in the literature. See [8] for an extensive survey.

In 2016, Olson [19] presented a generalization of Bulgarian solitaire which we call σ -Bulgarian solitaire, in which multiple cards may be picked from a single pile. Specifically, the number of cards to pick from a pile is some function $\sigma : \mathbb{Z}_+ \rightarrow \mathbb{N}$ of the pile size, such that you pick $\sigma(h) \leq h$ cards from a pile of size h . The case $\sigma(h) = 1$ for all h reduces to the ordinary Bulgarian solitaire.

Just like the ordinary Bulgarian solitaire, the σ -Bulgarian solitaire is a deterministic process on a finite space, so cycles must necessarily occur. Olson gives bounds on maximum cycle lengths and conditions associated with the occurrence of isolated cycles.

When a σ -Bulgarian solitaire has a fixed-point (i.e. a cycle of length one) we call the fixed-point partition a *stable configuration*. Generally, partitions in a cycle (of any length) are called *recurrent configurations*.

Let us call σ *well-behaved* if

1. $\sigma(1) = 1$,
2. $\sigma(h)$ is a non-decreasing function,
3. $\bar{\sigma}(h) := h - \sigma(h)$ is a non-decreasing function.

The first condition says that from a pile with just a single card, you pick that card. The second condition says that you never pick fewer cards from a larger pile than from a smaller pile. The third condition says that the

number of unpicked cards are never fewer in the larger pile than in a smaller pile.

Paper II considers σ -Bulgarian solitaires for well-behaved σ . Note that the function $\sigma(h) = 1$ for all h that yields the ordinary solitaire is well-behaved.

3.2 Results for well-behaved σ -Bulgarian solitaire

The σ -Bulgarian solitaire is a wide generalization of the ordinary solitaire and it produces behaviour that deviates significantly from the original version. However, for well-behaved σ it turns out that a number of properties that are well-known to hold for ordinary Bulgarian solitaire generalize to well-behaved σ -solitaire, namely the following.

- *The dominance property.* If a configuration $\lambda = (\lambda_1, \lambda_2, \dots)$ is dominated by another configuration $\kappa = (\kappa_1, \kappa_2, \dots)$, in the sense that $\lambda_i \leq \kappa_i$ for all i , then this dominance relation is preserved as the solitaire is played in parallel from the two configurations.
- *A geometric interpretation.* Recall the geometric interpretation of a move of ordinary Bulgarian solitaire depicted in Figure 5, where the bottom layer is picked in each move. When generalizing to well-behaved σ -Bulgarian solitaire, certain layers (determined from σ) are picked.
- *Stable configurations are unique for any n for which a stable configuration exists.* To illustrate, an example of σ that is not well-behaved is $\sigma(1) = \sigma(3) = \sigma(5) = 1$, $\sigma(2) = 2$, $\sigma(4) = 3$ for $n = 5$. With this choice of σ , both $(3, 2)$ and $(4, 1)$ are stable configurations. This cannot happen if we impose the well-behaved criterion on σ . The game graph for this σ -Bulgarian solitaire is shown in Figure 8.
- *Stable configurations are convex.* Define a configuration λ as *convex* if $\lambda_i - \lambda_{i+1} \geq \lambda_{i+1} - \lambda_{i+2}$ for all $i \geq 1$. Then it holds that a configuration λ is convex if and only if it is a stable configuration of a well-behaved σ -Bulgarian solitaire. In the ordinary solitaire, the only stable configuration is the staircase partition, which clearly is convex (with equality in the definition).

In order to study asymptotics as $n \rightarrow \infty$, paper II considers a sequence of well-behaved σ_n , for $n = 1, 2, \dots$. Recall from Section 2.1 that the ordinary Bulgarian solitaire has a stable configuration if and only if the total number

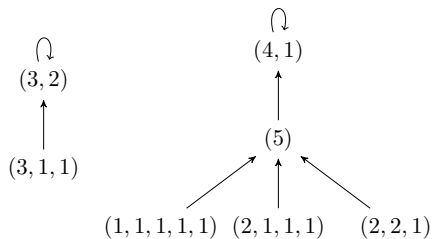


Figure 8: Game graph for a non-well-behaved σ -Bulgarian solitaire.

of cards n in the deck is a triangular number. Thus, the Young diagrams of stable configurations are staircase shaped and hence the limit shape as $n \rightarrow \infty$ is a triangle. When generalizing from ordinary Bulgarian solitaire to well-behaved σ_n -Bulgarian solitaire, the limit shapes that arise will not necessarily be triangular. In fact, in Theorem 3 we prove that *any* convex shape (with some properties) can be obtained as the limit shape of a suitably chosen family of well-behaved σ_n , for $n = 1, 2, \dots$:

Theorem 3. *Let $\phi : (0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ be a function and let $a_1, a_2, \dots \rightarrow \infty$ be any (positive) scaling factors such that a_n^2/n converges to some $c \geq 0$ as $n \rightarrow \infty$. Then the following are equivalent.*

- (a) *There is a sequence of well-behaved σ_n , $n = 1, 2, \dots$, such that ϕ is a stable-limit shape of (σ_n) under the scaling (a_n) .*
- (b) *ϕ is convex with $\int_0^\infty \phi(x) dx \leq 1$, and if $c > 0$ the right derivative $\phi'_R(x)$ is an integer multiple of c for any $x > 0$.*

One property of ordinary Bulgarian solitaire that does *not* generalize to well-behaved σ -Bulgarian solitaire is that, if a stable configuration exists, the game eventually reaches it. As we have seen, ordinary Bulgarian solitaire with a triangular number of cards always reaches the stable staircase partition. But the well-behaved σ -Bulgarian solitaire defined by $\sigma(h) = \lceil 3h/10 \rceil$ on $n = 11$ cards has the stable configuration $(5, 3, 2, 1)$ *as well as* the cycle

$$(6, 2, 2, 1) \mapsto (5, 4, 1, 1) \mapsto (6, 3, 2) \mapsto (4, 4, 2, 1) \mapsto (6, 2, 2, 1).$$

However, each of these recurrent configurations deviate from the stable configuration $(5, 3, 2, 1)$ with at most one card in each pile. Recall from Section 2.1 that the recurrent configurations in ordinary Bulgarian solitaire (with a non-triangular number of cards) deviate with at most one card per pile from a stable configuration. This led us to conclude that the limit shape

for ordinary Bulgarian solitaire is a straight line, as the deviations tend to zero as n tends to infinity.

For well-behaved Bulgarian solitaire, we believe that it holds in general that recurrent configurations' deviations also tend to zero as n tends to infinity, and that therefore the limit shape for recurrent configurations is the same as that for stable configurations. It is therefore left as a conjecture that the same limit shape holds for recurrent configurations, as for stable configurations:

Conjecture 1. *If ϕ is a limit shape of the stable configurations of a sequence of well-behaved σ_n , then ϕ is also a limit shape of the recurrent configurations.*

3.3 q_n -proportion Bulgarian solitaire

In order to calculate explicit limit shapes we make a canonical choice of a well-behaved σ , namely $\sigma(h) = \lceil qh \rceil$ for $q \in (0, 1]$. (It should be obvious that this function satisfies the conditions for being well-behaved; see the definition in Section 3.1.) In words, this form of σ defines a solitaire in which from each pile we pick a number of cards given by the proportion q of the pile size, rounded upward to the nearest integer. We will refer to this solitaire as *q -proportion Bulgarian solitaire*.

We may let the choice of q depend on n , in which case we write q_n . Note that for $q_n \leq 1/n$ only one card is picked in any pile. Thus by choosing $q_n \leq 1/n$ we obtain ordinary Bulgarian solitaire.

3.4 Results for q_n -proportion Bulgarian solitaire

In paper II, three different regimes for limit shapes of stable configurations of q_n -proportion Bulgarian solitaire are identified. A move of q -proportion Bulgarian solitaire involves rounding the number of picked cards in each pile to an integer. The three regimes differ in how much impact this rounding has on the result.

First, in case $nq_n^2 \rightarrow 0$ as $n \rightarrow \infty$, the rounding effect dominates and stable configurations have a triangular limit shape. This is a direct generalization of the limit shape result for the ordinary Bulgarian solitaire.

The second regime is when $nq_n^2 \rightarrow \infty$, in which case the rounding effect is negligible, and an exponential limit shape is obtained. With no rounding, pile sizes decrease geometrically with decay factor $1 - q_n$, which is the intuition behind this result.

The borderline regime when nq_n^2 tends to a constant C yields an infinite family of limit shapes (parameterized by C), which interpolate between the triangular shape of the first regime and the exponential shape of the second regime. See Figure 9.

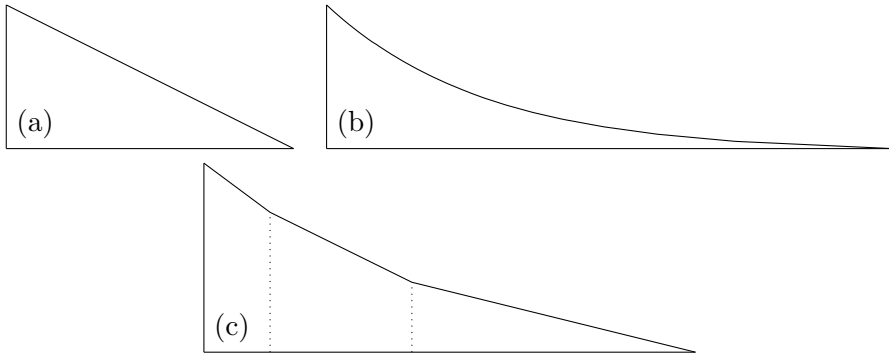


Figure 9: The three cases of limit shapes for stable configurations of q_n -proportion Bulgarian solitaire: (a) $q_n^2 n \rightarrow 0$ (triangular), (b) $q_n^2 n \rightarrow \infty$ (exponential), and (c) $q_n^2 n$ tends to a positive constant (a number of linear sections, here illustrated for three sections).

Although Conjecture 1 is not proved for general well-behaved σ -solitaire, we can prove the conjecture in the special cases of the two main regimes of q_n -proportion solitaire. In other words, we also prove in paper II that *recurrent* configurations of q_n -proportion solitaire have the same limit shape as *stable* configurations in the cases $q_n^2 n \rightarrow 0$ and $q_n^2 n \rightarrow \infty$.

3.5 Future research

A proof on Conjecture 1 remains to be found. In paper II we show that for well-behaved σ -Bulgarian solitaire, a configuration's total deviation from a stable configuration will often decrease but never increase during play, which is a first step towards a proof.

4 About paper III

4.1 p -random q -proportion Bulgarian solitaire

Popov [20] considered a random version of the ordinary Bulgarian solitaire defined by a probability $p \in (0, 1]$, such that one card from each pile is picked with probability p , independently of other piles. For $p = 1$ we obtain the ordinary Bulgarian solitaire.

In the spirit of Popov's p -random version of ordinary Bulgarian solitaire, paper III examines a p_n -random version of the q_n -proportion Bulgarian solitaire, in which the proportion q_n (rounded upward) of cards in a pile are only *candidates* to be picked, each of which is picked only with probability p_n , independently of all other candidate cards. This process is denoted by $\mathcal{B}(n, p_n, q_n)$. Note that in the special case of a fixed p and for $q_n \leq 1/n$, this process is equivalent to Popov's p -random Bulgarian solitaire.

The p_n -random q_n -proportion solitaire $\mathcal{B}(n, p_n, q_n)$ (with $p_n, q_n \in (0, 1]$) can be regarded as a Markov chain on the finite state-space $\mathcal{P}(n)$. Let us denote the sequence of visited states by $(\lambda^{(0)}, \lambda^{(1)}, \dots)$. In the truly random case of $p_n < 1$, it is straightforward to verify that this Markov chain is aperiodic and irreducible. It is well-known that an aperiodic and irreducible Markov chain on a finite state-space has a unique stationary distribution π and that starting from any initial state the distribution of the i th state $\lambda^{(i)}$ converges to π as i tends to infinity. We denote by π_{n, p_n, q_n} the stationary measure of the Markov chain $(\lambda^{(0)}, \lambda^{(1)}, \dots)$ on $\mathcal{P}(n)$ given by $\mathcal{B}(n, p_n, q_n)$ for $p_n < 1$. When we refer to a limit shape of the process $\mathcal{B}(n, p_n, q_n)$ for $p_n < 1$ as n grows to infinity, we shall mean the limit shape of the stationary measure π_{n, p_n, q_n} . The intuitive sense of this concept is that *when the solitaire is played on a sufficiently large number of cards for sufficiently long the configuration will almost surely be very close to the limit shape after suitable downscaling*.

4.2 Results

Recall from Section 3.4 the q_n -proportion Bulgarian solitaire, where the limit shape is triangular when $q_n^2 n \rightarrow 0$, exponential when $q_n^2 n \rightarrow \infty$ and an interpolation between the two when $q_n^2 n \rightarrow C > 0$.

The p_n -random q_n -proportion Bulgarian solitaire seems to share this property of three regimes of limit shapes.

In the p_n -random q_n -proportion solitaire, the number of candidate cards in a pile is the proportion q_n of the pile size rounded upwards to the nearest integer. As in the nonrandom version, the three regimes differ in how much impact this rounding has on the result. It turns out that when $p_n q_n^2 n \rightarrow \infty$, the effect of the rounding is negligible. With no rounding, expected pile sizes decrease geometrically with decay factor $1 - q_n$, so we expect an exponential limit shape.

The focus in paper III is the proof of the exponential limit shape of the p_n -random q_n -candidate Bulgarian solitaire, i.e. the case $p_n q_n^2 n \rightarrow \infty$ as $n \rightarrow \infty$. However, due to the employed proof technique we establish the

stronger statement that the limit shape holds even without sorting the piles of a configuration according to size to create a partition in $\mathcal{P}(n)$. We will instead require the stronger condition $p_n q_n^2 n / \log n \rightarrow \infty$ as $n \rightarrow \infty$:

Theorem 1. *For each positive integer n , pick q_n and p_n with $0 < p_n, q_n \leq 1$ and a (possibly random) initial configuration $\lambda^{(0)} \in \mathcal{P}(n)$. Let $(\lambda^{(0)}, \lambda^{(1)}, \dots)$ be the Markov chain on $\mathcal{P}(n)$ defined by $\mathcal{B}(n, p_n, q_n)$, and denote its stationary measure by π_{n, p_n, q_n} . Suppose*

$$p_n q_n \rightarrow 0 \quad \text{and} \quad \frac{p_n q_n^2 n}{\log n} \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

Then π_{n, p_n, q_n} has the limit shape e^{-x} under the scaling $a_n = (p_n q_n)^{-1}$.

As we see, the scaling factor in Theorem 1 is $(p_n q_n)^{-1}$. It is therefore natural to require $p_n q_n \rightarrow 0$, since $p_n q_n$ being bounded away from zero would mean that $(p_n q_n)^{-1}$ is bounded and hence cannot transform the jumpy boundary diagrams into a smooth limit shape.

The number of picked cards from the candidate cards in a pile is binomially distributed, and so it is not surprising that the proof of Theorem 1 relies heavily on the use of Chernoff bounds.

4.3 Future research

We conjecture that Theorem 1 holds also when the condition $np_n q_n^2 / \log n \rightarrow \infty$ is replaced by the weaker condition $np_n q_n^2 \rightarrow \infty$, as our simulations suggest that if we have $np_n q_n^2 \rightarrow \infty$ but not $np_n q_n^2 / \log n \rightarrow \infty$, limit shapes are still attained.

Also, paper III focuses on the exponential regime $p_n q_n^2 n \rightarrow \infty$. We also conjecture that the limit shapes in the p_n -random q_n -proportion Bulgarian solitaire are the same as in the deterministic q -proportion Bulgarian solitaire developed in paper II. Specifically, we conjecture the following.

- If $p_n q_n^2 n \rightarrow 0$ as $n \rightarrow \infty$, the limit shape of the p_n -random q_n -proportion Bulgarian solitaire is triangular.
- If $p_n q_n^2 n \rightarrow C$ as $n \rightarrow \infty$ for some constant $C > 0$, the limit shape of the p_n -random q_n -proportion Bulgarian solitaire is a piecewise linear shape that depends on the value of C .

Proofs of the above conjectures remain to be found.

5 About paper IV

5.1 Posets

I borrow the following definition from Stanley [26]: A *partially ordered set* (or *poset* for short) is a set P together with a binary relation denoted $\leq$ satisfying the following three axioms:

1. For all $t \in P$, $t \leq t$ (*reflexivity*).
2. If $s \leq t$ and $t \leq s$, then $s = t$ (*antisymmetry*).
3. If $s \leq t$ and $t \leq u$, then $s \leq u$ (*transitivity*).

We use the obvious notation $s < t$ to mean $s \leq t$ and $s \neq t$.

If $s, t \in P$ then we say that t *covers* s (or s is covered by t), denoted $s < t$ or $t > s$, if $s < t$ but there is no u with $s < u < t$.

Just like Young diagrams are convenient ways of representing integer partitions graphically, there is a convenient way of representing posets. The *Hasse diagram* of a poset P is the graph whose vertices are the elements of P , and whose edges are the cover relations and such that if $s < t$ then t is drawn “above” s (i.e., with a higher vertical coordinate).

The theory of posets plays an important role in enumerative combinatorics. To get some feeling for their structure, let us consider an example.

Example 1. Let $n \in \mathbb{N}$ and $[n] := \{1, 2, \dots, n\}$. We can make the set $2^{[n]}$ of all subsets of $[n]$ into a poset B_n by defining $S \leq T$ in B_n if $S \subseteq T$ as sets. One says that B_n consists of the subsets of $[n]$ “ordered by inclusion”. The Hasse diagram of B_4 can be seen in Figure 10.

Let $I \subseteq \mathbb{Z}$ be a (possibly infinite) interval of the integers. An *I -graded poset* Ω is a countable (or finite) poset together with a surjective map $\rho: \Omega \rightarrow I$, called the *rank function*, such that

- $u < v$ implies $\rho(u) < \rho(v)$, and
- $u < v$ implies $\rho(v) = \rho(u) + 1$.

We can partition $\Omega = \bigcup_{i \in I} \Omega_i$ into its *level sets* $\Omega_i = \rho^{-1}(i)$.

Clearly, the poset B_n in Example 1 is $\{0, 1, \dots, n\}$ -graded with rank function ρ being the set cardinality $\rho(u) = |u|$.

5.2 Young’s lattice and Hasse walks

One certain class of partially ordered sets are called *lattices*. See for example [26] for a formal definition.

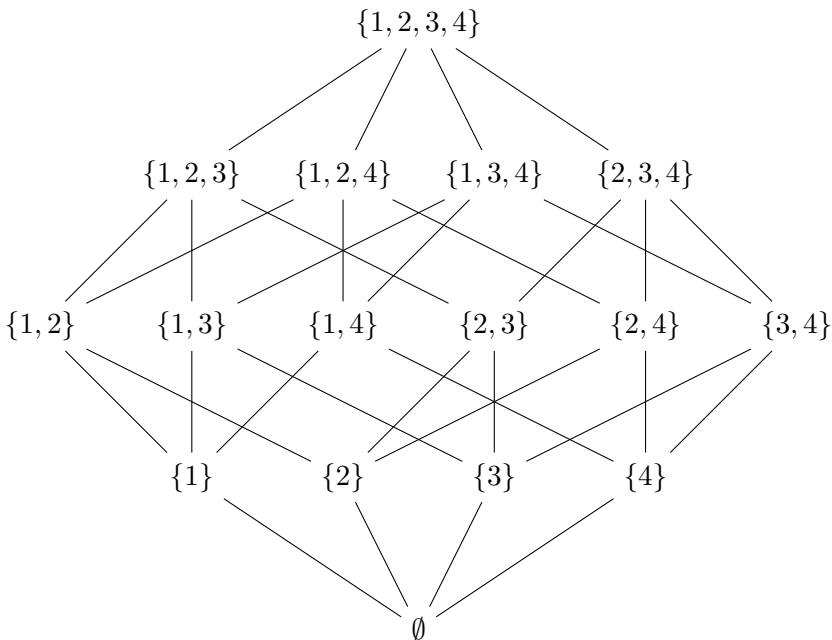


Figure 10: Hasse diagram of the poset B_4 .

Young's lattice $\mathcal{Y}$ is a poset formed by all integer partitions ordered by inclusion of their Young diagrams. For any integer partition λ , define $\lambda_i = 0$ for $i > N(\lambda)$. Let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_{N(\lambda)}, 0, 0, \dots)$ and $\mu = (\mu_1, \mu_2, \dots, \mu_{N(\mu)}, 0, 0, \dots)$ be two integer partitions. Then $\lambda \leq \mu$ in $\mathcal{Y}$ if $\lambda_i \leq \mu_i$ for any i , in other words, the Young diagram of λ fits inside that of μ . The first four levels of Young's lattice can be seen in Figure 11. Clearly, $\mathcal{Y}$ is a $\mathbb{Z}_+$ -graded poset with level sets $\Omega_i = \mathcal{P}(i)$, $i \in \mathbb{Z}_+$.

In this ordering, $\lambda < \mu$ if and only if μ can be obtained from λ by adding a square to any of its inner corners.

As mentioned in Section 1.4, the paper [9] studies stochastic processes on Young diagrams where each step entails either the birth of a square (a birth process) or the combined birth of one square and the death of another (a birth-and-death process). Births occur in inner corners and deaths in outer corners. A *Hasse walk* is a walk along the edges in a Hasse diagram, in other words, a walk along the covering relations in a poset [25, 26]. Therefore, the processes considered in [9] can be regarded as random Hasse walks on Young's lattice $\mathcal{Y}$. Walks that go steadily upwards in $\mathcal{Y}$ are birth processes and those that alternately go up and down are birth-and-death processes.

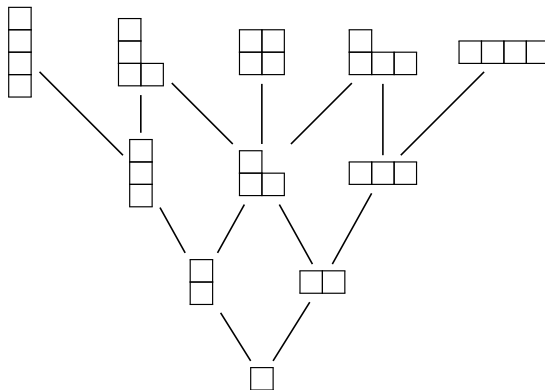


Figure 11: Hasse diagram of the first four levels of Young's lattice.

Paper IV develops a framework for such unidirected and alternatingly directed random Hasse walks on general graded posets.

5.3 Motivation

The motivation behind paper IV is a conjecture in Eriksson and Sjöstrand's paper [9]. We will now describe the birth step, called $\text{ROW}(\mu)$, and the death step, called DEROW , involved in this conjecture. Consider a current Young diagram λ .

- The action of the death step DEROW is defined by choosing a non-empty row i uniformly at random and removing the corresponding outer corner.
- The action of the birth step $\text{ROW}(\mu)$ is defined as follows: With probability μ create a new row of length 1. Otherwise make a uniformly random choice of a row and insert a new square at the corresponding inner corner.

The process induced by the birth step $\text{ROW}(\mu)$ will also be referred to as $\text{ROW}(\mu)$. The birth-and-death process with alternating $\text{ROW}(\mu)$ -births and DEROW -deaths will be referred to as $\text{DEROW-ROW}(\mu)$.

Eriksson and Sjöstrand established that, for any fixed n and μ_n , the limit shape for the distribution $\kappa^{(n)}$ on $\mathcal{P}(n)$ induced by starting with a single square and running $n - 1$ steps of $\text{ROW}(\mu_n)$ is $y = e^{-x}$ under the scaling $a_n = 1/\mu_n$ (where μ_n must fulfill some asymptotic properties as $n \rightarrow \infty$).

As for DEROW-ROW(μ), they proved explicit formulas for the stationary distributions π_n and π_{n-1} on $\mathcal{P}(n)$ and $\mathcal{P}(n-1)$, respectively, when DEROW-ROW(μ) toggles between $\mathcal{P}(n)$ and $\mathcal{P}(n-1)$. They conjectured that these stationary distributions have the same limit shape $y = e^{-x}$ as for ROW(μ). This conjecture inspired to compare the distributions $\kappa^{(n)}$ and π_n programmatically. The simulations suggested that $\kappa^{(n)}$ actually *coincides* with π_n for any n . If a proof of this were found, it would solve the limit shape conjecture for $\kappa^{(n)}$.

This observation motivated not only the efforts to find a proof for this, but also the investigation of the conditions under which this holds in general on $\mathcal{Y}$. In other words,

- Q1. *When does the distribution induced by a birth process on $\mathcal{Y}$ coincide with the corresponding stationary distribution of a birth-and-death process?*

5.4 Results

Although question Q1 is asked for processes on Young's lattice, paper IV develops a framework for general discrete-time Markov chains on graded posets where the transitions are taken along the covering relations in the poset. The answer to Q1 then follows by applying this framework to the case when the poset is Young's lattice. One way of formulating this answer is

- A1. *When the stationary distribution on level n for the birth-and-death process when toggling between levels n and $n-1$ coincides with the stationary distribution on level n when toggling between levels n and $n+1$, for each n .*

The one-directed Markov chains considered in paper IV are called *up chains* (a generalization of birth processes on Young's lattice) or *down chains*. The alternatingly directed Markov chains that toggle between two adjacent rank levels in the poset are called *up-and-down chains* (a generalization of birth-and-death processes on Young's lattice).

A birth step is called an *up rule* and a death step is called a *down rule* on general posets. Paper IV also settles the questions whether the reverse of an up chain is a down chain for some down rule and whether there exists an up or down chain at all if the rank function is not bounded.

5.5 Future research

Apart from the directions for future research outlined in Section 6 (Discussion) in paper IV, a possible application of this framework is the extension of the processes in [9] from integer partitions to integer compositions.

A *composition of n* is a nonempty sequence $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_N)$ of positive integers, of length $N = N(\alpha)$, satisfying $\sum_{i=1}^{N(\alpha)} \alpha_i = n$. For any positive integer n , let $\mathcal{C}_n = \{\text{compositions of } n\}$ and $\mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2 \cup \dots$.

One can turn $\mathcal{C}$ into a partially ordered set by introducing a cover relation $\triangleleft$ and obtaining the partial order by transitive closure of this covering relation. A number of such relations on $\mathcal{C}$ have been studied in the past. Bergeron, Bousquet-Mélou and Dulucq [5] were the first to study $\mathcal{C}$ as a poset. They use the cover relation that $\alpha \triangleleft \beta$ if β is obtained either by adding 1 to a part of α , or by adding a part of size 1 to α . The resulting poset is usually denoted $\mathfrak{B}\mathfrak{B}\mathfrak{D}$. (For other partial orders on $\mathcal{C}$, see for example [6, 23, 24].)

Young diagrams of integer partitions can be extended to *composition diagrams* where the composition diagram of α are left-justified rows of squares such that the i th row from the bottom has length α_i . For example, the composition diagram for $(1, 4, 2, 3)$ is



The birth and death steps on Young diagrams studied in [9] may be extended to integer compositions such that $\alpha \triangleleft \beta$ in $\mathfrak{B}\mathfrak{B}\mathfrak{D}$ if β can be reached from α by a birth step and α can be reached from β by a death step. For example, we may define the death step DEROW and the birth step ROW(μ) in the refined world of composition diagrams as follows. Assume a current composition diagram $\alpha = (\alpha_1, \dots, \alpha_{N(\alpha)})$.

DEROW: Choose a non-empty row i uniformly at random, and remove the right-most square in that row. If the chosen row has only one square (i.e. if $\alpha_i = 1$), contract that row.

ROW(μ): With probability μ create a new row of length 1 and insert this in any of the available $N(\alpha) + 1$ positions (i.e. above any of the current $N(\alpha)$ rows or before the first row), each with equal probability $\frac{1}{N(\alpha)+1}$.

Otherwise (i.e. with probability $1 - \mu$) make a uniformly random choice of a row i among the $N(\alpha)$ non-empty rows and insert a new square at the end of that row.

References

- [1] E. Akin and M. Davis, *Bulgarian solitaire*, The American Mathematical Monthly **92** (1985), no. 4, pp. 237–250.
- [2] G.E. Andrews, *The theory of partitions*, no. 2, Cambridge university press, 1998.
- [3] G.E. Andrews and K. Eriksson, *Integer partitions*, Cambridge University Press, 2004.
- [4] H.J. Bentz, *Proof of the Bulgarian solitaire conjectures*, Ars Combin. **23** (1987), 151–170.
- [5] F. Bergeron, M. Bousquet-Mélou, and S. Dulucq, *Standard paths in the composition poset*, Annales des sciences mathématiques du Québec, vol. 19, Université du Québec à Montréal, Département de mathématiques et informatique, 1995, pp. 139–151.
- [6] A. Björner and R.P. Stanley, *An analogue of Young’s lattice for compositions*, arXiv preprint math.CO/0508043 (2005).
- [7] J. Brandt, *Cycles of partitions*, Proc. Amer. Math. Soc. **85** (1982), no. 3, pp. 483–486.
- [8] V. Drensky, *The Bulgarian solitaire and the mathematics around it*, Math. and Education in Math., Proc. of the 44-th Spring Conf. of the Union of Bulgar. Mathematicians, SOK-Kamchia, April 2–6, 2015 (2015), 79–91.
- [9] K. Eriksson and J. Sjöstrand, *Limiting shapes of birth-and-death processes on Young diagrams*, Adv. in Appl. Math. **48** (2012), no. 4, 575 – 602.
- [10] G. Etienne, *Tableaux de Young et solitaire bulgare*, J. Combin. Theory Ser. A **58** (1991), no. 2, 181–197.
- [11] W.J. Ewens, *Mathematical population genetics 1: theoretical introduction*, vol. 27, Springer Science & Business Media, 2012.
- [12] C. Greene, A. Nijenhuis, and H.S. Wilf, *Another probabilistic method in the theory of young tableaux*, Journal of Combinatorial Theory, Series A **37** (1984), no. 2, 127 – 135.

- [13] J.R. Griggs and C.C. Ho, *The cycling of partitions and compositions under repeated shifts*, Adv. in Appl. Math. **21** (1998), no. 2, 205–227.
- [14] B. Hopkins, *30 years of Bulgarian solitaire*, College Math. J. **43** (2012), no. 2, 136–141.
- [15] B. Hopkins and J.A. Sellers, *Exact enumeration of Garden of Eden partitions*, Integers **7** (2007), no. 2, A19.
- [16] K. Igusa, *Solution of the Bulgarian solitaire conjecture*, Math. mag. **58** (1985), no. 5, 259–271.
- [17] W. Jockusch, J. Propp, and P. Shor, *Random domino tilings and the arctic circle theorem*, arXiv preprint math/9801068 (1998).
- [18] B.F. Logan and L.A. Shepp, *A variational problem for random young tableaux*, Advances in mathematics **26** (1977), no. 2, 206–222.
- [19] J.S. Olson, *Variants of Bulgarian solitaire*, Integers **16** (2016), A8.
- [20] S. Popov, *Random Bulgarian solitaire*, Random Structures Algorithms **27** (2005), no. 3, 310–330.
- [21] H. Rost, *Non-equilibrium behaviour of a many particle process: Density profile and local equilibria*, Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete **58** (1981), no. 1, 41–53.
- [22] T. Seppäläinen, *Hydrodynamic scaling, convex duality, and asymptotic shapes of growth models*, (1996).
- [23] J. Snellman, *A poset classifying non-commutative term orders*, Discrete Mathematics and Theoretical Computer Science **301** (2001), 314.
- [24] ———, *Saturated chains in composition posets*, arXiv preprint math/0505262 (2005).
- [25] R.P. Stanley, *Differential posets*, J. Amer. Math. Soc. **1** (1988), no. 4, 919–961.
- [26] ———, *Enumerative combinatorics, vol. 1*, Wadsworth and Brooks/Cole, Pacific Grove, CA, 1986; second printing, 1996.
- [27] M. Szalay and P. Turán, *On some problems of the statistical theory of partitions with application to characters of the symmetric group. i*, Acta Mathematica Academiae Scientiarum Hungaricae **29** (1977), no. 3–4, 361–379.

- [28] H.N.V. Temperley, *Statistical mechanics and the partition of numbers. ii. the form of crystal surfaces*, Proc. Cambridge Philos. Soc, vol. 48, Cambridge Univ Press, 1952, pp. 683–697.
- [29] N.B. Vassilev, V.L. Gutenmaher, J.M. Rabbot, and A.L. Toom, *Mathematical olympiads by mail, 2nd ed.*, Ph.D. thesis, Nauka, Moscow, 1986.
- [30] A. Vershik, *Statistical mechanics of combinatorial partitions, and their limit shapes*, Funct. Anal. Appl. **30** (1996), no. 2, 90–105.
- [31] A. Vershik and S. Kerov, *Asymptotics of the Plancherel measure of the symmetric group and the limiting form of young tables*, Soviet Math. Dokl., vol. 18, 1977, p. 38.
- [32] ———, *Asymptotic of the largest and the typical dimensions of irreducible representations of a symmetric group*, Funct. Anal. Appl. **19** (1985), no. 1, 21–31 (English).
- [33] A. Vershik, *Asymptotic combinatorics and algebraic analysis*, pp. 1384–1394, Birkhäuser Basel, Basel, 1995.
- [34] Y. Yakubovich, *Ergodicity of multiplicative statistics*, J. Combin. Theory, Series A **119** (2012), no. 6, 1250–1279.

Paper I

LEVEL SIZES OF THE BULGARIAN SOLITAIRE GAME TREE

HENRIK ERIKSSON AND MARKUS JONSSON

ABSTRACT. Bulgarian solitaire is a dynamical system on integer partitions of n which converges to a unique fixed point if $n = 1 + 2 + \dots + k$ is a triangular number. There are few results about the structure of the game tree, but when k tends to infinity the game tree itself converges to a structure that we are able to analyze. Its level sizes turns out to be a bisection of the Fibonacci numbers. The leaves in this tree structure are enumerated using Fibonacci numbers as well. We also demonstrate to which extent these results apply to the case when k is finite.

1. INTRODUCTION

The game of Bulgarian solitaire works as follows: First divide a finite deck of n identical cards into piles. A move consists in removing one card from each pile and forming a new pile. This operation is repeated over and over. We pay attention only to how many piles there are of each positive size, ignoring the locations of the piles. Thus, each position of the game is an integer partition of n whose parts are the pile-sizes, and each move is an operation on the set of partitions inducing a finite dynamical system. See for example [2, 4] for the fascinating history of Bulgarian solitaire.

In terms of Young diagrams of integer partitions, pile-sizes correspond to rows in the diagram and the move is to delete the first column and add it as a new row.

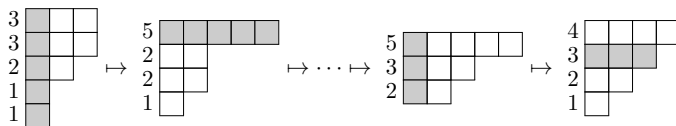


FIGURE 1. A sequence of positions converging to the fixed point.

The illustrating game for $n = 10$ in Figure 1 converges in twelve moves to the staircase shape of the partition $(4, 3, 2, 1)$, which is a fixed point. In the case when $n = 1 + 2 + \dots + k$ is a triangular number, all games will converge to the fixed point $(k, k - 1, \dots, 1)$ (see for example [1, 8]).

Let $\mathcal{P}(n)$ be the set of integer partitions of n . Let B be the operation on $\mathcal{P}(n)$ such that $B(\lambda)$ is the partition resulting from a move of Bulgarian solitaire on λ . The *game graph* is a directed graph where $\mathcal{P}(n)$ are the nodes and where there is an edge from $\lambda \in \mathcal{P}(n)$ to $\lambda' \in \mathcal{P}(n)$ if $B(\lambda) = \lambda'$. When the game graph is a tree, we call it a *game tree*.

For non-triangular n , the game has no fixed point. Instead the game graph has cycles, one for each connected component, and the game always reaches some cycle [1]. The number of cycles and their sizes are counted in [1]. The game graph analysis in [6] mainly characterizes and enumerates partitions with no preimage under B (called *Garden of Eden partitions*) and [5] investigates their relation to the cycles.

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In this paper we will consider the case when $n = 1 + 2 + \cdots + k$ is a triangular number. Since all games will converge to the staircase shaped fixed point, the game graph is a directed tree whose leaves are the Garden of Eden partitions and whose root is the staircase partition. The height of this tree is known [3] to be $k(k - 1)$, but to our knowledge the level size of the tree at a given level d has not been studied before. Paradoxically, the game tree's being finite is what makes further analysis difficult. We will show how a quasi-infinite version of the game, obtained by fixing the level d and letting $k \rightarrow \infty$, makes it possible to study the level sizes.

2. REVERSED BULGARIAN SOLITAIRE

By reversing all arrows in the directed game tree and labelling as below, we get *reversed Bulgarian solitaire*, starting with the staircase partition. A move in the reversed game is the following:

$\xrightarrow{i}$: Delete row i . Add it as the leftmost column.

Such a move can be made if and only if the length of the deleted row is greater than or equal to the height of the Young diagram after the deletion. In other words, if λ is a partition and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{N(\lambda)}$ are its parts, the move $\lambda \xrightarrow{i}$ is only possible if $\lambda_i \geq N(\lambda) - 1$.

The reversed game tree for $k = 4$ ($n = 10$) is shown in Figure 2. Recall that edge labels indicate which row is being played (i.e., which row is being deleted in the move)¹. For reasons of space, in Figure 2 we represent positions as horizontal vectors. In the following, we will always represent positions as vertical vectors, so as to match the notion of Young diagram row lengths in Figure 1. Deleting the first row in the staircase partition leads back to the staircase partition, so in Figure 2 the subtree reached by an initial $\xrightarrow{1}$ move is identical to the whole tree and the edge is a loop in disguise. However, we prefer to keep it a tree.

The reverse of the game in Figure 1 may be written as

$$\begin{array}{cccccccccccccccccccc} \langle 4 \rangle & \langle 5 \rangle & \langle 4 \rangle & \langle 4 \rangle & \langle 3 \rangle & \langle 4 \rangle & \langle 5 \rangle & \langle 4 \rangle & \langle 3 \rangle & \langle 4 \rangle & \langle 5 \rangle & \langle 4 \rangle & \langle 3 \rangle & \langle 4 \rangle & \langle 5 \rangle & \langle 3 \rangle \\ \langle 3 \rangle & 2 & \langle 3 \rangle & 1 & 3 & 1 & 2 & 1 & \langle 3 \rangle & 1 & \langle 4 \rangle & 1 & \langle 3 \rangle & 1 & \langle 3 \rangle & 1 & \langle 4 \rangle & 1 & 2 & 1 & 3 \\ 2 & \rightarrow & \langle 2 \rangle & \rightarrow & 1 & \rightarrow & 2 & \rightarrow & 1 & \rightarrow & 2 & \rightarrow & 1 & \rightarrow & 2 & \rightarrow & 1 & \rightarrow & 2 & \rightarrow & 1 & \rightarrow & 2 \\ 1 & & & & 1 & & 2 & & 1 & & 2 & & 1 & & 2 & & 1 & & 2 & & 1 & & 2 & & 1 \end{array} \quad (2.1)$$

Numbers within the angle brackets $\langle \rangle$ are playable, that is, at least $N(\lambda) - 1$, and we shall refer to them as *bracketed* numbers. Clearly, the start vector representing the staircase partition always has two bracketed numbers.

In our example in Figure 1 we chose a Garden of Eden partition $(3, 3, 2, 1, 1)$ as the starting position. Recall from the introduction that the height of the game tree is $k(k - 1)$ when the number of cards n is triangular, $n = 1 + 2 + \cdots + k$. Since the sequence in (2.1) has length $12 = 4 \cdot 3$, this sequence represents the tree height, in other words, λ is not only a Garden of Eden partition but also has the maximum number of moves to the fixed point (and is therefore at the bottommost level in the reversed game tree in Figure 2).

Using the introduced bracket notation, the move rule may now be expressed as follows.

Rule for $\xrightarrow{i}$ in reversed Bulgarian solitaire

1. Delete the bracketed i th number p .
 2. Increase all remaining numbers by 1.
 3. Add trailing 1s as necessary to make the vector length equal to p .
 4. Bracket all numbers greater than or equal to $p - 1$.
- (R1)

¹If the length of a playable row is repeated in a partition, then playing any of these rows is represented by playing the uppermost row of this length.

LEVEL SIZES OF THE BULGARIAN SOLITAIRE GAME TREE

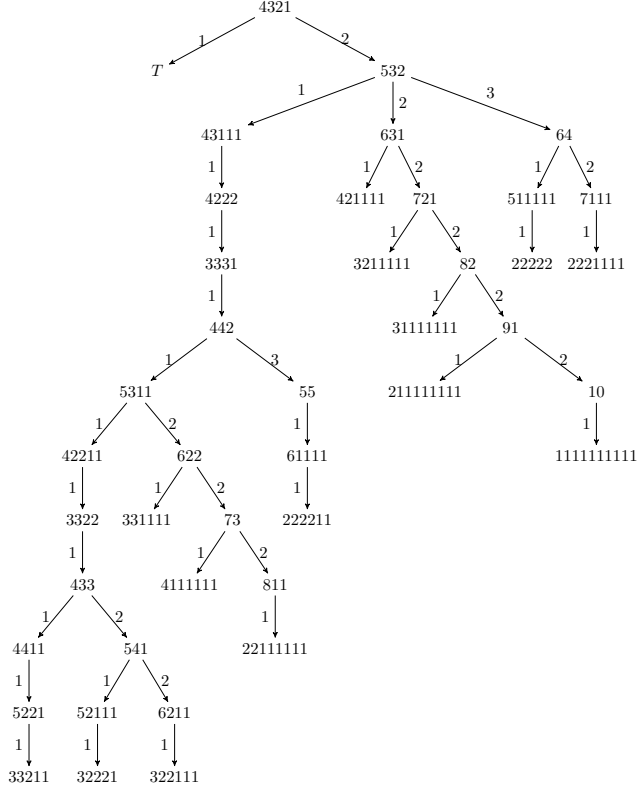


FIGURE 2. Reversed game tree T for $k = 4$.

We will denote a sequence of played rows by $[r_1, r_2, \dots]$ or simply $[r_1 r_2 \dots]$. Such a sequence may also be referred to as a *play sequence*. Also, we will use exponents to mean repeated play of a row number: $r^s = \underbrace{r, r, \dots, r}_{s \text{ times}}$. Thus, the play sequence in (2.1) is denoted $[2, 1^{11}]$.

In our small example, the partitions in the game soon lost all similarity with the staircase shape. Now we will investigate what happens as k grows. To this end, instead of writing down the lengths of the rows, we will note the difference: row length minus original row length in the staircase (where we append zeros as necessary to compute the difference).

This is what the sequence (2.1) looks like in this *difference notation*:

$$\begin{array}{cc} \langle 0 \rangle & \langle 1 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle -1 \rangle & \langle 0 \rangle & \langle 1 \rangle & \langle 0 \rangle & \langle -1 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 1 \rangle & -1 & \\ \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 1 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 1 \rangle & \langle 1 \rangle & \langle 0 \rangle & \langle 0 \rangle & \\ 0 & -1 & 0 & 1 & 0 & -1 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & \end{array}$$

To see more clearly what happens as k grows, below follows an example for a larger k .

Example 2.1. For $k = 8$ ($n = 36$), the play sequence $[2, 1^6]$ yields

$$\begin{array}{cccccccc}
 \langle 0 \rangle & \langle 1 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle & \langle 0 \rangle \\
 \langle 0 \rangle & \langle 0 \rangle & 0 & 0 & 0 & 0 & 0 & -1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & \xrightarrow{2} 0 & \xrightarrow{1} 0 & \xrightarrow{1} 0 & \xrightarrow{1} 0 & \xrightarrow{1} -1 & \xrightarrow{1} 0 & \xrightarrow{1} 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & -1 & -1 & 0 & 1 & 0 & 0 \\
 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\
 0 & & 1 & & & & &
 \end{array}$$

Each zero travel upwards, one step in each move, until it expires (either is deleted or is transformed into a 1). As we will see, it takes at least $k/2$ moves for the bottom zero in the start vector to expire, and neither the -1 below it or the numbers further down have any influence on the game so far (not being bracketed). Therefore, all game trees for $k \geq 8$ are identical up to and including level four. This observation brings us to introduce a quasi-infinite version of the game.

3. QUASI-INFINITE REVERSED BULGARIAN SOLITAIRE

In quasi-infinite difference notation, the start vector should now be an infinite sequence of zeros. However, we will write only two zeros and assume that there are infinitely many invisible non-bracketed zeros below them. Now, we can write the game $[21111 \dots]$ as

$$\begin{array}{ccccccc}
 \langle 0 \rangle & \xrightarrow{2} \langle 1 \rangle & \xrightarrow{1} \langle 0 \rangle & \xrightarrow{1} \langle 0 \rangle & \xrightarrow{1} \langle 0 \rangle & \xrightarrow{1} \dots \\
 \langle 0 \rangle & \langle 0 \rangle & 0 & \langle 0 \rangle & \langle 0 \rangle &
 \end{array}$$

With this notation, the game looks very different. In *quasi-infinite reversed Bulgarian solitaire* the start vector is $\begin{smallmatrix} \langle 0 \rangle \\ \langle 0 \rangle \end{smallmatrix}$ and the rules are a straightforward extension of the rules (R1) to quasi-infinite difference notation.

Rule for $\xrightarrow{i}$ in reversed quasi-infinite difference notation

1. Delete the bracketed i th number.
 2. Increase all numbers above it by 1 and make them bracketed.
 3. Bracket the new i th number (if there is one) if it differs at most 1 from the old one.
 4. If a zero was played, add zeros at the end so that there are two, and make them bracketed.
- (R2)

In the quasi-infinite version, the game $[2345]$ is represented as

$$\begin{array}{ccccccc}
 \langle 0 \rangle & \langle 1 \rangle & \langle 2 \rangle & \langle 3 \rangle & \langle 4 \rangle \\
 \langle 0 \rangle & \xrightarrow{2} \langle 0 \rangle & \xrightarrow{3} \langle 0 \rangle & \xrightarrow{4} \langle 1 \rangle & \xrightarrow{5} \langle 2 \rangle \\
 & & & \langle 0 \rangle & \langle 0 \rangle \\
 & & & & \langle 0 \rangle
 \end{array}$$

Note that the vectors are always strictly decreasing down to the zeros. This property follows by induction as it is invariant under the move rules.

Proposition 3.1. *The quasi-infinite reversed Bulgarian solitaire has the following properties.*

- (i) *Once 1 is played, only 1 can be played (until the start vector is reached).*
- (ii) *For $r \geq 2$, the play sequence $[234 \dots r 1^r]$ leads to the start vector. Together with $[1]$, these are the only play sequences with this property.*

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Proof. (i): It follows from (R2) that the position vector following a $\xrightarrow{1}$ move either has no bracketed number (so it is a leaf), one bracketed number (so another $\xrightarrow{1}$ must be played to reach the start vector) or is the start vector (in case a zero was played).

(ii): Playing $[234 \dots r]$, we reach

$$\begin{pmatrix} \langle r-1 \rangle \\ \langle r-2 \rangle \\ \vdots \\ \langle 1 \rangle \\ \langle 0 \rangle \\ \langle 0 \rangle \end{pmatrix},$$

and then $[1^r]$ deletes the top number r times, yielding the start vector. From property (i) it is clear that the start vector cannot be obtained in any other way. $\square$

Figure 3 shows what the quasi-infinite game tree looks like for the first three moves. As

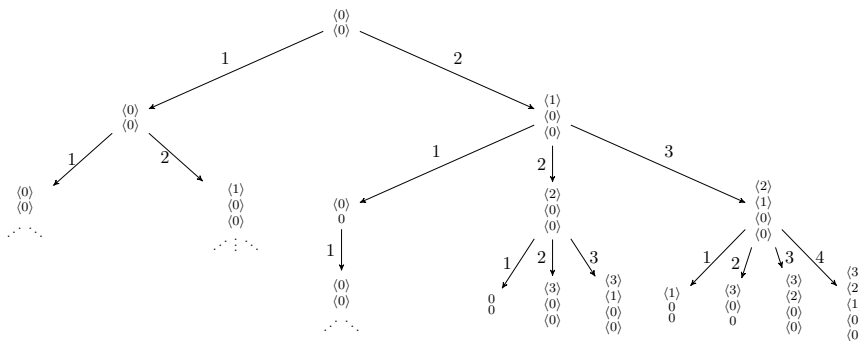


FIGURE 3. Quasi-infinite game tree for the first three moves. Every instance of a node $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ represents a new copy of the entire tree.

suggested by Proposition 3.1, this tree is recursive in several ways and we will use that to compute its *growth function*, $g = 1 + 2x + 5x^2 + 13x^3 + \dots$, the generating function for the size of its levels. Figure 3 shows only eight nodes on level three, but since the start vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ reappears after a 1 move, a copy of the whole tree hangs below it. This copy adds five nodes to the eight nodes shown on level three. Since a 1 move leads to the start vector, the set of all sequences starting with 1 contribute the term xg to the growth function g .

The start vector also reappears after the play sequence $[211]$, so another tree copy hangs below that node. Note that the reappearance does not mean that we are back to the staircase; this partition is different at the bottom. However, the quasi-infinite analysis is based on the assumption that k is large enough that this difference will not affect the game continuation in terms of number of playable rows. The level size of the quasi-infinite game tree will coincide with the finite case until a certain depth. This depth will be determined in Section 5.

We will now identify all other occurrences of the start vector in the game tree. By Proposition 3.1(ii) other games that lead to the start vector are $[231^3]$, $[2341^4]$, etc. The contribution to the growth function of these recursions is $x^3g + x^5g + \dots$.

Another recursiveness appears when we consider play sequences not including any 1 move. For these sequences, the game goes on as if the top row did not exist. The subtree of these play

sequences must therefore be isomorphic to the full tree, but only after the first move 2, which makes 3 playable. This gives us a term xg , corresponding to $[2(\dots \geq 2)]$, where $(\dots \geq r)$ denotes a (possibly empty) sequence of numbers r or greater.

The neglected top row is always playable, so the type $[2(\dots \geq 2)1]$, will contribute x^2g to the growth function.

Is it possible to end by more than one 1? Yes, if the two top rows are neglected until the end. Thus the type $[23(\dots \geq 3)11]$ contributes x^4g . Similarly for three or more top rows one gets $x^6g + x^8g + \dots$.

Table 3 presents the play sequences we have covered so far and their respective contribution to the growth function g .

Sequence	Contribution	Sequence	Contribution
$[1\dots]$	xg	$[2(\dots \geq 2)]$	xg
$[21^2]$	x^3g	$[2(\dots \geq 2)1]$	x^2g
$[231^3]$	x^5g	$[23(\dots \geq 3)1^2]$	x^4g
$[2341^4]$	x^7g	$[234(\dots \geq 4)1^3]$	x^6g
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$[234\dots j, 1^j]$	$x^{2j-1}g$	$[234\dots j(\dots \geq j)1^{j-1}]$	$x^{2(j-1)}g$

TABLE 3. Play sequences and their corresponding contribution to the growth function g . By virtue of Proposition 3.1(i), these sequences do in fact cover all possible play sequences. Adding all terms, together with the fact that level 0 (the root) has size 1, we obtain

$$g(x) = (2x + x^2 + x^3 + \dots)g(x) + 1 = \left(x + \frac{x}{1-x}\right)g(x) + 1. \quad (3.1)$$

We are now ready to formulate the main result, which involves the classical Fibonacci sequence ($F_0 = 0, F_1 = 1, F_2 = 1, F_3 = 2, \dots$).

Theorem 3.2. *The generating function $g(x) = g_0 + g_1x + g_2x^2 + \dots$ for the level sizes in the quasi-infinite reversed Bulgarian solitaire is*

$$g(x) = \frac{1-x}{1-3x+x^2} = 1 + 2x + 5x^2 + 13x^3 + \dots$$

where $g_m = F_{2m+1}$ is a bisection of the Fibonacci sequence and has the recurrence $g_m = 3g_{m-1} - g_{m-2}$ for $m \geq 2$.

Proof. The formula is a consequence of (3.1) and identifying coefficients yields the recurrence. This is the sequence A001519 in The On-Line Encyclopedia of Integer Sequences (OEIS) [7]. $\square$

4. HEIGHT AND LEAF COUNT

In the non-reversed game, the *height* of a position is defined as the number of moves until the fixed point is reached (for the first time). This corresponds to disregarding the left subtree in the reversed game (for example in Figure 2), i.e., considering the root and the subtree $[2\dots]$ only.

For the quasi-infinite tree in Figure 3, let $h(x) = h_0 + h_1x + h_2x^2 + \dots$ be the height generating function which counts the level sizes of the root node together with the subtree $[2\dots]$. (As pointed out before, in this subtree occurrences of the quasi-infinite start vector are not proper staircases, being different at the bottom.) It satisfies

$$g(x) = xg(x) + h(x).$$

LEVEL SIZES OF THE BULGARIAN SOLITAIRE GAME TREE

Using Theorem 3.2 we get

$$h(x) = \frac{(1-x)^2}{1-3x+x^2} = 1 + x + 3x^2 + 8x^3 + \dots$$

and identifying coefficients gives the recurrence

$$h_0 = 1, \quad h_1 = 1, \quad h_2 = 3, \quad h_m = 3h_{m-1} - h_{m-2}, m \geq 3.$$

We have $h_i = F_{2i}$, $i \geq 1$, so this is also a bisection of the Fibonacci sequence. It is A088305 in OEIS [7].

Recall that the *leaves* of a game tree are called Garden of Eden partitions. We now turn our attention to their enumeration. In Figure 2, we find one leaf on level 3, three leaves on level 4 and so on until level 12 where there are three leaves. Thus, $x^3 + 3x^4 + x^5 + \dots + 3x^{12}$ is the generating function that counts the leaves on each level in this tree. As before, we have disregarded the left subtree.

For the quasi-infinite tree in Figure 3, disregarding the left subtree, let $\ell(x) = \ell_0 + \ell_1 x + \ell_2 x^2 + \dots$ be the generating function for the number of leaves on each level. We have $\ell(x) = x^3 + 3x^4 + \dots$, where the first term counts [221] and the second term [2221], [2231], and [2321]. If we include the left subtree, we get another generating function $L(x) = L_0 + L_1 x + L_2 x^2 + \dots$ and as with g and h they are related by $L(x) = xL(x) + \ell(x)$.

All games are listed in Table 3 and we can count those that start with 2 and end in a leaf (no bracketed numbers). Games starting with [21²], [231³], [2341⁴], etc., may continue in any way to a leaf, so their contribution to the growth function is $x^3 L + x^5 L + \dots$. The game [2(... $\geq$ 2)1] ends in a leaf unless it is either [21] or of the type [23(... $\geq$ 3)1], so the contribution is $x^2 g - x^2 - x^3 g$. In the same way, the game type [23(... $\geq$ 3)11] contributes $x^4 g - x^4 - x^5 g$ etc. Adding all contributions, we get

$$\ell(x) = (x^3 + x^5 + \dots)L(x) + (x^2 - x^3 + x^4 - x^5 + \dots)g(x) - (x^2 + x^4 + x^6 + \dots).$$

Substituting $L(x) = \ell(x)/(1-x)$, using the expression for $g(x)$ in Theorem 3.2, and solving for $\ell(x)$ yields

$$\ell(x) = \frac{x^3 - x^4}{(1-x-x^2)(1-3x+x^2)} = x^3 + 3x^4 + 9x^5 + 25x^6 + \dots \quad (4.1)$$

The coefficient sequence $(\ell_1, \ell_2, \dots)$ is essentially A094292 in OEIS [7] (but with an additional initial zero) and has the explicit formula

$$\ell_i = \frac{F_{2i-2} - F_{i-1}}{2},$$

as is easily verified, for example by decomposing $\ell(x)$ in (4.1) into partial fractions and identifying coefficients in each part.

Since $L(x) = \ell(x)/(1-x)$, (4.1) yields $L(x) = x^3/((1-x-x^2)(1-3x+x^2))$. It is well known that dividing a generating function by $1-x$ gives the generating function for the cumulative sum. Thus, $L_m = \sum_{i=3}^m \ell_i$ is the number of leaves up to and including level m . The sequence $(L_0, L_1, \dots)$ is A056014 in OEIS and by decomposing $L(x)$ into partial fractions it is easily verified that

$$L_m = \sum_{i=3}^m \ell_i = \frac{F_{2m-1} - F_{m+1}}{2}.$$

5. THE FINITE CASE $n = k(k+1)/2$

We will now investigate what happens when we keep k fixed and increase the tree level d . In other words, we will see how far the results in the quasi-infinite case apply to the finite case, when the staircase partition has k rows. Playing 2 lengthens the top row by one and shortens the bottom row by one, but the other $k-2$ rows are unaffected. These will correspond to $k-2$ zeros in our representation. As we have seen in Section 3, only the two uppermost zeros can be involved in a move, so the number of zeros decreases by at most two every move. Therefore, up to and including level $\lfloor k/2 \rfloor$, the level sizes are the same as in the quasi-infinite game tree.

In the finite game tree, at level $\lfloor k/2 \rfloor$ the first $\lfloor k/2 \rfloor$ vector components are unaffected. For example, for $k = 6$, the first three components of the vectors at level three are as in Figure 3. The fourth component, however, is not zero but a negative number. We know that the sixth component at level 1 is -1 (cf. Example 2.1), which means that the fifth component at level 2 and the fourth component at level 3 are also negative. Thus, the play sequences $[2345]$, $[2344]$, $[2334]$ and $[2234]$ are illegal for $k = 6$ (k even). For $k = 7$ (k odd) the situation is similar. The first four vector components are the same as in the quasi-infinite game tree, but the fifth component in each vector at level 3 is negative and that makes $[2345]$ illegal. This observation generalizes to the following theorem.

Theorem 5.1. *When $n = k(k+1)/2$, the sizes of levels $0, 1, \dots, \lfloor k/2 \rfloor$ in the reversed Bulgarian solitaire game tree coincide with those of the quasi-infinite game tree but the next level, $\lfloor k/2 \rfloor + 1$, has fewer elements, 1 less for odd k and $1 + k/2$ less for even k .*

Thus, when $n = k(k+1)/2$, for $1 \leq m \leq \lfloor k/2 \rfloor$ the number of partitions with m moves of Bulgarian solitaire to the staircase partition is the Fibonacci number F_{2m} .

As we saw in Section 1, the game tree for the finite case has a finite height, $k(k-1)$. One may ask the question whether it is a subtree of the quasi-infinite tree. In other words, are all play sequences in the finite reversed game also legal in the quasi-infinite reversed game? The answer is no. The minimal counter-example is the play sequence $[211113]$ in the reversed game for $k = 4$ (which leads from the staircase partition $(4, 3, 2, 1)$ to the partition $(5, 5)$, see Figure 2). This play sequence is not legal in the quasi-infinite reversed game since $[21111]$ leads to the start vector in which the third row is not playable.

6. DISCUSSION

The current paper concerns Bulgarian solitaire in the case where the number of cards n is triangular, $n = k(k+1)/2$ for some positive integer k . Theorem 5.1 gives the number of partitions with m moves to the staircase partition for $1 \leq m \leq \lfloor k/2 \rfloor + 1$. A natural open question is the extension of this result to *all* m , i.e., to $m \leq k(k-1)$. Recall that the maximal play sequence length in this case is $k(k-1)$.

As mentioned in the introduction, when n is not triangular, the game graph has one cycle per connected component and the game eventually reaches a cycle of partitions. Note that the fixed point partition for triangular n also constitutes a cycle (of length one). The maximal sequence length to a cycle partition for any n is conjectured by Griggs and Ho [3].

For non-triangular n , one may also ask the question corresponding to the one addressed in the current paper: *How many* partitions are there with a given sequence length to a cycle partition? Hopkins and Jones [5] has data on the number of leaves (Garden of Eden partitions) in the game graph by distance to the cycle, but a full answer to this question is yet to be found.

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7. ACKNOWLEDGMENTS

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REFERENCES

- [1] J. Brandt, *Cycles of partitions*, Proc. Amer. Math. Soc. **85** (1982), no. 3, 483–486.
- [2] V. Drensky, *The Bulgarian solitaire and the mathematics around it*, Math. and Education in Math., Proc. of the 44-th Spring Conf. of the Union of Bulgar. Mathematicians, SOK-Kamchia, April 2–6, 2015, 79–91.
- [3] J.R. Griggs and C.C. Ho, *The cycling of partitions and compositions under repeated shifts*, Adv. Appl. Math. **21** (1998), no. 2, 205–227.
- [4] B. Hopkins, *30 years of Bulgarian solitaire*, College Math. J. **43** (2012), no. 2, 136–141.
- [5] B. Hopkins and M.A. Jones, *Shift-induced dynamical systems on partitions and compositions*, J. Comb. **13** (2006), no. 3, R80.
- [6] B. Hopkins and J.A. Sellers, *Exact enumeration of Garden of Eden partitions*, Integers **7** (2007), no. 2, A19.
- [7] OEIS Foundation Inc. (2017), The On-Line Encyclopedia of Integer Sequences, <http://oeis.org>.
- [8] N.B. Vassilev, V.L. Gutenmaher, J.M. Rabbot, and A.L. Toom, *Zaochnye matematicheskie olimpiady, 2-E Izd., Pererab., S II.* (Russian), Nauka, Moscow, 1986.

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Paper II

LIMIT SHAPES OF STABLE CONFIGURATIONS OF A GENERALIZED BULGARIAN SOLITAIRE

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Abstract

Bulgarian solitaire is played on n cards divided into several piles; a move consists of picking one card from each pile to form a new pile. In a recent generalization, σ -Bulgarian solitaire, the number of cards you pick from a pile is some function σ of the pile size, such that you pick $\sigma(h) \leq h$ cards from a pile of size h . Here we consider a special class of such functions. Let us call σ well-behaved if $\sigma(1) = 1$ and if both $\sigma(h)$ and $h - \sigma(h)$ are non-decreasing functions of h . Well-behaved σ -Bulgarian solitaire has a geometric interpretation in terms of layers at certain levels being picked in each move. It also satisfies that if a stable configuration of n cards exists it is unique. Moreover, if piles are sorted in order of decreasing size ($\lambda_1 \geq \lambda_2 \geq \dots$) then a configuration is convex if and only if it is a stable configuration of some well-behaved σ -Bulgarian solitaire. If sorted configurations are represented by Young diagrams and scaled down to have unit height and unit area, the stable configurations corresponding to an infinite sequence of well-behaved functions $(\sigma_1, \sigma_2, \dots)$ may tend to a limit shape ϕ . We show that every convex ϕ with certain properties can arise as the limit shape of some sequence of well-behaved σ_n . For the special case when $\sigma_n(h) = \lceil q_n h \rceil$ for $0 < q_n \leq 1$, these limit shapes are triangular (in case $q_n^2 n \rightarrow 0$), or exponential (in case $q_n^2 n \rightarrow \infty$), or interpolating between these shapes (in case $q_n^2 n \rightarrow C > 0$).

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1. Introduction

The game of Bulgarian solitaire is played with a deck of n identical cards divided arbitrarily into several piles. A move consists of picking a card from each pile and letting these cards form a new pile. This move is repeated over and over again. For information about the earlier history of the Bulgarian solitaire and a summary of subsequent research, see reviews by Hopkins [9] and Drensky [4].

Let $\mathcal{P}(n)$ denote the set of integer partitions of n . An integer partition of n is a $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$ such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell > 0$ and $\sum_{i=1}^{\ell} \lambda_i = n$. For $i > \ell(\lambda)$ it will be convenient to define $\lambda_i = 0$. The number of non-zero parts of the partition λ is denoted by $\ell = \ell(\lambda)$. If piles of cards are sorted in order of decreasing size, any configuration of n cards can be regarded as an integer partition of n . A geometric shape arises when a configuration λ is represented by a Young diagram of unit squares in the first quadrant of a coordinate system for the real plane, such that the i th column has height λ_i . A move of the Bulgarian solitaire then has the geometric interpretation of picking the first (i.e., bottom) layer of the diagram and making it the new first column, left-shifting cards if needed so that the configuration remains sorted.

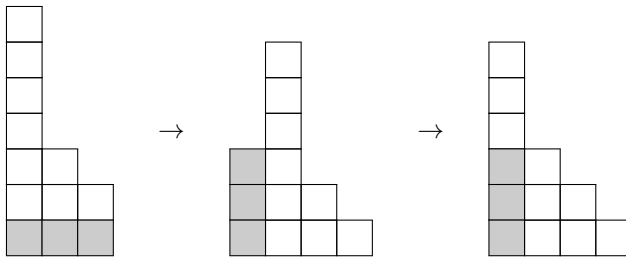


Figure 1: A move in Bulgarian solitaire from $\lambda = (7, 3, 2, 0, 0, \dots) \in \mathcal{P}(12)$: The bottom layer is picked to form a new pile with three cards and the cards are then left-shifted.

Olson [10] recently introduced a generalization of Bulgarian solitaire in which the number of cards picked from a pile of size h is given by $\sigma(h)$, where $\sigma : \mathbb{Z}_+ \rightarrow \mathbb{N}$ can be any function such that $\sigma(h) \leq h$ for all $h \in \mathbb{Z}_+$. The ordinary Bulgarian solitaire is obtained for the constant function $\sigma(h) = 1$. Olson studied cycle lengths, proving a general upper bound on cycle lengths for any specification of σ .

The key results known to hold for ordinary Bulgarian solitaire do not generalize to Olson's σ -Bulgarian solitaire. For instance, it is well-known that ordinary Bulgarian solitaire has the property that if a stable (i.e., fixpoint) configuration exists it is unique [3]. This uniqueness property of stable configurations does not generally hold for the σ -Bulgarian solitaire; a simple counter-example is obtained by defining $\sigma(1) = 1$, $\sigma(2) = 1$, and $\sigma(3) = 3$, in which case both $(2, 1)$ and (3) are stable

configurations of three cards. To avoid such pathological cases we impose some additional conditions on σ :

Definition 1. *A σ -Bulgarian solitaire is said to be well-behaved if the following three conditions on σ are satisfied:*

1. $\sigma(1) = 1$,
2. $\sigma(h)$ is a non-decreasing function,
3. $\bar{\sigma}(h) := h - \sigma(h)$ is a non-decreasing function.

The first condition in the definition says that from a pile with just a single card, you pick that card. The second condition says that you never pick fewer cards from a larger pile than from a smaller pile. The third condition says that the number of unpicked cards are never fewer in the larger pile than in a smaller pile.

The aim of this paper is to show how a number of properties that are well-known to hold for ordinary Bulgarian solitaire generalize to well-behaved σ -Bulgarian solitaire. A simple example is the dominance property: If a configuration λ is dominated by another configuration κ , in the sense that $\lambda_i \leq \kappa_i$ holds for all i , then this dominance relation is preserved as the solitaire is played in parallel from the two configurations. Let $\lambda \leq \kappa$ denote the dominance relation and let λ^{new} and κ^{new} denote the configurations obtained from playing one move of σ -Bulgarian solitaire from configurations λ and κ , respectively.

Theorem 1. *The implication $\lambda \leq \kappa \Rightarrow \lambda^{\text{new}} \leq \kappa^{\text{new}}$ holds in σ -Bulgarian solitaire if both σ and $\bar{\sigma}$ are non-decreasing functions. In particular, the implication holds for well-behaved σ .*

Proof. If $\bar{\sigma}$ is non-decreasing, what remains of the old piles of λ will be dominated by what remains of the old piles of κ . If σ is non-decreasing, the new pile formed from λ will be dominated by the new pile formed from κ . This pilewise dominance clearly remains when the piles in each configuration are sorted by size. By Definition 1, all well-behaved σ satisfy that both σ and $\bar{\sigma}$ are non-decreasing functions. $\square$

We now outline the other properties to be examined in this paper. In section 2 we show that moves of any well-behaved σ -solitaire have a geometric interpretation in terms of picking certain layers of cards. In section 3 we demonstrate that stable configurations of any well-behaved σ -solitaire are unique for any n for which a stable configuration exists. In section 4 we characterize stable configurations of well-behaved σ -solitaires as convex (i.e., $\lambda_i - \lambda_{i+1} \geq \lambda_{i+1} - \lambda_{i+2}$ for all $i \geq 1$). In sections 5 and 6 we define limit shapes of stable configurations for an infinite sequence of well-behaved σ_n -solitaires and show that any convex shape can be obtained as a limit shape of such a sequence.

In section 7 we define the surplus and deficit of a configuration with respect to a given stable configuration and show that the total surplus and total deficit can decrease but never increase as a well-behaved σ -solitaire is played. This property suggests that as a well-behaved σ -solitaire is played the configurations tend to converge toward stable configurations. We conjecture that recurrent configurations are so close to stable configurations that they have the same limit shape. In sections 8 and 9 we prove this conjecture in the special case when the σ_n -solitaires are given by $\sigma_n(h) = \lceil q_n h \rceil$ for $q_n \in (0, 1]$. The limit shapes of stable and recurrent configurations are then triangular in case $q_n^2 n \rightarrow 0$, and exponential in case $q_n^2 n \rightarrow \infty$.

2. A geometric interpretation of well-behaved σ -Bulgarian solitaires

The moves of a well-behaved σ -Bulgarian solitaire has an intuitive geometric interpretation. First note that σ is well-behaved if and only if it satisfies the boundary condition $\sigma(1) = 1$ and the condition that for all pile sizes $h > 0$ the difference $\Delta\sigma(h) := \sigma(h) - \sigma(h-1)$ equals either 1 or 0. We can then record the values of h for which $\Delta\sigma(h) = 1$ as a (finite or infinite) sequence $H_1 = 1, H_2, H_3, \dots$. For any index i for which H_i is defined it follows from the definition that $H_i = \min\{h : \sigma(h) = i\}$. A move of the σ -Bulgarian solitaire can now be described as a move on the Young diagram of the configuration in which the cards in layers number $H_1 = 1, H_2, H_3, \dots$ (counted from the bottom) are removed to form a new column; sorting is then achieved by left-shifting the cards in every layer. See Figure 2 for an example. Note that ordinary Bulgarian solitaire is the special case in which only row number $H_1 = 1$ is removed to form a new column.

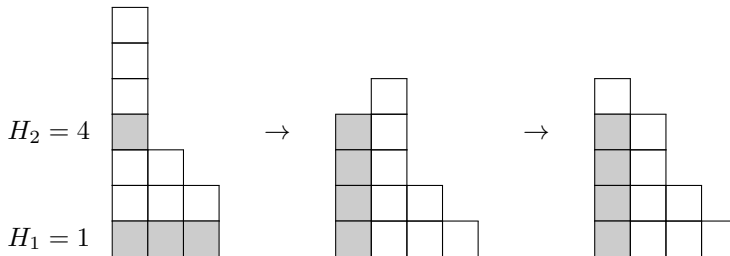


Figure 2: A move from the partition $\lambda = (7, 3, 2, 0, 0, \dots) \in \mathcal{P}(12)$ in a well-behaved σ -Bulgarian solitaire in which layers number $H_1 = 1$ and $H_2 = 4$ are picked to form a new pile with four cards.

Observation 1. *In the geometric interpretation of a well-behaved σ -Bulgarian solitaire, $\sigma(h)$ equals the number of picked layers up to layer h .*

In a move of a well-behaved σ -Bulgarian solitaire, the layers at levels $H_1 =$

$1, H_2, H_3, \dots$ are picked and any other layer, say at level h , drops $\sigma(h)$ levels down (with one of the picked card inserted at the left end). A layer may continue to drop down for several moves but must eventually reach one of the levels $H_1 = 1, H_2, H_3, \dots$ and there be picked. Thus, any card currently at level h will eventually be picked at some level $P(h)$ that can be calculated recursively by

$$P(h) = \begin{cases} h & \text{if } h \in \{H_1 = 1, H_2, H_3, \dots\}, \\ P(h - \sigma(h)) & \text{otherwise.} \end{cases}$$

The following observation then follows immediately by induction.

Observation 2. *In the geometric interpretation of a well-behaved σ -Bulgarian solitaire, it holds for any level h that the set*

$$\{P(h), P(h-1), P(h-2), \dots, P(h - \sigma(h) + 1)\}$$

is some permutation of the set $\{H_1 = 1, H_2, \dots, H_{\sigma(h)}\}$.

3. Uniqueness of stable configurations

Recall that a configuration is said to be stable with respect to σ -Bulgarian solitaire if a move in the solitaire leaves the configuration invariant.

Lemma 1. *On the condition that $\bar{\sigma}$ is a non-decreasing function, λ is a stable configuration with respect to σ -Bulgarian solitaire if and only if $\lambda_{i+1} = \bar{\sigma}(\lambda_i)$ for all $i \geq 1$.*

Proof. A move of the solitaire decreases the size of any pile from λ_i to $\bar{\sigma}(\lambda_i)$ and then creates a new pile such that the sum of all pile sizes stays constant at n , the total number of cards. Because $\bar{\sigma}$ is assumed to be a non-decreasing function, the decreased piles will still satisfy $\bar{\sigma}(\lambda_i) \geq \bar{\sigma}(\lambda_{i+1})$ for all $i \geq 1$, that is, they will not need to be reordered. Therefore λ is a stable configuration if and only if $\bar{\sigma}(\lambda_i) = \lambda_{i+1}$ for all $i \geq 1$, as the new pile will then necessarily have size λ_1 . $\square$

Theorem 2. *If $\bar{\sigma}$ is a non-decreasing function, there is at most one stable configuration with respect to σ -Bulgarian solitaire for a given number n of cards.*

Proof. When $\bar{\sigma}$ is a non-decreasing function it follows from Lemma 1 that each stable configuration is completely determined by the choice of λ_1 , the size of the largest pile. Let λ be a stable configuration with n cards and consider another stable configuration λ' with $\lambda_1 > \lambda'_1$. It follows immediately by induction that $\lambda_i \geq \lambda'_i$ for all $i \geq 1$, and consequently that the total number of cards in these two configurations are different. $\square$

We can also bound the difference in the total number of cards between consecutive stable configurations in a well-behaved σ -solitaire.

Corollary 1. *Assume that $\sigma(1) = 1$ and that both σ and $\bar{\sigma}$ are non-decreasing functions and let λ and λ' be the stable configurations (with respect to σ -Bulgarian solitaire) determined by first piles of size λ_1 and $\lambda'_1 = \lambda_1 + 1$, respectively. Then the difference in the total number of cards between λ' and λ is at most $\ell(\lambda) + 1$.*

Proof. As we noted in the introduction, the assumption that both σ and $\bar{\sigma}$ are non-decreasing functions implies that for any pile size h we have that $\sigma(h+1) - \sigma(h)$ equals either 1 or 0. Starting from the relation $\lambda'_1 = \lambda_1 + 1$, it follows immediately by induction that as long as $\sigma(\lambda_i + 1) - \sigma(\lambda_i) = 0$ we will also have $\lambda'_{i+1} = \lambda_{i+1} + 1$. The first time we instead have $\sigma(\lambda_i + 1) - \sigma(\lambda_i) = 1$, we will obtain $\lambda'_{i+1} = \lambda_{i+1}$, and from that point on the pile sizes will be identical in the two configurations. Thus, the difference in the total number of cards is equal to the number of piles that differed in size, which is at most the number of piles in the larger configuration λ' . Because each of its piles is at most one larger than the corresponding piles in the smaller configuration λ , it can have at most one pile more. Hence, the difference in the total number of cards is bounded by $\ell(\lambda) + 1$. $\square$

4. Characterization of stable configurations

We shall now characterize what stable configurations of well-behaved σ -solitaires look like. Define a configuration λ as *convex* if $\lambda_i - \lambda_{i+1} \geq \lambda_{i+1} - \lambda_{i+2} \geq 0$ for all $i \geq 1$.

Lemma 2. *A configuration λ is convex if and only if it is a stable configuration of a well-behaved σ -Bulgarian solitaire.*

Proof. First assume that λ is a stable configuration of a well-behaved σ -Bulgarian solitaire. Then Lemma 1 says that $\lambda_i - \lambda_{i+1} = \sigma(\lambda_i) \geq 0$ for all $i \geq 1$. As a well-behaved σ is non-decreasing, this inequality implies that λ is convex.

To prove the converse, assume that λ is a convex configuration with ℓ nonzero piles. Then for each $i \geq 1$ we can choose a subset of $(\lambda_i - \lambda_{i+1}) - (\lambda_{i+1} - \lambda_{i+2})$ layers in the interval of layers $(\lambda_{i+1}, \lambda_i]$. Note that this means all layers in the interval $(0, \lambda_\ell]$ are chosen, in particular layer 1. A well-behaved σ -Bulgarian solitaire is therefore defined by picking the chosen layers in each move. Moreover, for all $i \geq 1$ the corresponding σ will satisfy $\sigma(\lambda_i) = \lambda_i - \lambda_{i+1}$ as the latter expression equals the number of picked layers up to layer λ_i (see Observation 1). Thus, λ is a stable configuration of this well-behaved σ -Bulgarian solitaire. $\square$

5. The concept of limit shapes of stable and recurrent configurations

We shall consider a sequence of well-behaved σ_n , for $n = 1, 2, \dots$. Let $f_{\sigma_n} : \mathcal{P}(n) \rightarrow \mathcal{P}(n)$ denote the map on integer partitions of n defined by a move of the σ_n -Bulgarian solitaire. For any n , the σ_n -Bulgarian solitaire can be regarded as the deterministic process on $\mathcal{P}(n)$ defined by the iteration of the map f_{σ_n} .

Given a process on $\mathcal{P}(n)$ defined by the iteration of any map $f_n : \mathcal{P}(n) \rightarrow \mathcal{P}(n)$, we say that a configuration $\lambda \in \mathcal{P}(n)$ is *stable* with respect to this process in case $f(\lambda) = \lambda$, and that the configuration is *recurrent* if there exists a positive integer k such that $f_n^k(\lambda) = \lambda$. Thus, the stable configurations constitute a subset of the recurrent configurations. Note that the set of all configurations, $\mathcal{P}(n)$, is finite. Regardless of choice of starting configuration, the process must therefore inevitably enter the set of recurrent configurations after a finite number of moves. We shall now define what we mean by limit shapes of stable or recurrent configurations.

5.1. Downscaling of diagram-boundary functions

For any partition λ , define its *diagram-boundary function* as the nonnegative, weakly decreasing and piecewise constant function $\partial\lambda : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ given by

$$\partial\lambda(x) = \lambda_{\lfloor x \rfloor + 1}.$$

To illustrate, Figure 3 depicts the function graph $y = \partial\lambda(x)$ for the partition $\lambda = (4, 4, 2, 1, 1, 0, 0, \dots)$.

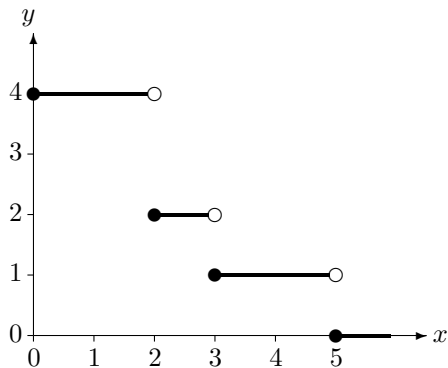


Figure 3: Function graph $y = \partial\lambda(x)$ for the partition $\lambda = (4, 4, 2, 1, 1) \in \mathcal{P}(12)$.

To achieve limiting behavior of such function graphs as n grows we need to rescale diagrams depending on the value of n . Following [6] and [12] we apply a scaling factor $a_n > 0$ such that all row lengths are multiplied by $1/a_n$ and all column heights are multiplied by a_n/n , yielding a constant area of 1. To avoid having to

specify the scaling factor we shall consistently make the choice $a_n = n/\lambda_1$, such that the height of the diagram is also scaled to 1. Thus, given a partition λ we define the *downscaled* diagram-boundary function of λ as the positive, real-valued, weakly decreasing and piecewise constant function $\partial\lambda : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ given by

$$\partial\lambda(x) = \frac{1}{\lambda_1} \partial\lambda(xn/\lambda_1) = \frac{1}{\lambda_1} \lambda_{\lfloor xn/\lambda_1 \rfloor + 1}. \quad (1)$$

5.2. Limit shapes of recurrent configurations

Given an infinite family of maps on configurations, $\{f_n : \mathcal{P}(n) \rightarrow \mathcal{P}(n)\}$ for $n = 1, 2, 3, \dots$, we say that $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a *limit shape of recurrent configurations* of the corresponding family of processes if the downscaled diagrams of recurrent configurations with respect to $f_1, f_2, f_3, \dots$ converge to ϕ in the following sense:

$$\partial\lambda^{(n)}(x) \rightarrow \phi(x) \text{ as } n \rightarrow \infty \quad (2)$$

for all $x > 0$, where $\lambda^{(n)}$ is any recurrent configuration of f_n .

5.3. Limit shapes of stable configurations of well-behaved σ_n -solitaires

Consider a sequence of well-behaved σ_n for $n = 1, 2, \dots$. For each value of n , consider σ_n -Bulgarian solitaire and let

$$n^* = n^*(n) := \max\{n' \leq n : \text{there exists a stable configuration of } n' \text{ cards}\}.$$

The number n^* is well-defined as there always exists a stable configuration on a single card. Define $\lambda^{(n^*)}$ as the stable configuration on n^* cards; this is well-defined according to Theorem 2. According to Corollary 1, the choice $\lambda^{(n^*(n))}$ as the stable configuration corresponding to n will be the same for at most $\ell(\lambda^{(n^*)}(n)) + 1$ values of n . Thus, $n^*(n)$ tends to infinity as n tends to infinity. In line with the limit shape concept defined in equation (2), we define a *limit shape of stable configurations* for the sequence of well-behaved σ_n , $n = 1, 2, \dots$, as a function $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\partial\lambda^{(n^*(n))}(x) \rightarrow \phi(x) \text{ as } n \rightarrow \infty \quad (3)$$

for all $x > 0$.

6. Characterization of limit shapes of stable configurations of well-behaved σ_n -solitaires

It is well-known that the Bulgarian solitaire has a stable configuration if and only if the total number of cards in the deck is a triangular number, $n = 1 + 2 + \dots + k$ for some positive integer k , in which case the unique stable configuration has one

pile of each integer size from k down to 1 [3]. Thus, the Young diagrams of stable configurations are staircase shaped and hence the limit shape, as n tends to infinity, is a triangle. By the convention that the height and area are scaled to unity, the limit shape will be a triangle of height 1 and width 2.

When generalizing from ordinary Bulgarian solitaire to well-behaved σ_n -Bulgarian solitaire, the limit shapes that arise will not necessarily be triangular. Indeed, in Theorem 3 we prove that any convex shape (with some properties) can be obtained as the limit shape of a suitably chosen family of well-behaved σ_n , for $n = 1, 2, \dots$. First we need a lemma.

Lemma 3. *Let $f_n : (0, \infty) \rightarrow \mathbb{R}$ be convex functions for $n = 1, 2, \dots$, and suppose there is a function $f : (0, \infty) \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for any $x > 0$. Then we also have pointwise convergence of right derivatives: $\lim_{n \rightarrow \infty} (f_n)'_R(x) = f'_R(x)$ for any $x > 0$.*

Proof. Clearly, f is convex. Therefore the left derivative f'_L and the right derivative f'_R exists in all points in $(0, \infty)$. By the definition of right derivative, for any $\varepsilon > 0$ there exists an $h > 0$ such that

$$\frac{f(x+h) - f(x)}{h} < f'_R(x) + \varepsilon$$

for all $x > 0$. Since $\lim_{n \rightarrow \infty} f_n(x) = f(x)$, there exists an $N \in \mathbb{N}$ such that

$$\frac{f_n(x+h) - f_n(x)}{h} < f'_R(x) + \varepsilon \quad \text{for all } n > N.$$

By the convexity of f_n , we have $\frac{f_n(x+h) - f_n(x)}{h} \geq f'_n(x)$. Therefore,

$$f'_n(x) < f'_R(x) + \varepsilon \quad \text{for all } n > N.$$

An analogous reasoning for the left-derivative gives $f'_n(x) > f'_L(x) - \varepsilon$. Thus, $f'_L(x) - \varepsilon < f'_n(x) < f'_R(x) + \varepsilon$ for all $n > N$. As a consequence we must have $\lim_{n \rightarrow \infty} f'_n(x) = f'(x)$, in particular $\lim_{n \rightarrow \infty} (f_n)'_R(x) = f'_R(x)$ for any $x > 0$. $\square$

Theorem 3. *Let $\phi : (0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ be a function and let $a_1, a_2, \dots \rightarrow \infty$ be any (positive) scaling factors such that a_n^2/n converges to some $c \geq 0$ as $n \rightarrow \infty$. Then the following are equivalent.*

- (a) *There is a sequence of well-behaved σ_n , $n = 1, 2, \dots$, such that ϕ is a stable-limit shape of (σ_n) under the scaling (a_n) .*
- (b) *ϕ is convex with $\int_0^\infty \phi(x) dx \leq 1$, and if $c > 0$ the right derivative $\phi'_R(x)$ is an integer multiple of c for any $x > 0$.*

Proof. To prove that (a) implies (b), suppose ϕ is a stable-limit shape of (σ_n) under the scaling (a_n) . Let $\lambda^{(n^*)}$ denote the stable configuration in the σ_n -Bulgarian solitaire with $n^*(n)$ cards.

For each n , define a piecewise linear function $\phi_n : (0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ by letting

$$\phi_n(x) = \frac{a_n}{n} ((1 - t(x))\lambda_{\lfloor a_n x \rfloor + 1}^{(n^*)} + t(x)\lambda_{\lfloor a_n x \rfloor + 2}^{(n^*)}),$$

where $t(x) := a_n x - \lfloor a_n x \rfloor$. Since each ϕ_n is convex, so is ϕ , and by Fatou's lemma

$$\int_0^\infty \phi(x) dx \leq \liminf_{n \rightarrow \infty} \int_0^\infty \phi_n(x) dx \leq 1.$$

Now suppose $c > 0$. By Lemma 3, $(\phi_n)'_R(x) \rightarrow \phi'_R(x)$ for any $x > 0$, so

$$\frac{(\phi_n)'_R(x)}{a_n^2/n} \rightarrow \phi'_R(x)/c$$

as $n \rightarrow \infty$. But

$$\frac{(\phi_n)'_R(x)}{a_n^2/n} = (\lambda_{\lfloor a_n x \rfloor + 2}^{(n^*)} - \lambda_{\lfloor a_n x \rfloor + 1}^{(n^*)}),$$

which is an integer, so it follows that $\phi'_R(x)/c$ is an integer.

For the other direction, suppose (b) holds true and that $c = 0$. Since ϕ is convex, it has a right derivate ϕ'_R . Let $s_1, s_2, \dots$ be a sequence of positive real numbers such that $s_n \rightarrow \infty$ but $s_n a_n^2/n \rightarrow 0$ as $n \rightarrow \infty$, and such that $s_n a_n$ is an integer for any n .

Define a partition $\lambda^{(n)}$ by letting

$$\lambda_k^{(n)} = \sum_{i=k+1}^{s_n a_n} \lfloor -\frac{n}{a_n^2} \phi'_R(i/a_n) \rfloor$$

for $k = 1, 2, \dots$. By the convexity of ϕ , we have

$$\begin{aligned} \lambda_k^{(n)} &\leq -\frac{n}{a_n} \int_{k/a_n}^\infty \phi'_R(x) dx = \frac{n}{a_n} \phi(k/a_n), \\ \lambda_k^{(n)} &\geq -\frac{n}{a_n} \int_{(k+1)/a_n}^{s_n} \phi'_R(x) dx - s_n a_n = \frac{n}{a_n} (\phi((k+1)/a_n) - \phi(s_n)) - s_n a_n. \end{aligned}$$

From the first of these inequalities it follows that

$$\sum_{k=1}^\infty \lambda_k^{(n)} \leq a_n \int_0^\infty \frac{n}{a_n} \phi(x) dx \leq n.$$

Now, let $\mu_k^{(n)} = \lambda_k^{(n)}$ for $k = 2, 3, \dots$ but choose $\mu_1^{(n)}$ so that $\mu_1^{(n)} + \mu_2^{(n)} + \dots = n$. Clearly, $\mu^{(n)}$ is monotonic convex, so by Lemma 2 it is a stable configuration of a

σ_n -Bulgarian solitaire for some well-behaved rule σ_n . Since ϕ is continuous, for any $x > 0$ we have

$$\frac{a_n}{n} \lambda_{\lceil a_n x \rceil} \leq \phi(\lceil a_n x \rceil / a_n) \rightarrow \phi(x)$$

and, since $s_n \rightarrow \infty$ and $s_n a_n^2 / n \rightarrow 0$ as $n \rightarrow \infty$, for any $x > 0$,

$$\frac{a_n}{n} \lambda_{\lceil a_n x \rceil} \geq \phi(\lceil a_n x + 1 \rceil / a_n) - \phi(s_n) - \frac{s_n a_n^2}{n} \rightarrow \phi(x),$$

in other words, $\frac{a_n}{n} \lambda_{\lceil a_n x \rceil} \rightarrow \phi(x)$ for any $x > 0$, establishing that ϕ is the desired stable-limit shape by the definition (3).

Now suppose (b) holds true and $c > 0$. Define a partition $\lambda^{(n)}$ by letting

$$\lambda_k^{(n)} = -\frac{1}{c} \sum_{i=k+1}^{\infty} \phi'_R(i/\sqrt{cn})$$

for $k = 1, 2, \dots$. Since ϕ is convex, we have

$$\begin{aligned} \lambda_k^{(n)} &\leq -\frac{\sqrt{cn}}{c} \int_{k/\sqrt{cn}}^{\infty} \phi'_R(x) dx = \sqrt{\frac{n}{c}} \phi(k/\sqrt{cn}), \\ \lambda_k^{(n)} &\geq -\frac{\sqrt{cn}}{c} \int_{(k+1)/\sqrt{cn}}^{\infty} \phi'_R(x) dx = \sqrt{\frac{n}{c}} \phi((k+1)/\sqrt{cn}). \end{aligned}$$

From the first of these inequalities it follows that

$$\sum_{k=1}^{\infty} \lambda_k^{(n)} \leq \sqrt{cn} \int_0^{\infty} \sqrt{\frac{n}{c}} \phi(x) dx \leq n.$$

Now, let $\mu_k^{(n)} = \lambda_k^{(n)}$ for $k = 2, 3, \dots$ but choose $\mu_1^{(n)}$ so that $\mu_1^{(n)} + \mu_2^{(n)} + \dots = n$. Clearly, $\mu^{(n)}$ is monotonic convex, so by Lemma 2 it is a stable configuration of a σ_n -Bulgarian solitaire for some well-behaved rule σ_n . Finally, since $a_n^2/n \rightarrow c$ as $n \rightarrow \infty$, and since ϕ is continuous, for any $x > 0$ we have

$$\phi(x) \leftarrow \frac{a_n}{\sqrt{cn}} \phi(\lceil a_n x + 1 \rceil / \sqrt{cn}) \leq \frac{a_n}{n} \lambda_{\lceil a_n x \rceil} \leq \frac{a_n}{\sqrt{cn}} \phi(\lceil a_n x \rceil / \sqrt{cn}) \rightarrow \phi(x),$$

and hence ϕ is the desired limit shape. $\square$

Note that for a finite n a Young diagram will have unit area under our conventional scaling with scaling factors (a_n) . The reason why we have $\int_0^{\infty} \phi(x) dx \leq 1$ in Theorem 3 is that the largest pile (or a few of the largest piles) may be arbitrarily large without affecting the limit shape ϕ . Our limit shape definitions (2) and (3) do not include $x = 0$ as to allow for $\lim_{x \rightarrow 0+} \phi(x)$ to be infinite.

7. A conjecture on limit shapes of recurrent configurations of well-behaved σ_n -solitaires

When a stable configuration exists, ordinary Bulgarian solitaire eventually reaches it. This does not hold in general for well-behaved σ -Bulgarian solitaire. One counter-example is given by the well-behaved σ -Bulgarian solitaire defined by $\sigma(h) = \lceil 3h/10 \rceil$ on $n = 11$ cards, as this game has both a stable configuration $(5, 3, 2, 1)$ and a non-trivial cycle

$$(6, 2, 2, 1) \mapsto (5, 4, 1, 1) \mapsto (6, 3, 2) \mapsto (4, 4, 2, 1) \mapsto (6, 2, 2, 1).$$

However, it is worth noting that the pile sizes in these recurrent configurations never deviated by more than one card from the corresponding pile sizes in the stable configuration. This is akin to the ordinary Bulgarian solitaire in the case when n is not a triangular number so that no stable configuration exists; in that case the solitaire will eventually reach a cycle of recurrent configurations, and these are close to staircase shaped in the sense that they can all be constructed by starting with some staircase configuration $(k, k-1, \dots, 1)$ and adding at most one card to each pile, and possibly adding one more pile of size 1 [1, 2, 7, 8]. As n grows to infinity and the diagram is rescaled such that its height and area are both equal to 1, the deviation of recurrent configurations from the perfect staircase tends to zero. Thus, ordinary Bulgarian solitaire has a limit shape, namely the triangle of height 1 and width 2.

We believe that it is generally true that recurrent configurations must be sufficiently close to a stable configuration for a limit shape of stable configurations to also be a limit shape of recurrent configurations.

Conjecture 1. *If ϕ is a limit shape of the stable configurations of a sequence of well-behaved σ_n , then ϕ is also a limit shape of the recurrent configurations.*

We leave this conjecture as an open problem. A first step toward its proof is that a configuration's total deviation from a stable configuration will often decrease but never increase during play, as we show below.

7.1. Deviations, surplus, and deficit, with reference to a stable configuration

Represent configurations by infinite vectors of pile sizes (a finite number of non-zero piles and an infinite tail of zeros). For a given σ -Bulgarian solitaire, fix some stable configuration λ^* to be used as a reference and let n^* denote the number of cards of λ^* .

For any configuration λ of n cards we can calculate the component-wise difference to the reference configuration λ^* to obtain an infinite *deviation* vector $d(\lambda) := \lambda - \lambda^*$. The sum of the elements of the deviation vector must equal the difference in the

number of cards in the two configurations: $d_1 + d_2 + \dots = n - n^*$, suppressing the dependence on λ to avoid cumbersome notation.

The deviation vector can be decomposed as $d(\lambda) = d^+(\lambda) - d^-(\lambda)$ where the *surplus* vector $d^+(\lambda)$ is given by $d_i^+ = \max(d_i, 0)$ for all $i \geq 1$, and the *deficit* vector $d^-(\lambda)$ is given by $d_i^- = \max(-d_i, 0)$ for all $i \geq 1$. Denote the total surplus and total deficit in by $d_{\text{tot}}^+(\lambda)$ and $d_{\text{tot}}^-(\lambda)$, respectively. Then, obviously,

$$d_{\text{tot}}^+(\lambda) - d_{\text{tot}}^-(\lambda) = n - n^*.$$

Thus, the difference between total surplus and total deficit is invariant under play.

7.2. The total surplus never decreases in well-behaved σ -Bulgarian solitaires

It is well-known and easy to see that ordinary Bulgarian solitaire has the property that a move can never increase the total surplus: $d_{\text{tot}}^+(\lambda) \geq d_{\text{tot}}^+(\lambda^{\text{new}})$ will always hold. This property generalizes to well-behaved σ -Bulgarian solitaire. To prove this we shall extend the geometric interpretation of the solitaire by marking some cards as *plus-cards* or *minus-cards* with special properties.

Given a reference stable configuration λ^* of n^* cards, any configuration λ of n cards can be transformed to a marked version of λ in the following two steps:

1. *Marking plus-cards*: For any pile index i such that $\lambda_i > \lambda_i^*$ (i.e., such that the pile has a surplus $d_i^+ = \lambda_i - \lambda_i^* > 0$), mark the d_i^+ top cards of the pile as plus-cards.
2. *Creating minus-cards*: For any pile index i such that $\lambda_i < \lambda_i^*$ (i.e., such that the pile has a deficit $d_i^- = \lambda_i^* - \lambda_i > 0$), add an additional d_i^- minus-cards to the top of the pile.

Observation 3. *In the marked version of λ the plus-cards and unmarked cards together make up λ , while the minus-cards and unmarked cards together make up λ^* . It follows that the number of plus-cards equals the total surplus and the number of minus-cards equals the total deficit.*

Recall the geometric interpretation of well-behaved σ -Bulgarian solitaire described in section 2: Remove the cards in layers number $H_1 = 1, H_2, H_3, \dots$ (counted from the bottom) and let these cards form a new first pile, then sort the configuration by left-shifting the cards in every layer. The rules of play on marked configurations follow this interpretation, with the following special rules for marked cards.

Cancellation within the new pile: When a new first pile is formed, all minus-cards and plus-cards in this pile float to the top of the pile. If there are both minus-cards

and plus-cards in the pile then cancellation occurs: Repeatedly replace one minus-card and one plus-card by a single unmarked card until either there are no more minus-cards or there are no more plus-cards in the pile.

Cancellation within a layer: Left-shifting layers clearly corresponds to minus-cards in the first pile swapping places with any cards to the right in the same layer. If at any point a minus-card that started in the first pile swaps places with a plus-card, replace the minus-card and the plus-card by a single unmarked card; then continue left-shifting as usual.

We shall refer to this process as the *marked σ -Bulgarian solitaire*.

Lemma 4. *If a well-behaved σ -Bulgarian solitaire from λ is played in parallel with the marked σ -Bulgarian solitaire from the marked version of λ , Observation 3 will continue to hold.*

Proof. First restrict attention to the movement of the minus-cards and unmarked cards. Together they make up the stable configuration λ^* so, precisely because this is a stable configuration, these cards will still make up λ^* after a move. Now restrict attention to the movement of plus-cards and unmarked cards. Together they make up λ so clearly after one move they will make up λ^{new} . Finally, different types of cards interact only in the form of cancellations within the new pile or within a layer. In both forms of cancellation, the configuration made up of minus-cards and unmarked cards together is unchanged as in effect a minus-card is replaced by an unmarked card. The same goes for the configuration made up of plus-cards and unmarked cards together. $\square$

From Lemma 4 the desired result follows immediately.

Theorem 4. *Total surplus and total deficit may decrease but never increase during play of a well-behaved σ -solitaire.*

Proof. From Lemma 4 and the rules of the marked σ -Bulgarian solitaire it follows that the total surplus equals the number of plus-cards and the total deficit equals the number of minus-cards, which decrease when cancellations occur but never increase. $\square$

In order to prove Conjecture 1 we now only need to prove that whenever the deviation from the closest stable configuration is large, a cancellation of a plus-card against a minus-card must eventually occur.

A reason why it is likely that a cancellation will eventually occur is that when plus-cards start over from the first pile they obviously do so from a higher level than minus-cards. It will therefore generally take a greater number of moves for a plus-card than for a minus-card from when it starts over until it reaches the level of a picking layer where it starts over again. Thus minus-cards have shorter periods than plus-cards and should eventually catch up with them. For ordinary

Bulgarian solitaire, which has only one picking layer, this argument is sufficient to prove that cancellations must occur until no more minus-cards remain. In general, however, well-behaved σ -solitaire has several picking layers and this makes it difficult to perform a rigorous study of periods. However, we still believe that it may be possible to overcome these technical difficulties to achieve a proof of Conjecture 1.

8. Limit shapes of stable configurations of q_n -proportion Bulgarian solitaire

In order to calculate explicit limit shapes we make a canonical choice of a well-behaved σ , namely $\sigma(h) = \lceil qh \rceil$ for $q \in (0, 1]$. (It should be obvious that this function satisfies the conditions for being well-behaved, see Definition 1.) In words, this form of σ defines a solitaire in which from each pile we pick a number of cards given by the proportion q of the pile size, rounded upward to the closest integer. We will refer to this solitaire as *q -proportion Bulgarian solitaire*. Following the geometric interpretation of well-behaved σ -Bulgarian solitaire, the cards picked in q -proportion Bulgarian solitaire can be seen as layers number $H_1, H_2, H_3, \dots$, where for any $i > 0$ we have $H_i = \min\{h : \lceil q_n h \rceil = i\}$. Thus, q -proportion Bulgarian solitaire is a well-behaved σ -Bulgarian solitaire in which the picked layers are approximately equidistant.

We may let the choice of q depend on n , in which we write q_n . Note that for $q_n \leq 1/n$ only one card is picked in any pile. Thus by choosing $q_n \leq 1/n$ we obtain ordinary Bulgarian solitaire.

Thanks to Lemma 1 we can determine a unique stable configuration of a q -proportion solitaire by choosing the size of the largest part and then obtain the other parts by repeatedly applying the function $\bar{\sigma}(h) = h - \lceil qh \rceil$. This makes it easy to determine the limit shapes of stable configurations. Specifically, we identify three different regimes defined by the asymptotic behavior of nq_n^2 .

First, in case nq_n^2 tends to zero as n tends to infinity, stable configurations have a triangular limit shape. This is a direct generalization of the limit shape result for the ordinary Bulgarian solitaire.

The second regime is when nq_n^2 tends to infinity, in which case an exponential limit shape is obtained. The borderline regime when nq_n^2 tends to a constant C yields an infinite family of limit shapes (parameterized by C), which interpolate between the triangular shape of the first regime and the exponential shape of the second regime.

A move of q -proportion Bulgarian solitaire involves rounding the number of picked cards in each pile to integers. The three regimes differ in how much impact this rounding has on the result. The following couple of lemmas estimate the impact of rounding.

Lemma 5. *After a move in q -proportion Bulgarian solitaire from a configuration with m non-empty piles, the size of the new pile is $nq + r$ where $0 \leq r < m$.*

Proof. By the definition of q -proportion Bulgarian solitaire, the contribution to the new pile from any old non-empty pile λ_i is $\lceil q\lambda_i \rceil$, which is bounded from below and from above by

$$\lambda_i q \leq \lceil q\lambda_i \rceil < \lambda_i q + 1.$$

The lemma follows from summing over all m non-empty piles. $\square$

Lemma 6. *The number of moves in q -proportion Bulgarian solitaire until a pile of size h disappears is at most*

$$\frac{\ln(qh) + 1}{q}.$$

Proof. In each move a pile decreases at least by a factor of $(1 - q)$. A pile starting at size h will have gone down to size $1/q$ after at most $\ln(1/(qh))/\ln(1 - q)$ moves, which (using the MacLaurin expansion of the denominator) in turn is bounded by $\ln(qh)/q$. From size $1/q$ and onward the pile will, due to rounding, lose exactly 1 card per move for $1/q$ moves at which point the pile has disappeared. $\square$

We shall now derive the limit shape of stable configurations depending on the asymptotic behavior of nq_n^2 . See Figure 4.

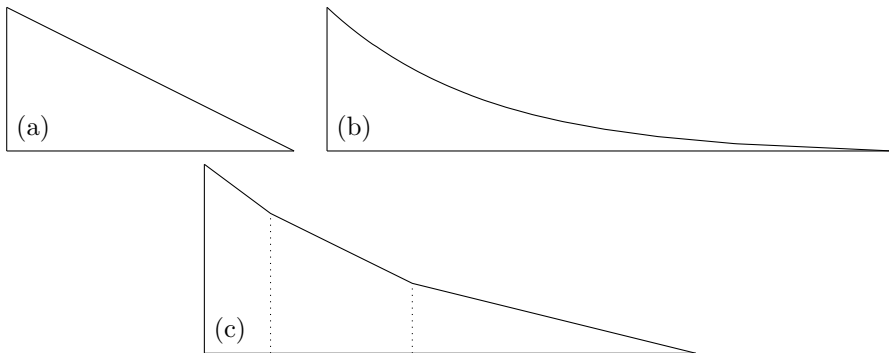


Figure 4: The three cases of limit shapes in Theorem 5: (a) triangular, (b) exponential, and (c) interpolating with J linear sections, here illustrated for $J = 3$.

First, recall from Section 5.1 that the scaling factor we employ is $a_n = n/\lambda_1 = \frac{1}{q_n}$. Thus, if q_n is bounded away from zero, then the scaling $\frac{1}{q_n}$ is bounded and hence cannot transform the jumpy boundary diagrams into a smooth limit shape. Therefore we shall require

$$q_n \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Theorem 5. *There are three cases for limit shapes of stable configurations of q -proportion Bulgarian solitaire, depending on the asymptotic behavior of nq_n^2 :*

- (a) *In case $nq_n^2 \rightarrow 0$, there is a triangular limit shape. Under the standard scaling we apply, by which the height of the diagram is scaled to 1, the downward slope of the triangle will be $1/2$.*
- (b) *In case $nq_n^2 \rightarrow \infty$, there is an exponential limit shape.*
- (c) *Interpolating between the two previous cases is the case $nq_n^2 \rightarrow C > 0$. Define $z > 0$ by the equation*

$$2C = \frac{z^2 + \lceil z \rceil^2}{\lceil z \rceil} - \sum_{i=0}^{\lceil z \rceil - 1} \frac{1}{\lceil z \rceil - i}$$

and set $A_0 = \frac{z}{C} \frac{1+z-\lceil z \rceil}{\lceil z \rceil}$ and $A_k = \frac{z}{C} \frac{1}{\lceil z \rceil - k}$ for $1 \leq k \leq \lceil z \rceil - 1$. The limit shape approximates the exponential shape using $Z := \lceil z \rceil$ linear sections such that the first section has width A_0 and every subsequent section, numbered $k = 1, 2, \dots, Z - 1$, has width A_k . The slope of the i th section is $\frac{C(Z-i)}{z^2}$ for all $i = 0, 1, \dots, Z - 1$.

Proof. (a) In case $nq_n^2 \rightarrow 0$ we shall see that the effect of rounding dominates in a move from the stable configuration. Specifically, for all sufficiently large n we have $q_n\sqrt{2n} < 1$ and hence $\lceil q_n h \rceil = 1$ for all $0 < h < \sqrt{2n}$. Now, assume that the largest pile of a configuration is $\lambda_1 \approx \sqrt{2n}$. Then the size of this pile will decrease by 1 card per move until after $\lceil \lambda_1 \rceil \approx \sqrt{2n}$ moves the pile has disappeared. By Lemma 1 the corresponding triangular configuration $\bar{\lambda}$ is stable. The number of cards in this configuration is approximately n , which confirms that a stable configuration of n cards (if it exists) will indeed have a largest pile of size $\approx \sqrt{2n}$. Downscaling (where the vertical scaling factor is $1/\lambda_1 = 1/\sqrt{2n}$ and horizontal scaling factor $\lambda_1/n = \sqrt{2/n}$) yields a boundary diagram $\partial\bar{\lambda}$ where column k has height $1 - (k-1)/\sqrt{2n}$, $k = 1, 2, \dots$. Since the width of each column is $\sqrt{2/n}$, the boundary diagram function for $\bar{\lambda}$ is $\partial\bar{\lambda}(x) = 1 - \lfloor x/\sqrt{2/n} \rfloor / \sqrt{2n} \rightarrow \infty$ as $n \rightarrow \infty$. By definition (3), the proposed limit shape is $y = 1 - x/2$ for $x \geq 0$.

(b) In case $nq_n^2 \rightarrow \infty$ the effect of rounding turns out to be negligible in a move from the stable configuration. Let λ_1 denote the largest pile in the stable configuration and let m denote the number of non-empty piles. In view of Lemma 1 it follows from sequential application of Lemma 5 and Lemma 6 that

$$\frac{\lambda_1}{nq_n} = 1 + O\left(\frac{m}{nq_n}\right) = 1 + O\left(\frac{\ln(q_n\lambda_1) + 1}{nq_n^2}\right).$$

Under the assumption $nq_n^2 \rightarrow \infty$ it is easy to see that the second term, which estimates the total effect of rounding after downscaling, will tend to zero. With no

rounding, pile sizes decrease geometrically with decay factor $1 - q_n$. Thus, after downscaling the stable configuration is asymptotically equal to a configuration $\bar{\lambda}$ with a first pile of size 1 and subsequent piles of size $1 - q_n, (1 - q_n)^2, \dots$. Since the horizontal scaling factor is $\lambda_1/n = q_n$, the width of each column in the rescaled boundary diagram $\partial\bar{\lambda}$ is q_n . Thus $\partial\bar{\lambda}(x) = (1 - q_n)^{\lfloor x/q_n \rfloor}$ for $x \geq 0$. As $n \rightarrow \infty$, $q_n \rightarrow 0$ and $\partial\bar{\lambda}(x) \rightarrow e^{-x}$, proving the proposed limit shape.

(c) For the remaining case, the crucial observation is that the rate by which a pile melts away depends on how the pile size relates to multiples of $1/q_n$. Any pile size can be expressed on the form y/q_n for some $y > 0$. From a pile of that size, a move will take away the amount $\lceil y \rceil$. Thus, a pile starting at a size of z/q_n will initially melt away at a slope of $Z = \lceil z \rceil$ per move for $\left\lceil \frac{1+z-Z}{Zq_n} \right\rceil$ moves, i.e. until the pile size reaches the threshold $(Z - 1)/q_n$. At this point the slope decreases to $Z - 1$ per move for $\left\lceil \frac{1}{q_n(Z-1)} \right\rceil$ moves until the pile size reaches the next threshold, $(Z - 2)/q_n$, etc. This pattern ends with a section of slope 1 per move for $\lceil 1/q_n \rceil$ moves. See Figure 5. By Lemma 1 this sequence of pile sizes constitutes a stable configuration Λ . It is elementary, although somewhat tedious, to verify that the total amount of cards n' in this configuration is

$$n' = \sum_{i=0}^{Z-1} \left[\frac{zB_i}{q_n} - (B_i - 1)(Z - i) \left(\frac{B_i}{2} + \sum_{j=i+1}^{Z-1} B_j \right) \right]$$

where $B_0 = \lceil \frac{1+z-Z}{q_n} \rceil$ and $B_k = \lceil \frac{1}{q_n(Z-k)} \rceil$ for $1 \leq k \leq Z - 1$ are the lengths of the Z sections with constant slope in the configuration Λ . Under the assumptions $nq_n^2 \rightarrow C$ and $q_n \rightarrow 0$ as $n \rightarrow \infty$, it is straightforward (but again somewhat tedious) to verify that, as $n \rightarrow \infty$, we have

$$q_n^2 n' \rightarrow \frac{1}{2} \left(\frac{z^2 + Z^2}{Z} - \sum_{i=0}^{Z-1} \frac{1}{Z - i} \right) = C$$

where the equality comes from using the value of z defined in the theorem. The actual total amount of cards that we play with is n . Thus, the two amounts of cards n and n' are asymptotically equal under the assumption $nq_n^2 \rightarrow C$, in which case the actual stable configuration is asymptotically equal to the stable configuration Λ described above.

Let A_i be the length of the i th section, $0 \leq i \leq Z - 1$, after downscaling Λ with our standard scaling factors (horizontal scaling by $\frac{z}{nq_n}$ and vertical scaling by $\frac{q_n}{z}$). Then

$$A_0 = \frac{z}{nq_n} \left\lceil \frac{1+z-Z}{q_n Z} \right\rceil \rightarrow \frac{z(1+z-Z)}{CZ} \text{ and}$$

$$A_k = \frac{z}{nq_n} \left\lceil \frac{1}{q_n(Z-k)} \right\rceil \rightarrow \frac{z}{C(Z-k)}, \quad 1 \leq k \leq Z - 1$$

as $n \rightarrow \infty$. The proposed slopes of the sections follow immediately. Analogously to the proof in case (a), it follows that the above describes the limit shape. $\square$

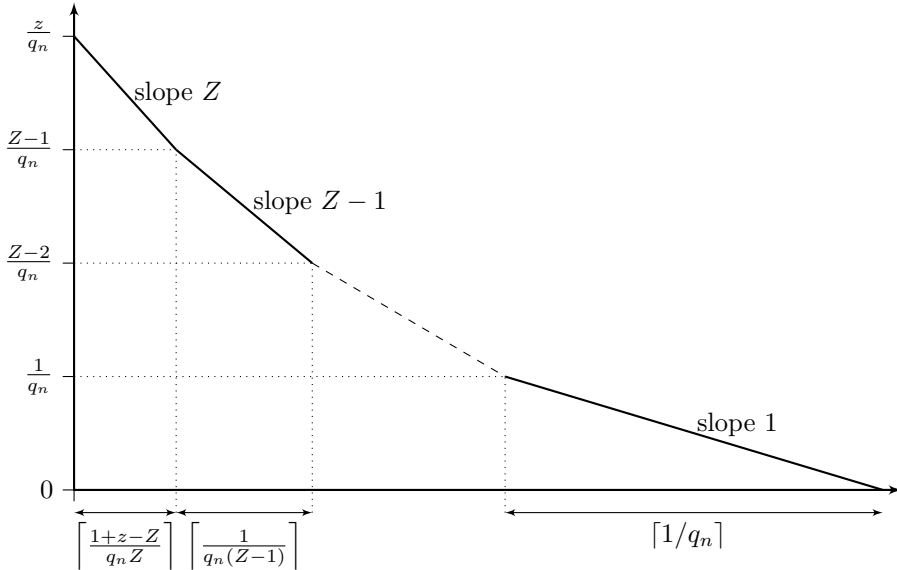


Figure 5: The stable configuration Λ in case (c) of the proof of Theorem 5.

9. Limit shapes of recurrent configurations of q_n -proportion Bulgarian solitaire

Although we have not been able to prove Conjecture 1 for general well-behaved σ -solitaire, we can prove the conjecture in the special cases of the two main regimes of q_n -proportion solitaire.

Lemma 7. *After n moves of q_n -proportion solitaire on n cards there are at most $2\sqrt{n}$ nonempty piles and the largest pile is at most of size $nq + O(\sqrt{n})$.*

Proof. Every nonempty pile decreases by at least one card in each move. As all pile sizes are bounded by n , all original piles must have died out after n moves. Moreover, because there are n cards in total there are always at most $\sqrt{n}$ piles of size greater than $\sqrt{n}$. After $\sqrt{n}$ moves all other piles will have died and $\sqrt{n}$ new piles will have been created, hence there will then be at most $2\sqrt{n}$ nonempty piles. From then on, when new piles are formed they will have size $nq + O(\sqrt{n})$, where the latter term is the contribution from the number of picked cards in each pile being rounded upwards to the closest integer. $\square$

9.1. The limit shape of recurrent configurations in the case $q_n^2 n \rightarrow 0$

In case $q_n^2 n \rightarrow 0$, Lemma 7 implies that after n moves the largest pile size is $O(\sqrt{n})$. Then the number of picked cards in each pile is bounded by $\lceil q_n O(\sqrt{n}) \rceil$. This number equals 1 for sufficiently large n . From then on the solitaire is therefore equivalent to ordinary Bulgarian solitaire. As the recurrent configurations of ordinary Bulgarian solitaire are known to converge to a triangular limit shape, it follows that the recurrent configurations of q_n -proportion Bulgarian solitaire do too in this case.

9.2. The limit shape of recurrent configurations in the case $q_n^2 n \rightarrow \infty$

Finally, we shall prove that in the case $q_n^2 n \rightarrow \infty$, the recurrent configurations of q_n -proportion solitaire have an exponential limit shape. We do this by showing that regardless of which configuration we start at we must eventually reach configurations that are close to the exponential shape. Our proof works with piles sorted by time of creation rather than by size. Thus, as mathematical objects the configurations are then compositions rather than partitions. However, as we prove in the companion paper [5], if a sequence of compositions has a decreasing limit shape then the same limit shape is obtained by the corresponding partitions.

Lemma 7 implies that after n moves the largest pile is always of size $nq_n + O(\sqrt{n}) = n(q_n + o(1))$ and that after an additional $2\sqrt{n}$ moves all non-empty piles will be stemming from piles of that size. At this point, let α_i denote the size of the pile that was created i moves ago ($i = 1, 2, \dots$) and has since been decreased $i - 1$ times. Thus $\alpha_i = (1 - q_n)^{i-1} n(q_n + o(1)) - O(i)$ where the latter term is the contribution from rounding downward in each move.

Let $x = x(i)$, $x \geq 0$, be the x -coordinate of the column corresponding to α_i in the composition diagram of α , and y the height of this column, after downscaling:

$$x = (i - 1)(q_n + o(1)), \quad y = \frac{(1 - q_n)^{i-1} n(q_n + o(1)) - O(i)}{n(q_n + o(1))}. \quad (4)$$

Then $i = 1 + \frac{x}{q_n + o(1)}$ and

$$\begin{aligned} y(x) &= \frac{(1 - q_n)^{x/(q_n + o(1))} n(q_n + o(1)) - O(\frac{x}{q_n + o(1)})}{n(q_n + o(1))} \\ &= (1 - q_n)^{x/(q_n + o(1))} - O\left(\frac{x}{n(q_n + o(1))^2}\right) \\ &= (1 - q_n)^{x/(q_n + o(1))} - o(1), \end{aligned}$$

where we used $q_n^2 n \rightarrow \infty$ as $n \rightarrow \infty$ in the last step. Since $(1 - q_n)^{x/q_n} \rightarrow e^{-x}$, we therefore have $y(x) \rightarrow e^{-x}$. Recall that $q_n \rightarrow 0$ as $n \rightarrow \infty$. Thus, by (4), for a fixed i we have $x = x(i) \rightarrow 0$ as $n \rightarrow \infty$. In other words, the width of each column in the

composition diagram for α (after downscaling) is $q_n + o(1) \rightarrow 0$. This means that the limit $y(x) \rightarrow e^{-x}$ holds for any $x \geq 0$, which, by the definition (2), establishes the limit shape for the q_n -proportion Bulgarian solitaire.

Finally, thanks to the abovementioned result from [5], the same limit shape is obtained when the piles of the weak compositions are reordered to form partitions.

10. Discussion

In this paper we have discussed limit shape results for stable and recurrent configurations. Popov [11] studied the limit shape of the configurations drawn from the stationary distribution of a random version of Bulgarian solitaire, in which a card is picked from a pile only with probability p (and independently of other piles). He found that also this random version yields a triangular limit shape, in the sense that the probability of deviations larger than some $\varepsilon > 0$ tends to zero as n tends to infinity. In a companion paper [5] to the present paper we study a random version of q_n -proportion Bulgarian solitaire and the conditions under which it has an exponential limit shape.

References

- [1] E. Akin and M. Davis, *Bulgarian solitaire*, The American Mathematical Monthly **92** (1985), no. 4, pp. 237–250.
- [2] H.J. Bentz, *Proof of the Bulgarian solitaire conjectures*, Ars Combin. **23** (1987), 151–170.
- [3] J. Brandt, *Cycles of partitions*, Proc. Amer. Math. Soc. **85** (1982), no. 3, pp. 483–486.
- [4] V. Drensky, *The Bulgarian solitaire and the mathematics around it*, arXiv:1503.00885v1 [math.CO] (2015).
- [5] K. Eriksson, M. Jonsson, and J. Sjöstrand, *An exponential limit shape of random q -proportion Bulgarian solitaire*, arXiv:1703.07102 [math.PR] (2017).
- [6] K. Eriksson and J. Sjöstrand, *Limiting shapes of birth-and-death processes on Young diagrams*, Adv. in Appl. Math. **48** (2012), no. 4, 575 – 602.
- [7] G. Etienne, *Tableaux de Young et solitaire bulgare*, J. Combin. Theory Ser. A **58** (1991), no. 2, 181–197.

- [8] J.R. Griggs and C.C. Ho, *The cycling of partitions and compositions under repeated shifts*, Adv. in Appl. Math. **21** (1998), no. 2, 205–227.
- [9] B. Hopkins, *30 years of Bulgarian solitaire*, College Math. J. **43** (2012), no. 2, 136–141.
- [10] J.S. Olson, *Variants of Bulgarian solitaire*, Integers **16** (2016), A8.
- [11] S. Popov, *Random Bulgarian solitaire*, Random Structures Algorithms **27** (2005), no. 3, 310–330.
- [12] A. Vershik, *Statistical mechanics of combinatorial partitions, and their limit shapes*, Funct. Anal. Appl. **30** (1996), no. 2, 90–105.

Paper III

AN EXPONENTIAL LIMIT SHAPE OF RANDOM q -PROPORTION BULGARIAN SOLITAIRE

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Abstract

We introduce p_n -random q_n -proportion Bulgarian solitaire ($0 < p_n, q_n \leq 1$), played on n cards distributed in piles. In each pile, a number of cards equal to the proportion q_n of the pile size rounded upward to the nearest integer are candidates to be picked. Each candidate card is picked with probability p_n , independently of other candidate cards. This generalizes Popov's random Bulgarian solitaire, in which there is a single candidate card in each pile. Popov showed that a triangular limit shape is obtained for a fixed p as n tends to infinity. Here we let both p_n and q_n vary with n . We show that under the conditions $q_n^2 p_n n / \log n \rightarrow \infty$ and $p_n q_n \rightarrow 0$ as $n \rightarrow \infty$, the p_n -random q_n -proportion Bulgarian solitaire has an exponential limit shape.

1. Introduction

The game of Bulgarian solitaire has received a great deal of attention, see reviews by Hopkins [10] and Drensky [2]. The Bulgarian solitaire is played with a deck of n identical cards divided arbitrarily into a number of piles. A move consists of picking a card from each pile and letting these cards form a new pile. If piles are sorted

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in order of decreasing size, every position in the solitaire is equivalent to a Young diagram of an integer partition of n .

Popov [15] considered a random version of Bulgarian solitaire defined by a probability $p \in (0, 1]$, such that one card from each pile is picked with probability p , independently of the other piles. We will refer to this stochastic process on configurations as *p-random Bulgarian solitaire*. The probabilities of configurations converge to a stationary distribution. Popov showed that as n grows to infinity and configuration diagrams are downscaled by $\sqrt{n}$ in both dimensions, the stationary probability of the set of configurations that deviate from a triangle with slope p by more than $\varepsilon > 0$ tends to zero. In this sense, random configurations has a limit shape.

The objective of the present paper is to study such limit shapes in a generalization of random Bulgarian solitaire.

1.1. q_n -proportion Bulgarian solitaire

Olson [13] introduced a generalization of Bulgarian solitaire in which the number of cards that are picked from a pile of size h is given by some non-negative valued function $\sigma(h)$. Eriksson, Jonsson and Sjöstrand [3] recently studied the special case when σ is well-behaved in the sense that $\sigma(1) = 1$ and both $\sigma(h)$ and $h - \sigma(h)$ are non-decreasing functions of h . In particular, they studied a special case that they called *q_n -proportion Bulgarian solitaire*, defined by the rule $\sigma(h) = \lceil q_n h \rceil$. This means that from each pile we pick a number of cards given by the proportion q_n of the pile size rounded upward to the nearest integer. To illustrate the effect of the parameter q_n , set it to 0.3 and consider the configuration $(6, 2, 2, 1)$. From the first pile we pick $\lceil 0.3 \times 6 \rceil = 2$ cards; similar calculations give that 1 card is picked from each of the other three piles. Note that for $q_n \leq 1/n$ exactly one card is always picked from each pile, retrieving the ordinary Bulgarian solitaire.

As n tends to infinity, Eriksson, Jonsson and Sjöstrand [3] determined limit shapes of stable configurations of q_n -proportion Bulgarian solitaire: In case $q_n^2 n \rightarrow 0$, the limit shape is triangular, which generalizes the limit shape result for the ordinary Bulgarian solitaire. For other asymptotic behavior of q_n , other limit shapes were obtained. Specifically, in case $q_n^2 n \rightarrow \infty$, the limit shape is exponential. The intermediate case $q_n^2 n \rightarrow C > 0$ produces a family of limit shapes interpolating between the triangular and the exponential shape.

1.1.1. p_n -random q_n -proportion Bulgarian solitaire

We shall examine a p_n -random version of q_n -proportion Bulgarian solitaire, in which the proportion q_n (rounded upward) of cards in a pile are only *candidates* to be picked, each of which is picked only with probability p_n , independently of all other candidate cards. This process will be denoted by $\mathcal{B}(n, p_n, q_n)$. Note that in the

special case of a fixed p and for $q_n \leq 1/n$, this process is equivalent to Popov's p -random Bulgarian solitaire.

Our focus will be on establishing a regime in which p_n -random q_n -proportion Bulgarian solitaire has an exponential limit shape.

2. The concept of limit shapes

In this section we give the precise definitions of the limit shapes we consider. Let $\mathcal{P}(n)$ be the set of integer partitions of n . For any partition $\lambda \in \mathcal{P}(n)$ with $N = N(\lambda)$ positive parts $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N > 0$, define $\lambda_i = 0$ for $i > N(\lambda)$, and the *diagram* of λ as the Young diagram oriented such that the parts of λ are represented by left and bottom aligned columns, weakly decreasing in height from left to right. For example, $\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}$ is the diagram of the partition $(2, 1, 1)$. We define the *diagram-boundary function* of λ as the nonnegative, weakly decreasing and piecewise constant function $\partial\lambda : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ describing the boundary of λ , given by

$$\partial\lambda(x) = \lambda_{\lfloor x \rfloor + 1}.$$

Following [4] and [16], the diagram is downscaled using some *scaling factor* $a_n > 0$ such that all row lengths are multiplied by $1/a_n$ and all column heights are multiplied by a_n/n , yielding a constant area of 1. Following [3], we shall consistently make the choice $a_n = n/\lambda_1$, such that the height of the diagram is scaled to 1.

Thus, given a partition λ , define the *rescaled diagram-boundary function* of λ as the nonnegative, real-valued, weakly decreasing and piecewise constant function $\partial\lambda : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ given by

$$\partial\lambda(x) = \frac{1}{\lambda_1} \partial\lambda(xn/\lambda_1) = \frac{1}{\lambda_1} \lambda_{\lfloor xn/\lambda_1 \rfloor + 1}. \quad (1)$$

The p_n -random q_n -proportion Bulgarian solitaire $\mathcal{B}(n, p_n, q_n)$ (with $p_n, q_n \in (0, 1]$) can be regarded as a Markov chain on the finite state-space $\mathcal{P}(n)$. Let us denote the sequence of visited states by $(\lambda^{(0)}, \lambda^{(1)}, \dots)$. In the truly random case of $p_n < 1$, it is straightforward to verify that this Markov chain is aperiodic and irreducible. It is well-known that an aperiodic and irreducible Markov chain on a finite state-space has a unique stationary distribution π and that starting from any initial state the distribution of the i th state $\lambda^{(i)}$ converges to π as i tends to infinity. We denote by π_{n, p_n, q_n} the stationary measure of the Markov chain $(\lambda^{(0)}, \lambda^{(1)}, \dots)$ on $\mathcal{P}(n)$ given by $\mathcal{B}(n, p_n, q_n)$ for $p_n < 1$.² When we refer to a limit shape of the

²Readers acquainted with the limit shape literature may wonder whether the stationary measure has the property of being *multiplicative*, in the sense of interpretable as the product measure on the space of integer sequences restricted to a certain affine subspace [7]. The multiplicative property is useful in limit shape problems and related problems [1, 5, 8, 14, 16]. However, such techniques will not be used here as π_{n, p_n, q_n} is unlikely to be multiplicative in general.

process $\mathcal{B}(n, p_n, q_n)$ for $p_n < 1$ as n grows to infinity, we shall mean the limit shape of the stationary measure π_{n, p_n, q_n} . The intuitive sense of this concept is that *when the solitaire is played on a sufficiently large number of cards for sufficiently long the configuration will almost surely be very close to the limit shape after suitable downscaling*. Following Vershik [16], a sequence $\{\pi_n\}$ of probability distributions on $\mathcal{P}(n)$ is said to have a limit shape ϕ if the downscaled diagrams approach ϕ in probability as n grows to infinity. The exact condition for convergence can vary. Consistent with Yakubovich [17] and Eriksson and Sjöstrand [4], we shall use the definition that

$$\lim_{n \rightarrow \infty} \pi_n \{ \lambda \in \mathcal{P}(n) : |\partial^{a_n} \lambda(x) - \phi(x)| < \varepsilon \} = 1 \quad (2)$$

for all $x > 0$ and all $\varepsilon > 0$.³

3. The approach of ordering piles by time of creation

It will sometimes be useful to explicitly order piles by time of creation rather than by size. Here we develop this approach.

When parts are not sorted by size, a configuration is not represented by an integer partition but by a *weak integer composition*: an infinite sequence $\alpha = (\alpha_1, \alpha_2, \dots)$, not necessarily decreasing, of nonnegative integers adding up to n . Let $\mathcal{W}(n)$ denote the set of weak compositions of n . We define the *diagram*, the *diagram-boundary function* $\partial\alpha$, and the *rescaled diagram-boundary function* $\partial\alpha$ of a weak composition α in exact analogy to the way we defined them for partitions in Section 2. For example, the diagram of $\alpha = (3, 0, 2, 4, 1, 0, 0, \dots)$ and the corresponding function graph $y = \partial\alpha(x)$ are shown in Figure 1. Also, for a weak composition $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_N, 0, 0, \dots)$ we define the number of parts $N = N(\alpha)$ disregarding the trailing zeros.

3.1. Connecting the limit shapes of compositions and partitions

We shall now connect compositions with partitions. For any $\alpha \in \mathcal{W}(n)$, define the operator ord as the ordering operator that arranges the parts of α in descending order, thus yielding a partition. We shall now prove that such sorting of the piles by size respects the convergence to a limit shape. The proof uses some basic theory

³Vershik [16] and Erlihson and Granovsky [6] used a stronger condition for convergence toward a limit shape, namely that

$$\lim_{n \rightarrow \infty} \pi_n \left\{ \lambda \in \mathcal{P}(n) : \sup_{x \in [a, b]} |\partial^{a_n} \lambda(x) - \phi(x)| < \varepsilon \right\} = 1$$

should hold for any compact interval $[a, b]$, and any $\varepsilon > 0$.

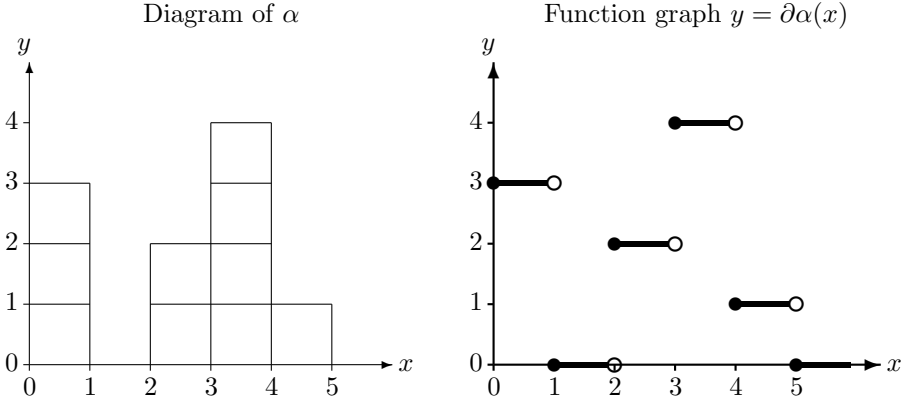


Figure 1: The composition $\alpha = (3, 0, 2, 4, 1, 0, 0, \dots) \in \mathcal{W}(10)$.

of symmetric-decreasing rearrangements, see for example [9, Ch. 10] or [11, Ch. 3]. For any measurable function $f: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ such that $\lim_{x \rightarrow \pm\infty} f(x) = 0$, there is a unique function $f^*: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, called the *symmetric-decreasing rearrangement* of f , with the following properties:

- f^* is symmetric, that is, $f^*(-x) = f^*(x)$ for all x ,
- f^* is weakly decreasing on the interval $[0, \infty)$,
- f^* and f are equimeasurable, that is,

$$\mathcal{L}(\{x : f(x) > t\}) = \mathcal{L}(\{x : f^*(x) > t\})$$

for all $t > 0$, where $\mathcal{L}$ denotes the Lebesgue measure,

- f^* is lower semi-continuous.

In particular, if f is a symmetric function that is weakly decreasing and right-continuous on $[0, \infty)$ and tends to 0 at infinity, then $f^* = f$.

Lemma 1. *Let $\alpha \in \mathcal{W}(n)$ be a weak composition of n and let $f: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be a right-continuous and weakly decreasing function such that $f(x) \rightarrow 0$ as $x \rightarrow \infty$. The downscaled diagram-boundary functions before and after sorting of the weak composition satisfy the inequality*

$$\|\partial \text{ord } \alpha - f\|_{\infty} \leq \|\partial \alpha - f\|_{\infty}.$$

where $\|\cdot\|_{\infty}$ denotes the max-norm $\|f\|_{\infty} = \sup \{|f(x)| : x \geq 0\}$.

Proof. The intuition of the lemma should be obvious from Figure 2. To be able to use the standard machinery of symmetric rearrangements, we consider the functions

f , $\partial\alpha$, and $\partial\text{ord}\alpha$ as being defined on the entire real axis by letting $f(x) = f(|x|)$ and analogously for $\partial\alpha$, and $\partial\text{ord}\alpha$.

Since $f(x) \rightarrow 0$ as $x \rightarrow \infty$, its symmetric-decreasing rearrangement f^* is defined and, since f is weakly decreasing and lower semi-continuous, we have $f^* = f$. Similarly, $\partial\text{ord}\alpha(x) \rightarrow 0$ as $x \rightarrow \infty$ and is weakly decreasing, so $(\partial\text{ord}\alpha)^* = \partial\text{ord}\alpha$. Moreover, $(\partial\alpha)^* = \partial\text{ord}\alpha$ must hold because the operator ord arranges the composition parts in descending order.

Now, since symmetric rearrangements decrease L^p -distances for any $1 \leq p \leq \infty$ (see for example [11], Section 3.4), we obtain

$$\|\partial\text{ord}\alpha - f\|_\infty = \|(\partial\alpha)^* - f^*\|_\infty \leq \|\partial\alpha - f\|_\infty.$$

□

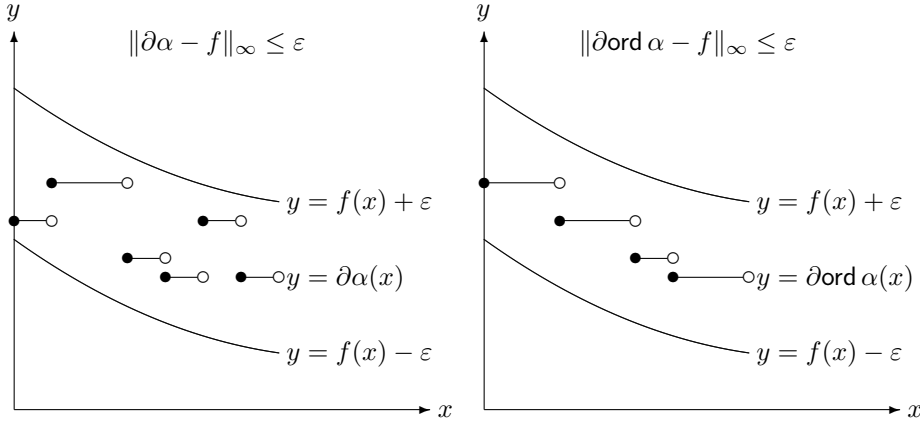


Figure 2: An example of a composition α and a decreasing function f showing that if $\partial\alpha(x)$ is enclosed between $f(x) - \varepsilon$ and $f(x) + \varepsilon$, then so is $\partial\text{ord}\alpha(x)$, an immediate consequence of Lemma 1.

Clearly, Lemma 1 holds true also when the max-norm is replaced by the weaker convergence condition used in our limit shape definition (2).

Lemma 2. For any distribution π_n on $\mathcal{W}(n)$, define a corresponding distribution $\tilde{\rho}^{(n)}$ on $\mathcal{P}(n)$ by

$$\tilde{\rho}^{(n)}(\lambda) = \sum_{\substack{\alpha \in \mathcal{W}(n) \\ \text{ord } \alpha = \lambda}} \pi_n(\alpha). \quad (3)$$

If ϕ is a limit shape of π_n on $\mathcal{W}(n)$ then ϕ is also a limit shape of $\tilde{\rho}^{(n)}$ on $\mathcal{P}(n)$.

Proof. The assumption that ϕ is a limit shape of the distribution π_n on $\mathcal{W}(n)$ means that

$$\lim_{n \rightarrow \infty} \pi_n \{ \alpha \in \mathcal{W}(n) : |\partial\alpha(x) - \phi(x)| < \varepsilon \} = 1.$$

for all $x > 0$. By virtue of Lemma 1 we can replace α with $\text{ord } \alpha$ in this formula:

$$\lim_{n \rightarrow \infty} \pi_n \{ \alpha \in \mathcal{W}(n) : |(\partial \text{ord } \alpha)(x) - \phi(x)| < \varepsilon \} = 1. \quad (4)$$

The set $A := \{ \alpha \in \mathcal{W}(n) : |(\partial \text{ord } \alpha)(x) - \phi(x)| < \varepsilon \text{ for all } x > 0 \}$ can be written as a disjoint union of equivalence classes with respect to sorting:

$$A = \bigcup_{\lambda \in L} \{ \alpha \in \mathcal{W}(n) : \text{ord } \alpha = \lambda \}$$

where $L = \{ \lambda \in \mathcal{P}(n) : |\partial \lambda(x) - \phi(x)| < \varepsilon \text{ for all } x > 0 \}$. The π_n -probability measure of A is

$$\begin{aligned} \pi_n(A) &= \pi_n \left(\bigcup_{\lambda \in L} \{ \alpha \in \mathcal{W}(n) : \text{ord } \alpha = \lambda \} \right) \\ &= \sum_{\lambda \in L} \pi_n \{ \alpha \in \mathcal{W}(n) : \text{ord } \alpha = \lambda \} \\ &= \sum_{\lambda \in L} \sum_{\substack{\alpha \in \mathcal{W}(n) \\ \text{ord } \alpha = \lambda}} \pi_n(\alpha) \\ &= \sum_{\lambda \in L} \tilde{\rho}^{(n)}(\lambda) \quad (\text{by (3)}) \\ &= \tilde{\rho}^{(n)}(L). \end{aligned}$$

From (4) we have that $\lim_{n \rightarrow \infty} \pi_n(A) = 1$. Because $\pi_n(A) = \tilde{\rho}^{(n)}(L)$, we can conclude that also $\lim_{n \rightarrow \infty} \tilde{\rho}^{(n)}(L) = 1$, that is,

$$\lim_{n \rightarrow \infty} \tilde{\rho}^{(n)} \{ \lambda \in \mathcal{P}(n) : |\partial \lambda(x) - \phi(x)| < \varepsilon \text{ for all } x > 0 \} = 1.$$

This means that ϕ is a limit shape of the distribution π_n on $\mathcal{P}(n)$. $\square$

4. Three regimes

Recall from Section 1.1 the q_n -proportion Bulgarian solitaire developed in [3], where the limit shape is triangular when $q_n^2 n \rightarrow 0$, exponential when $q_n^2 n \rightarrow \infty$ and an interpolation between the two when $q_n^2 n \rightarrow C > 0$.

The p_n -random q_n -proportion Bulgarian solitaire seems to share this property of three regimes of limit shapes. Specifically, in Section 7 we conjecture the limit shape to be triangular when $p_n q_n^2 n \rightarrow 0$, exponential when $p_n q_n^2 n \rightarrow \infty$ and an interpolation between the two (a piecewise linear function graph that depends on C) when $p_n q_n^2 n \rightarrow C > 0$.

The focus in this paper is the exponential regime of the p_n -random q_n -candidate Bulgarian solitaire, i.e. the case $p_n q_n^2 n \rightarrow \infty$ as $n \rightarrow \infty$. However, with the proof technique we employ we will prove the stronger statement that the limit shape holds even when the configurations are considered elements of $\mathcal{W}(n)$, i.e. even without sorting the piles of a configuration according to size to create a partition in $\mathcal{P}(n)$. We will instead require the stronger condition $p_n q_n^2 n / \log n \rightarrow \infty$ as $n \rightarrow \infty$. By virtue of Lemma 2, the limit shape will also hold for partitions.

5. The exponential limit shape

Here we investigate the limit shape of configurations in the p_n -random q_n -candidate Bulgarian solitaire $\mathcal{B}(n, p_n, q_n)$ in the regime

$$\frac{p_n q_n^2 n}{\log n} \rightarrow \infty \text{ as } n \rightarrow \infty. \quad (5)$$

Our main result, Theorem 1, says that, under the additional asymptotic property $p_n q_n \rightarrow 0$ as $n \rightarrow \infty$, the boundary function of the diagram, downscaled, will resemble the exponential shape e^{-x} asymptotically almost surely (a.a.s.), i.e. with a probability that tends to 1 as $n \rightarrow \infty$. See Figure 3. Throughout this section, “a.a.s.” can be read as “with a probability that tends to 1 as $n \rightarrow \infty$ ”. Also, the asymptotic notations o and O will always be with respect to $n \rightarrow \infty$.

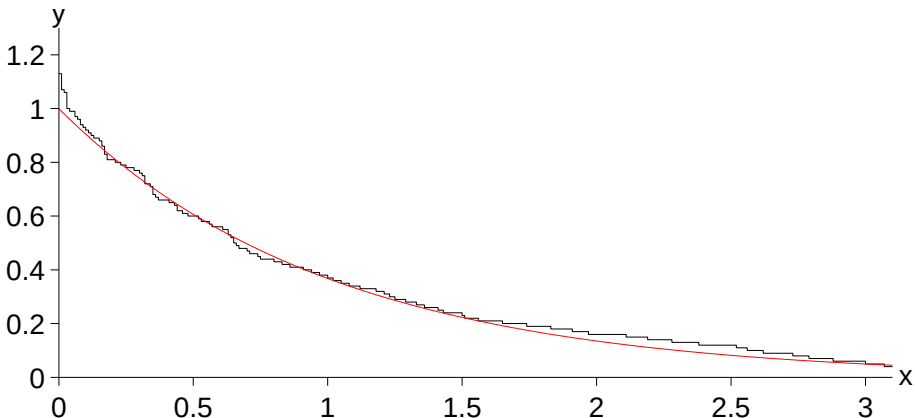


Figure 3: The result of a computer simulation after 200 moves of p_n -random q_n -proportion Bulgarian solitaire in the case $q_n = 1$, with $n = 10^5$ cards and $p_n = 0.01$, starting from a triangular configuration. The jagged curve is the rescaled diagram-boundary function of the resulting configuration and the smooth curve is the limit shape $y = e^{-x}$.

We shall see that the condition $p_n q_n^2 n / \log n \rightarrow \infty$ implies that the rounding effect in computing the number of candidate cards is negligible. Thus, the number of candidate cards will tend to $q_n n$ as $n \rightarrow \infty$. This in turn means that the expected number of picked cards will eventually be close to $p_n q_n n$, thus $\lambda_1 \approx p_n q_n n$ is the size of the pile created in a move of the solitaire. Recall from Section 2 that the scaling factor we employ is $a_n = n / \lambda_1 = \frac{1}{p_n q_n}$. Thus, if $p_n q_n$ is bounded away from zero, then the scaling $\frac{1}{p_n q_n}$ is bounded and hence cannot transform the jumpy boundary diagrams into a smooth limit shape. Therefore, we also require

$$p_n q_n \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (6)$$

On the other hand, if $p_n q_n$ tends to zero too fast, the pile sizes will be small and their random fluctuations will be large. For instance, the new pile after each move has a size drawn from the binomial distribution $\text{Bin}(K, p_n)$, where $K \approx q_n n$ is the number of candidate cards, with relative standard deviation $\sim 1/\sqrt{p_n q_n n}$. The requirement (5) guarantees that $p_n q_n$ does not tend to zero too fast.

Theorem 1. *For each positive integer n , pick q_n and p_n with $0 < p_n, q_n \leq 1$ and a (possibly random) initial configuration $\lambda^{(0)} \in \mathcal{P}(n)$. Let $(\lambda^{(0)}, \lambda^{(1)}, \dots)$ be the Markov chain on $\mathcal{P}(n)$ defined by $\mathcal{B}(n, p_n, q_n)$, and denote its stationary measure by π_{n, p_n, q_n} . Suppose*

$$p_n q_n \rightarrow 0 \quad \text{and} \quad \frac{p_n q_n^2 n}{\log n} \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

Then π_{n, p_n, q_n} has the limit shape e^{-x} under the scaling $a_n = (p_n q_n)^{-1}$.

The proof of Theorem 1 heavily relies on the following version of Chernoff bounds. For a proof, see for example [12].

Chernoff Bound. *For $n \geq 1$ and $0 < p \leq 1$, let $X \sim \text{Bin}(n, p)$ and set $\mu = E(X) = np$. Then, for any $0 < \gamma < \mu$,*

$$P(|X - \mu| \geq \gamma) \leq 2 \exp\left(-\frac{\gamma^2}{3\mu}\right). \quad (7)$$

The idea of the proof of Theorem 1 is the following.

We will use the approach developed in Section 3, i.e. card configurations in the solitaire will be represented by weak integer compositions and the piles are ordered with respect to creation time, i.e. if $\alpha \in \mathcal{W}(n)$ is the current configuration in the solitaire, then α_1 was the last formed pile, α_2 the pile that was formed two moves ago, etc. With this representation, some piles may be empty, so one may imagine each pile being placed in a bowl and the bowls are lined up in a row on the table. In each move of the solitaire, the new (possibly empty) pile is put in a new bowl to the left of all old bowls. As mentioned in Section 4, we shall prove Theorem 1 as a limit

shape result for diagram-boundary functions of *compositions*. Thus, throughout this section, each configuration of n cards will be represented by an element of $\mathcal{W}(n)$. Also, in the following we may abbreviate $p = p_n$ and $q = q_n$ unless the dependency on n is crucial.

Assume a configuration $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_N, 0, 0, \dots)$ of n cards with $N = N(\alpha)$ piles (so that $\sum_{i=1}^N \alpha_i = n$) in the solitaire $\mathcal{B}(n, p, q)$. The number of candidate cards in the next move is $\kappa := \sum_{i=1}^N \lceil q\alpha_i \rceil$. We denote the rounding effect in pile $1 \leq i \leq N$ by $R_i := \lceil q\alpha_i \rceil - q\alpha_i$ and the total rounding effect by $R := \kappa - qn = \sum_{i=1}^N R_i$.

Clearly, $R < N$ (since $R_i < 1$ for any i), i.e. the total rounding effect is bounded above by the number of piles. The first thing we will do is to make sure that after a sufficient number D of moves from the initial configuration $\alpha^{(0)}$, the number of piles $N(\alpha^{(D)})$ in the resulting configuration $\alpha^{(D)}$ is much smaller than qn a.s. (so that the number of candidate cards κ is approximately qn and thus the total rounding effect R is negligible). In Lemma 4 we show that it is possible to choose such a D , namely $D = c \frac{\log n}{pq}$ for any $c \geq 14$.

We also need to make sure that the number of piles stays $o(qn)$ for sufficiently many additional moves M , long enough to establish the convergence of the overall shape. Lemma 4 will also guarantee that $M = \lceil n^2/p \rceil$ suffices for this purpose.

Thus, in the following we shall use

$$D = \left\lceil 14 \frac{\log n}{pq} \right\rceil \quad \text{and} \quad M = \left\lceil \frac{n^2}{p} \right\rceil. \quad (8)$$

If the number of piles stays $o(qn)$ during M moves so that the number of candidate cards stays approximately qn , the newly formed pile in each of these moves will have expected size pqn . Our proof technique involves studying the evolution of such a pile (which will follow an exponential decay in size). Therefore we need to additionally make sure that no old piles (which could potentially be much larger than pqn) remain after these M moves. Lemma 3 shows that, in fact, after M moves all piles in the starting configuration have disappeared a.s.

Lemma 3. *Let M be given by (8). From any initial configuration $\alpha \in \mathcal{W}(n)$, after M moves in the solitaire $\mathcal{B}(n, p_n, q_n)$, all piles in α have been consumed a.s.*

Proof. Consider a pile of size n . The size of this pile after M moves is statistically dominated by $\max(n - X, 0)$ where $X \sim \text{Bin}(M, p_n)$ whose expected value is $E(X) = Mp_n = \lceil n^2/p_n \rceil p_n > n$. Therefore, the probability that the pile remains after M moves is $P(X < n)$ with the bound

$$\begin{aligned} P(X < n) &\leq P(|X - Mp_n| > |Mp_n - n|) \leq 2 \exp \left(-\frac{(Mp_n - n)^2}{3Mp_n} \right) \\ &\leq 2 \exp \left(-\frac{n^2}{3} (1 + o(1)) \right), \end{aligned}$$

where we used the Chernoff bound (7). Thus, since any given pile in α has size $\leq n$, and the number of piles (of any size in any configuration) is $\leq n$, the probability that *all* piles in α have been consumed after M moves is at least

$$1 - 2n \exp\left(-\frac{n^2}{3}(1 + o(1))\right) \rightarrow 1$$

which concludes the proof. $\square$

Lemma 4. *Let n, p_n, q_n and an initial configuration $\alpha^{(0)}$ be given in the solitaire $\mathcal{B}(n, p_n, q_n)$. Then*

$$\frac{1}{q_n n} \max \left\{ N(\alpha^{(D+1)}), \dots, N(\alpha^{(D+M)}) \right\} \rightarrow 0 \text{ in probability,}$$

where D and M are given by (8).

Proof. Let us abbreviate $p = p_n$ and $q = q_n$. We will first prove that all piles of size at most $q^{-1} \log n$ disappear with high probability after D moves, making sure that there are not many small piles in $\alpha^{(D)}$. Consider a pile of size at most $q^{-1} \log n$ in $\alpha^{(0)}$. Note that every nonempty pile decreases by at least 1 with probability at least p in each move. Therefore, after D moves the number of picked cards from this pile statistically dominates $X \sim \text{Bin}(D, p)$ with expected value $Dp = 14q^{-1} \log n$. Using the Chernoff bound (7), the probability that this pile remains after D moves is at most

$$\begin{aligned} P_1 &:= P\left(X < \frac{\log n}{q}\right) \leq P\left(|X - Dp| > \left|Dp - \frac{\log n}{q}\right|\right) \\ &\leq 2 \exp\left(-\frac{\left(Dp - \frac{\log n}{q}\right)^2}{3Dp}\right) = 2 \exp\left(-\frac{\left(\frac{14 \log n}{q} - \frac{\log n}{q}\right)^2}{3 \cdot 14 \frac{\log n}{q}}\right) = 2n^{-\frac{13^2}{42} \frac{1}{q}} < 2n^{-4}. \end{aligned}$$

Since there can be at most n piles of size at most $q^{-1} \log n$, the probability that not *all* piles of size at most $q^{-1} \log n$ have disappeared after D moves is bounded by

$$P_2 := nP_1 = 2n^{-3}.$$

Let us now turn our attention to *the number of* piles after these D moves. By the above, all piles smaller than $q^{-1} \log n$ have disappeared with high probability. Clearly, the number of piles *larger than* $q^{-1} \log n$ can never be more than $\frac{n}{q^{-1} \log n} = \frac{qn}{\log n}$. Also, during the process of these D moves, at most D new piles have been formed. (Exactly D piles have been formed but some may have disappeared in the process.) Thus, for the total number of piles $N(\alpha^{(D)})$ in the configuration $\alpha^{(D)}$ after D moves, with probability at least $1 - 2n^{-3}$, we have

$$N(\alpha^{(D)}) \leq \frac{qn}{\log n} + D = \frac{qn}{\log n} + 14 \frac{\log n}{pq} = qn \left(\frac{1}{\log n} + 14 \frac{\log n}{pq^2 n} \right) = o(qn),$$

where we used the assumption (5) in the last step. It follows that, for any $\varepsilon > 0$,

$$\frac{1}{qn} \max \left\{ N(\alpha^{(D+1)}), \dots, N(\alpha^{(D+M)}) \right\} < \varepsilon$$

with probability at least $1 - 2n^{-3}M \geq 1 - \frac{2}{pn} \rightarrow 1$ since $pn \rightarrow \infty$. (That $pn \rightarrow \infty$ is also a consequence of the assumption (5).) $\square$

Lemma 4 asserts that the number of piles remains to be $o(qn)$ during the M moves from $\alpha^{(D)}$ to $\alpha^{(D+M)}$, hence the number of candidate cards remains to be qn (a.a.s.) during the same moves. Therefore the number of picked cards (which equals the size of the newly formed pile), remains of *expected* size pqn . In Lemma 5 we prove that the *actual* number of picked cards in each of these M moves does not deviate (relatively) from pqn .

Lemma 5. *Let n, p_n, q_n and an initial configuration $\alpha^{(0)}$ be given in the solitaire $\mathcal{B}(n, p_n, q_n)$. Let D and M be given by (8). Then*

$$\max_{k \in [D+1, D+M]} \frac{|\alpha_1^{(k)} - p_n q_n n|}{p_n q_n n} \rightarrow 0 \text{ in probability}$$

as $n \rightarrow \infty$.

Proof. Let us abbreviate $p = p_n$ and $q = q_n$. Let $\varepsilon > 0$ and let κ be the number of candidate cards in $\alpha^{(k-1)}$ for some $k = D+1, \dots, D+M$. Recall that the total rounding effect in computing the number of candidate cards is bounded above by the number of piles. It therefore follows from Lemma 4 that $\kappa = nq(1 + o(1))$. The new pile size is $\alpha_1^{(k)} \sim \text{Bin}(\kappa, p)$. Then, using the triangle inequality and the Chernoff bound (7) we have

$$\begin{aligned} P_3 &:= P(|\alpha_1^{(k)} - pqn| > \varepsilon pqn) \leq P(|\alpha_1^{(k)} - \kappa p| > \varepsilon pqn - |\kappa p - pqn|) \\ &< 2 \exp \left(- \frac{(\varepsilon pqn - |\kappa p - pqn|)^2}{3\kappa p} \right) \\ &= 2 \exp \left(- \frac{\varepsilon^2}{3} pqn(1 + o(1)) \right) \\ &= o(1/M) \end{aligned}$$

where the last equality is derived as follows. By (5), $\log n = o(pqn)$ and hence $\log n^a = o(pqn)$ for any $a \geq 1$. Since $pqn \rightarrow \infty$, this means that $\exp(-pqn)$ tends to zero faster than $\exp(-\log n^a)$, i.e., $\exp(-pqn) = o(1/n^a)$. Since $np \rightarrow \infty$, we therefore also have $\exp(-pqn) = o(p/n^a) = o(1/M)$. The next to the last equality follows from the fact that εpqn dominates over $|pqn - \kappa p|$ (since $|pqn - \kappa p| = |pqn - nq(1 + o(1))p| = pqn \cdot o(1)$).

Therefore, the probability that $|\alpha_1^{(k)} - pqn| > \varepsilon pqn$ for *any* k during the entire process of M moves is bounded by $MP_3 = M \cdot o(1/M) = o(1)$. $\square$

While playing the solitaire, there is a possibility that at some point there will be too many piles, and thereby the number of candidate cards will be bigger than qn (and thus the size of the newly formed pile will be bigger than pqn). Lemmas 3 and 4 ensures that this never happens a.a.s. during the entire process of M moves from $\alpha^{(D)}$ to $\alpha^{(D+M)}$.

There is also a risk that, even if there are suitably many (qn) candidate cards, the number of *picked* cards among them will deviate from pqn due to random fluctuations (and thereby the size of the newly formed pile will deviate from pqn). Lemma 5 ensures that this never happens a.a.s. during the same period of M moves.

Therefore, after $m := D + M$ moves we have the following a.a.s.

- all current piles have been formed during the last M moves, and
- all current piles had size pqn when they were formed.

At this point, i.e. in the configuration $\Gamma := \alpha^{(m)}$, the leftmost pile (of size Γ_1) was formed one move ago, the second pile from the left (of size Γ_2) was formed two moves ago, and so on. We shall prove that the size Γ_k of the pile that was formed k moves ago for any $k = 1, 2, \dots, m$ is $\Gamma_k = \Gamma_1(1 - pq)^k = pqn(1 - pq)^k$ a.a.s., i.e. the size decreases exponentially with k with decay factor $1 - pq$.

We will now consider the evolution of a given pile of size A_1 during $r \geq 1$ steps in the p -random q -proportion Bulgarian solitaire in the following way. We will need to keep track of each individual card in this pile. To this end, we label the cards $1, 2, \dots, A_1$ starting from the top, and each card will keep their label throughout the process. Let $X_{i,k} \in \{0, 1\}$ where $i = 1, \dots, A_1$ and $k = 1, \dots, r$ be independent Bernoulli random variables with $P(X_{i,k} = 1) = p$.

Consider the following process. Let A_{k+1} be the number of cards after k moves. In each move $k = 1, 2, \dots, r$, we remove the card with label i if $X_{i,k} = 1$ and this card belongs to the candidate cards, i.e., the $\lceil qA_k \rceil$ top-most remaining cards. We will call this process a q -process. This process describes the evolution of a pile of size A_1 in the p -random q -proportion Bulgarian solitaire.

Using the same Bernoulli variables, for any real number $0 \leq s \leq 1$, we define an s -threshold process in the following way. In each move $k = 1, 2, \dots, r$, we remove the card with label i if $X_{i,k} = 1$ and $i \leq sA_1$. In this process, we let $A_{k+1}^{[s]}$ denote the number of remaining cards after k moves. When it is relevant to indicate the initial pile size, an s -threshold process is called an (s, A_1) -threshold process and the number of remaining cards after k moves is denoted by $A_{k+1}^{[s, A_1]}$.

In the proof of Theorem 1, we will use two different s -threshold processes (for two different values of s) to over- and underestimate the sizes of $r + 1$ consecutive piles in Γ (corresponding to the r steps in an s -threshold process). Both these processes will have the same desired limit shape and thus the limit shape of our solitaire will follow by the squeeze theorem. We first need a combinatorial lemma giving sufficient conditions for overestimation and for underestimation.

Lemma 6. (i) If $\lceil sA_1 \rceil \leq \lceil qA_1 \rceil$, then $A_k^{[s]} \geq A_k$ for $k = 1, \dots, r+1$.
(ii) If $(1-q)A_{r+1}^{[s]} \geq A_1 - \lceil sA_1 \rceil$, then $A_k^{[s]} \leq A_k$ for $k = 1, \dots, r+1$.

Proof. (i) A card that is removed at some step ℓ during the s -threshold process must have label $i \leq \lceil sA_1 \rceil$, so in the q -process it belongs to the $\lceil qA_1 \rceil$ candidate cards in the initial pile and hence it belongs to the candidate cards also at step ℓ and will be removed. Thus, every card removed in the s -threshold process is removed in the q -process too, and it follows that $A_k^{[s]} \geq A_k$ for $k = 1, \dots, r+1$.

(ii) We show by induction over r that, after r steps, the remaining cards in the s -threshold process is a subset of the remaining cards in the q -process. Suppose $(1-q)A_{r+1}^{[s]} \geq A_1 - \lceil sA_1 \rceil$. Since $A_r^{[s]} \geq A_{r+1}^{[s]}$ we have $(1-q)A_r^{[s]} \geq A_1 - \lceil sA_1 \rceil$ which by the induction hypothesis implies that $A_k^{[s]} \leq A_k$ for $1 \leq k \leq r$. It follows that $(1-q)A_r \geq A_1 - \lceil sA_1 \rceil$ which in turn implies that $A_r - (A_1 - \lceil sA_1 \rceil) \geq \lceil qA_r \rceil$. This latter inequality means that the $\lceil qA_r \rceil$ topmost cards before step r in the q -process all have labels larger than $A_1 - \lceil sA_1 \rceil$. Thus, if a card is removed in step r in the q -process it is also removed in step r or earlier in the s -threshold process. This concludes the induction step. The base step $r = 0$ is trivial. $\square$

Recall that we are considering the configuration $\Gamma = \alpha^{(m)}$ after $m = M + D$ moves in the solitaire from the initial configuration $\alpha^{(0)}$. We will compare the sizes of $r+1$ consecutive piles in Γ to the $r+1$ pile sizes in an s -threshold process. In order to make the comparison for *all* piles in Γ , this will be done for $r+1$ consecutive piles (which we will call an r -*chunk*) at a time. In each r -chunk the initial pile size is the corresponding pile size in the solitaire. In other words, $\Gamma_1, \Gamma_2, \dots, \Gamma_{r+1}$ will be compared to the pile sizes in an (s, Γ_1) -threshold process (with initial pile size Γ_1); and $\Gamma_{r+2}, \Gamma_{r+3}, \dots, \Gamma_{2(r+1)}$ will be compared to the pile sizes in an (s, Γ_{r+2}) -threshold process (with initial pile size Γ_{r+2}), and so on. Let us call the resulting union of s -threshold processes an (r, s) -*union* process. Thus, if we denote the pile sizes in this (r, s) -union process by $U_1, U_2, \dots$, we have

$$\begin{aligned} U_1 &= \Gamma_1 = A_1^{[s, \Gamma_1]}, & U_2 &= A_2^{[s, \Gamma_1]}, & \dots, & U_{r+1} &= A_{r+1}^{[s, \Gamma_1]}, \\ U_{r+2} &= \Gamma_{r+2} = A_1^{[s, \Gamma_{r+2}]}, & U_{r+3} &= A_2^{[s, \Gamma_{r+2}]}, & \dots, & U_{2(r+1)} &= A_{r+1}^{[s, \Gamma_{r+2}]}, \dots \end{aligned}$$

We intend to use the (r, s) -union process to estimate the pile sizes in Γ . In an s -threshold process, starting with a pile of size A_1 , the number of remaining cards B above the level $A_1(1-s)$ after r moves is binomially distributed: $B \sim \text{Bin}(A_1 s, (1-p)^r)$. See Figure 4. Therefore we need to choose $r = r_n$ and $s = s_n$ in such a way that we have the following in each s -threshold process:

- I The pile size A_{k+1} is close to $A_1(1-pq)^k$ a.a.s. for all $k = 1, \dots, r$, which we need to establish the wanted limit shape.

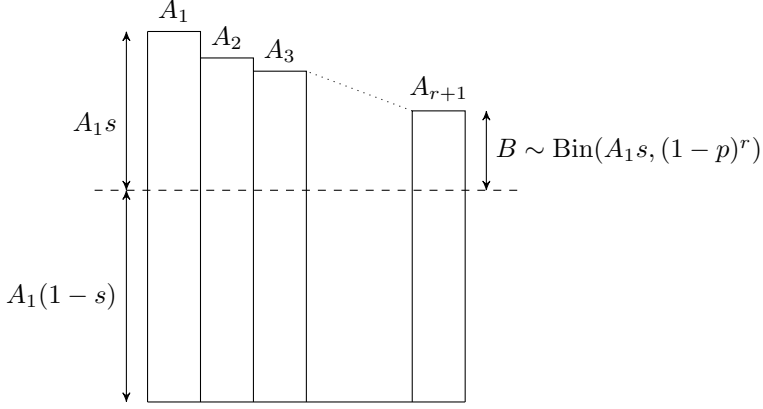


Figure 4: The r steps of an (s, A_1) -threshold process.

II At the same time s must be close enough to q to make the over- and underestimations tight enough.

To accomplish (I), clearly $r = r_n$ can at least not be chosen bigger than $1/p_n$, in fact we shall require $p_n r_n \rightarrow 0$ as $n \rightarrow \infty$, in order for the variance in the size of the last pile (after r_n moves) in an s -threshold process to be small with high probability. However, we shall see that $p_n r_n$ may not tend to zero too fast. We will require

$$(p_n r_n)^2 \frac{p_n q_n^2 n}{\log n} = \frac{p_n^3 q_n^2 n r_n^2}{\log n} \rightarrow \infty \text{ as } n \rightarrow \infty. \quad (9)$$

(Recall from (5) that $p_n q_n^2 n / \log n \rightarrow \infty$.) However, since r_n is a positive integer for any n , if $p_n \not\rightarrow 0$ we cannot have $p_n r_n \rightarrow 0$, but will see that $r_n = 1$ suffices in the case $p_n \not\rightarrow 0$. In other words, we will require

$$p_n(r_n - 1) \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (10)$$

To accomplish (II) we shall see that $s = q$ will suffice for the overestimation and $s = q(1 + 2pr) = q(1 + o(1))$ for the underestimation.

One way of choosing r_n such that (9) and (10) are fulfilled is

$$r_n = \left\lceil \rho_n^{-1/3} p_n^{-1} \right\rceil \text{ where } \rho_n = \frac{p_n q_n^2 n}{1 + \log n}. \quad (11)$$

This choice fulfills (10) since $(r_n - 1)p_n < r_n p_n \leq \rho_n^{-1/3} \rightarrow 0$. That (9) is fulfilled is easily verified:

$$(p_n r_n)^2 \frac{p_n q_n^2 n}{\log n} > \rho_n^{-2/3} \frac{p_n q_n^2 n}{\log n} = \frac{(p_n q_n^2 n)^{1/3}}{(1 + \log n)^{-2/3} \log n} > \left(\frac{p_n q_n^2 n}{\log n} \right)^{1/3} \rightarrow \infty$$

as $n \rightarrow \infty$ by (5).

Our next lemma, Lemma 7, will bound the probability P' that an initial pile of size $I_n := O(pqn)$ will, after r_n moves in an s_n -threshold process, deviate from the expected size assuming exponential decay, when $s_n = q_n(1 + o(1))$.

Since the number of piles is $\approx (p_n q_n)^{-1}$, the number of r -chunks is $\approx (p_n q_n r_n)^{-1}$. When using Lemma 7 we need the bound P' to hold for *each* chunk during *all* M moves (where M is given by (8)), specifically $P'M/(p_n q_n r_n) \rightarrow 0$ as $n \rightarrow \infty$. The probability in Lemma 7 is therefore bounded by $o(p_n q_n r_n / M) = o(p_n^2 q_n r_n / n^2)$. This is also why the pile size deviation $\varepsilon n p_n q_n$ is scaled with the number of chunks, resulting in the deviation $(\varepsilon p_n q_n n)(p_n q_n r_n) = \varepsilon p_n^2 q_n^2 n r_n$.

Lemma 7. *Let $(p_n)_n$ and $(q_n)_n$ be real sequences such that $0 < p_n, q_n \leq 1$ and $p_n q_n \rightarrow 0$ as $n \rightarrow \infty$. For each n , let also $B_n \sim \text{Bin}(F_n s_n, (1 - p_n)^{r_n})$ where $(F_n)_n$ and $(s_n)_n$ are real sequences such that*

$$F_n = O(p_n q_n n) \quad \text{and} \quad s_n = q_n(1 + o(1)), \quad (12)$$

and $F_n s_n$ is an integer for any n . Let also $(r_n)_n$ be the sequence of positive integers in (11). Then, for all $\varepsilon > 0$ we have

$$P(|B_n + F_n(1 - s) - F_n(1 - p_n q_n)^{r_n}| > \varepsilon p_n^2 q_n^2 n r_n) = o(p_n^2 q_n r_n / n^2).$$

Proof. Let us abbreviate $F = F_n$, $B = B_n$, $p = p_n$, $q = q_n$, $r = r_n$ and $s = s_n$. Thus, we want to prove that

$$P := P(|B + F(1 - s) - F(1 - pq)^r| > \varepsilon p^2 q^2 n r) = o(p^2 q r / n^2).$$

We first note that the expected value $E(B) = Fs(1 - p)^r$. Using the triangle inequality $|B + F(1 - s) - F(1 - pq)^r| \leq |B - E(B)| + |E - F(1 - pq)^r + F(1 - s)|$ we obtain

$$P \leq P(|B - E(B)| > \varepsilon p^2 q^2 n r - |E(B) - F(1 - pq)^r + F(1 - s)|).$$

By the Chernoff bound (7) we get

$$P \leq 2 \exp \left(- \frac{(\varepsilon p^2 q^2 n r - |E(B) - F(1 - pq)^r + F(1 - s)|)^2}{3E(B)} \right). \quad (13)$$

For the indices n for which $r_n > 1$ we have $rp \leq 2(r - 1)p \rightarrow 0$ and hence

$$(1 - p)^r = 1 - pr + o(pr) \quad \text{and} \quad (1 - pq)^r = 1 - pqr + o(pqr). \quad (14)$$

For the indices n for which $r_n = 1$, the relations in (14) are trivially true.

This means

$$\begin{aligned}
|E(B) - F(1 - pq)^r + F(1 - s)| &= |Fs(1 - p)^r - F(1 - pq)^r + F(1 - s)| \\
&= |Fs(1 - pr + o(pr)) - F(1 - pqr + o(pqr)) + F(1 - s)| \\
&= F\left(pr(q - s) + s \cdot o(pr) + o(pqr)\right) \\
&= o(Fpqr) = o(p^2 q^2 nr). \tag{by (12)}
\end{aligned}$$

Thus the numerator in (13) can be written $[(\varepsilon + o(1))p^2 q^2 nr]^2$. By the assumptions in (12), the denominator in (13) can be written

$$3E(B) = 3Fs(1 - p)^r = 3 \cdot O(pqn) \cdot q(1 + o(1)) \cdot O(1) = O(pq^2 n).$$

Putting these together, the bound (13) on P can be written

$$-\frac{1}{\log P} = O\left(\frac{O(pq^2 n)}{[(\varepsilon + o(1))p^2 q^2 nr]^2}\right) = O\left(\frac{1}{p^3 q^2 nr^2}\right) = o\left(\frac{1}{\log n}\right)$$

where (9) was used in the last step. Since $pqn r \rightarrow \infty$ (also by (9)) and $pqr \rightarrow 0$ (by (6) and (10)), we have $\frac{1}{pqr} = o(n)$ and hence $\log \frac{1}{pqr} = o(\log n)$. Therefore

$$-\frac{1}{\log P} = o\left(\frac{1}{\log n + \log \frac{1}{pqr}}\right) = o\left(\frac{1}{\log \frac{n}{pqr}}\right).$$

From this follows

$$\log P = o\left(\log \frac{pqr}{n}\right) = o\left(\log \frac{p^2 qr}{n^2}\right),$$

thus $P = o(p^2 qr/n^2)$. □

Note that Lemma 7 concerns an s -threshold process, i.e. only r steps. In other words, it asserts that

$$P\left(|A_{r+1} - A_1(1 - p_n q_n)^{r_n}| > \varepsilon p_n^2 q_n^2 n r_n\right) = o(p_n^2 q_n r_n / n^2), \tag{15}$$

where $A_1 = O(p_n q_n n)$ is the first pile size in an r -chunk and $A_{r+1} = (1 - s_n)A_1 + B_n$ the last (see Figure 4). However, the deviation and the probability were chosen in such a way that they can be added over all r -chunks. This is done in Lemma 8 which bounds the probability for deviation for the *entire* union process. Specifically, we will show that, for any $C > 0$, the piles in Γ formed at most $\frac{C}{pq}$ moves ago, i.e. Γ_k for $k \leq \frac{C}{pq}$, will follow an exponential decay a.a.s. The sizes of the piles formed more than $\frac{C}{pq}$ moves ago ($k > \frac{C}{pq}$) will be shown to be sufficiently small to be close enough to the tail in the exponential limit shape.

Lemma 8. *Let $U_1, U_2, \dots$ be the pile sizes in an (r_n, s_n) -union process corresponding to $\mathcal{B}(n, p_n, q_n)$, where the initial pile size is $U_1 = O(p_n q_n n)$, and r_n is given by (11) and $s_n = q_n(1 + o(1))$. Let $M = \lceil n^2/p_n \rceil$. Then*

$$\forall C, \varepsilon > 0 : \forall k < \frac{C}{p_n q_n} : P(|U_{k+1} - U_1(1 - p_n q_n)^k| > \varepsilon p_n q_n n) = o(1/M) = o(p_n/n^2).$$

Proof. As in the proof of Lemma 7, for the simplicity of notation we do not indicate in p, q, r and s the dependence on n . Let $C, \varepsilon > 0$ and $\varepsilon' = \varepsilon/C$. By the triangle inequality,

$$\begin{aligned} |U_{k+r+1} - U_1(1 - pq)^{k+r}| & \leq |U_{k+1} - U_1(1 - pq)^k|(1 - pq)^r + |U_{k+r+1} - U_{k+1}(1 - pq)^r| \\ & \leq |U_{k+1} - U_1(1 - pq)^k| + |U_{k+r+1} - U_{k+1}(1 - pq)^r|. \end{aligned}$$

Lemma 7 is now applicable for the first pile in each r -chunk (since $U_1 \geq U_2 \geq \dots$ and $U_1 = O(p_n q_n n)$), so by its formulation (15), $|U_{k+r+1} - U_{k+1}(1 - pq)^r| < \varepsilon' p^2 q^2 r n$ with probability $1 - o(p^2 q r / n^2)$. Thus,

$$|U_{k+r+1} - U_1(1 - pq)^{k+r}| < |U_{k+1} - U_1(1 - pq)^k| + \varepsilon' p^2 q^2 r n \quad (16)$$

with probability $1 - o(p^2 q r / n^2)$. We now note that the first term in the right hand side has the same form as the left hand side, only shifted with r piles. Thus, by induction it follows that, for any positive integer d , we have

$$|U_{dr+1} - U_1(1 - pq)^{dr}| < d\varepsilon' p^2 q^2 r n$$

with probability $1 - o(p^2 q r / n^2)$. Thus, adding the probabilities for deviation for $k = r, 2r, \dots, \eta r$, where $\eta = \lfloor \frac{C}{pqr} \rfloor$ we get

$$\begin{aligned} P(\forall k \in \{r, 2r, \dots, \eta r\} : |U_{k+1} - U_1(1 - pq)^k| > \eta \varepsilon' p^2 q^2 r n \geq \varepsilon p q n) \\ = \eta \cdot o(p^2 q r / n^2) = o(p/n^2). \end{aligned} \quad (17)$$

We have thereby proved the claim in the lemma for $k = r, 2r, \dots, \eta r$. If k is not a multiple of r , suppose $dr < k < (d+1)r$ for some positive integer d . Then, since $pqr \rightarrow 0$ as $n \rightarrow \infty$ (which follows from (5) and (11)), we have $(1 - pq)^r = 1 - pqr + o(pqr)$ and hence

$$|U_1(1 - pq)^{(d+1)r} - U_1(1 - pq)^{dr}| = O(pqn)(1 - pq)^{dr}|pqr + o(pqr)| < \varepsilon pqn.$$

The lemma then follows by (17) and the fact that $U_{dr} \leq U_k \leq U_{(d+1)r}$. $\square$

6. Proof of Theorem 1

Below follows the proof of Theorem 1, stated in Section 5.

Proof. First, as in the previous section, let us consider $\mathcal{B}(n, p_n, q_n)$ as a process on $\mathcal{W}(n)$ rather than on $\mathcal{P}(n)$, and let $\alpha^{(0)} \in \mathcal{W}(n)$ be the weak composition representing the initial configuration of cards in the solitaire. Let also M and D be given by (8).

Let $(r_n)_n$ be the sequence of positive integers given by (11) and let $(s_n)_n$ be the sequence $s_n = q_n(1 + 2p_nr_n)$. By Lemma 3 applied on $\alpha^{(D)}$, all piles present in $\alpha^{(D)}$ have disappeared in $\Gamma := \alpha^{(D+M)}$ a.a.s. Let $\Gamma_k = \alpha_k^{(D+M)}$ for $1 \leq k \leq M$ be the number of cards in the pile that was formed k moves ago. By Lemma 5, each of these piles had size $O(np_nq_n)$ a.a.s. when they were formed. Let $F_n := O(np_nq_n)$ be a sequence such that F_ns_n is an integer for each n . Let $0 < \varepsilon < 1$ and choose C_n such that $C_n > \frac{p_nq_n \log \varepsilon}{\log(1-p_nq_n)}$.

Let $\check{U}_1, \check{U}_2, \dots$ be the pile sizes in the (r_n, s_n) -union process with initial pile size Γ_1 . Using the fact that $p_nq_nr_n \rightarrow 0$, it is a straightforward computation to show that $s_n = q_n(1 + 2p_nr_n)$ implies $(1 - q_n)((1 - p_nr_n)^{r_n} - \varepsilon p_nq_nr_n) > 1 - s_n$ and therefore also

$$(1 - q_n)(A(1 - p_nr_n)^{r_n} - \varepsilon p_nq_nr_n) > (1 - s_n)A \quad \text{for any } A > 0.$$

By Lemma 7, the probability that $\check{U}_1(1 - p_nr_n)^{r_n} - \varepsilon \check{U}_1 p_nq_nr_n < \check{U}_{r_n+1}$ is $P_1 := 1 - o(p_n^2 q_n r_n / n^2)$. Thus, with probability P_1 we have $(1 - q_n)\check{U}_{r_n+1} > (1 - s_n)\check{U}_1 \geq \check{U}_1 - \lceil s\check{U}_1 \rceil$ so by Lemma 6(ii), the pile sizes $\check{U}_1, \check{U}_2, \dots, \check{U}_{r_n+1}$ in the first r -chunk of the (r_n, s_n) -process underestimate the pile sizes $\Gamma_1, \Gamma_2, \dots, \Gamma_{r_n+1}$ with probability P_1 . In the next chunk, we have a new absolute threshold $s\Gamma_{r_n+2} = s\check{U}_{r_n+2}$. Since $\Gamma_{r_n+2} \leq \Gamma_1$, we have $(1 - q_n)\check{U}_{2r_n+2} > (1 - s_n)\check{U}_{r_n+2}$ with probability at least P_1 , making Lemma 6(ii) applicable also for the second chunk to conclude that $\check{U}_{r_n+2}, \dots, \check{U}_{2r_n+2}$ underestimate $\Gamma_{r_n+2}, \dots, \Gamma_{2r_n+2}$ with probability at least P_1 . Continuing in the same manner for the first $C_n(p_nq_nr_n)^{-1}$ chunks, we conclude that the (r_n, s_n) -union process underestimates the solitaire with high probability:

$$P(\check{U}_k > \Gamma_k \text{ for all } k < \frac{C_n}{p_nq_n}) < (1 - P_1)C_n(p_nq_nr_n)^{-1} = o(p_n/n^2).$$

Let $\hat{U}_1, \hat{U}_2, \dots$ be the pile sizes in the (r_n, q_n) -union process with initial pile size Γ_1 . By Lemma 6(i) (with $s_n = q_n$), the (r_n, q_n) -union process surely overestimates the solitaire in each chunk.

Taking the results for the (r_n, s_n) -union process and the (r_n, q_n) -union process together we have

$$P(\check{U}_k \leq \Gamma_k \leq \hat{U}_k \text{ for all } k < \frac{C_n}{p_nq_n}) > 1 - o(p/n^2).$$

Now, applying Lemma 8 to both the pile sizes $\check{U}_k$ and to the pile sizes $\hat{U}_k$ and using the squeeze theorem, we obtain

$$\forall \varepsilon > 0 : \forall k < \frac{C_n}{p_n q_n} : P(|\Gamma_k - \Gamma_1(1 - p_n q_n)^k| > \varepsilon n p_n q_n) < o(p_n/n^2).$$

Thus, the probability that $|\Gamma_k - \Gamma_1(1 - p_n q_n)^k| < \varepsilon n p_n q_n$ for the first $\frac{C_n}{p_n q_n}$ piles throughout *all* M moves from $\alpha^{(D)}$ to $\alpha^{(D+M)}$ is $1 - M \cdot o(1/M) = 1 - o(1)$.

For piles $k > \frac{C_n}{p_n q_n}$, the exponential decrease (with decay factor $1 - pq$) in pile size will yield piles smaller than $n p_n q_n (1 - p_n q_n)^{\frac{C_n}{p_n q_n}} < \varepsilon n p_n q_n$ (by our choice of C_n). Thus, the pile sizes themselves are below $\varepsilon n p_n q_n$.

In summary; playing sufficiently many moves of $\mathcal{B}(n, p_n, q_n)$, the resulting composition diagram will a.a.s. be arbitrarily close to the boundary diagram of the composition α where $\alpha_k = n p_n q_n (1 - p_n q_n)^{k-1}$ for all $k = 1, 2, \dots$. The corresponding boundary function is $\partial\alpha(x) = n p_n q_n (1 - p_n q_n)^{\lfloor x \rfloor}$. The corresponding rescaled boundary function, with the given scaling factor $a_n = (p_n q_n)^{-1}$, is

$$\partial^{a_n} \alpha(x) = (1 - p_n q_n)^{\frac{x}{p_n q_n}} \rightarrow e^{-x}$$

since $p_n q_n \rightarrow 0$ as $n \rightarrow \infty$.

Setting $m := D + M$, and letting π_n^m denote the probability distribution on $\mathcal{W}(n)$ for $\alpha^{(m)}$, we have

$$\lim_{n \rightarrow \infty} \pi_n^m \{ \alpha \in \mathcal{W}(n) : |\partial^{a_n} \alpha(x) - e^{-x}| < \varepsilon \} = 1,$$

for all $\varepsilon > 0$ and all $x > 0$, in accordance with (2). By virtue of Lemma 2, the same limit shape holds when configurations in the solitaire $\mathcal{B}(n, p_n, q_n)$ are represented by partitions $\mathcal{P}(n)$.

Since π_{n, p_n, q_n} is the stationary distribution of the Markov chain $(\lambda^{(0)}, \lambda^{(1)}, \dots)$, if we start with a partition $\lambda^{(0)}$ sampled from π_{n, p_n, q_n} and play m moves, the resulting partition $\lambda^{(m)}$ will also be sampled from π_{n, p_n, q_n} . Thus, the theorem follows by choosing $\lambda^{(0)}$ as a stochastic partition sampled from the stationary distribution. $\square$

7. Conjectures

Recall that Theorem 1 was proved with $\mathcal{B}(n, p_n, q_n)$ being considered a process on $\mathcal{W}(n)$, and by virtue of Lemma 2 it also holds in $\mathcal{P}(n)$. We imposed the condition $\frac{p_n q_n^2 n}{\log n} \rightarrow \infty$. Here we conjecture that the weaker condition $p_n q_n^2 n \rightarrow \infty$ suffices in order for Theorem 1 to hold in $\mathcal{P}(n)$.

Conjecture 1. *Theorem 1 holds also when the condition $n p_n q_n^2 / \log n \rightarrow \infty$ is replaced by the weaker condition $n p_n q_n^2 \rightarrow \infty$.*

The reason for this conjecture can be understood by considering the example $q_n = 1$ and $p_n n = \log(\log n)$. For this example it is easy to prove that there is no limit shape when sorting is not performed. Since $q_n = 1$, the number of picked cards in each move and thus the expected size of a new pile is $\text{Bin}(n, p_n)$ with expected value np_n and standard deviation $\sigma \approx \sqrt{np_n}$. A pile of size np_n will after $1/p_n$ moves have the expected size $np_n(1 - p_n)^{1/p_n} \rightarrow e^{-1}np_n$ as $n \rightarrow \infty$.

Thus, the probability for a “visible” deviation (i.e. greater than $d = \sqrt{np_n}$ standard deviations) is $P(\text{deviation} \geq d\sigma) = e^{-np_n}$, so for $1/p_n$ piles, the probability for a visible deviation anywhere is $\frac{e^{-np_n}}{p_n} = \frac{n}{\log n \log(\log n)} \rightarrow \infty$ as $n \rightarrow \infty$, i.e. the expected number of such large deviations tends to infinity as n tends to infinity. This makes it impossible to achieve a convergence in probability towards a limit shape. However, from simulations we have reason to believe that the process converges towards a limit shape when sorting is performed.

Further, recall from Section 4 the other regimes $np_n q_n^2 \rightarrow 0$ and $np_n q_n^2 \rightarrow C$ from some constant $C > 0$. We conjecture that the limit shapes in the p_n -random q_n -proportion Bulgarian solitaire in these regimes are the same as in the deterministic q -proportion Bulgarian solitaire developed in [3].

Conjecture 2. *If $p_n q_n^2 n \rightarrow 0$ as $n \rightarrow \infty$, the limit shape of the p_n -random q_n -proportion Bulgarian solitaire is triangular.*

Conjecture 3. *If $p_n q_n^2 n \rightarrow C$ as $n \rightarrow \infty$ for some constant $C > 0$, the limit shape of the p_n -random q_n -proportion Bulgarian solitaire is a piecewise linear shape that depends on the value of C .*

References

- [1] S. Corteel, B. Pittel, C.D. Savage, and H.S. Wilf, *On the multiplicity of parts in a random partition*, Random Structures Algorithms **14** (1999), no. 2, 185–197.
- [2] V. Drensky, *The Bulgarian solitaire and the mathematics around it*, arXiv:1503.00885v1 [math.CO] (2015).
- [3] K. Eriksson, M. Jonsson, and J. Sjöstrand, *Limit shapes of stable configurations of a generalized Bulgarian solitaire*, arXiv:1703.07099 [math.CO] (2017).
- [4] K. Eriksson and J. Sjöstrand, *Limiting shapes of birth-and-death processes on Young diagrams*, Adv. in Appl. Math. **48** (2012), no. 4, 575 – 602.
- [5] M. Erlihson and B. Granovsky, *Reversible coagulation–fragmentation processes and random combinatorial structures: Asymptotics for the number of groups*, Random Structures Algorithms **25** (2004), no. 2, 227–245.

- [6] ———, *Limit shapes of Gibbs distributions on the set of integer partitions: The expansive case*, Ann. Inst. H. Poincaré Probab. Stat. **44** (2008), 915–945.
- [7] B. Fristedt, *The structure of random partitions of large integers*, Trans. Amer. Math. Soc. **337** (1993), no. 2, 703–735.
- [8] W.M.Y. Goh and P. Hitczenko, *Random partitions with restricted part sizes*, Random Structures Algorithms **32** (2008), no. 4, 440–462.
- [9] G.H. Hardy, J.E. Littlewood, and G. Pólya, *Inequalities*, Cambridge University Press, Cambridge, 1934.
- [10] B. Hopkins, *30 years of Bulgarian solitaire*, College Math. J. **43** (2012), no. 2, 136–141.
- [11] E.H. Lieb and M. Loss, *Analysis, volume 14 of graduate studies in mathematics*, vol. 4, American Mathematical Society, Providence, RI, 2001.
- [12] M. Mitzenmacher and E. Upfal, *Probability and computing: Randomized algorithms and probabilistic analysis*, Cambridge University Press, 2005.
- [13] J.S. Olson, *Variants of Bulgarian solitaire*, Integers **16** (2016), A8.
- [14] B. Pittel, *On a likely shape of the random Ferrers diagram*, Adv. in Appl. Math. **18** (1997), no. 4, 432–488.
- [15] S. Popov, *Random Bulgarian solitaire*, Random Structures Algorithms **27** (2005), no. 3, 310–330.
- [16] A. Vershik, *Statistical mechanics of combinatorial partitions, and their limit shapes*, Funct. Anal. Appl. **30** (1996), no. 2, 90–105.
- [17] Y. Yakubovich, *Ergodicity of multiplicative statistics*, J. Combin. Theory, Series A **119** (2012), no. 6, 1250–1279.

Paper IV

MARKOV CHAINS ON GRADED POSETS: COMPATIBILITY OF UP-DIRECTED AND DOWN-DIRECTED TRANSITION PROBABILITIES

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Abstract

We consider two types of discrete-time Markov chains where the state space is a graded poset and the transitions are taken along the covering relations in the poset. The first type of Markov chain goes only in one direction, either up or down in the poset (an *up chain* or *down chain*). The second type toggles between two adjacent rank levels (an *up-and-down chain*).

We introduce two compatibility concepts between the up-directed transition probabilities (an *up rule*) and the down-directed (a *down rule*), and we relate these to compatibility between up-and-down chains. This framework is used to prove a conjecture about a limit shape for a process on Young's lattice.

Finally, we settle the questions whether the reverse of an up chain is a down chain for some down rule and whether there exists an up or down chain at all if the rank function is not bounded.

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1. Introduction

A Hasse walk is a walk along the covering relations in a poset [9, 10]. In [5], Eriksson and Sjöstrand discussed how several famous models of stochastic processes can be regarded as random Hasse walks on Young's lattice, either walks that go steadily upwards (e.g. Simon's model of urban growth) or that alternately go up and down (e.g. the Moran model in population genetics). The aim of the present paper is to develop a general framework of such unidirected and alternatingly directed random Hasse walks on graded posets.

Let $I \subseteq \mathbb{Z}$ be a (possibly infinite) interval of the integers. An I -graded poset Ω is a countable (or finite) poset together with a surjective map $\rho: \Omega \rightarrow I$, called the *rank function*, such that

- $u < v$ implies $\rho(u) < \rho(v)$, and
- $u \lessdot v$ implies $\rho(v) = \rho(u) + 1$,

where $u \lessdot v$ means that v covers u , that is $u < v$ but there is no w with $u < w < v$. We can partition $\Omega = \bigcup_{i \in I} \Omega_i$ into its *level sets* $\Omega_i = \rho^{-1}(i)$.

We will now describe two types of stochastic processes on Ω .

1.1. Up chains and down chains

An assignment of a probability $T(u \rightarrow v)$ to any pair $(u, v) \in \Omega \times \Omega$ is an *up rule* if

- $T(u \rightarrow v) > 0 \Leftrightarrow u \lessdot v$, and
- $\sum_{v \in \Omega} T(u \rightarrow v) = 1$ for any u with non-maximal rank.

Analogously, it is a *down rule* if

- $T(u \rightarrow v) > 0 \Leftrightarrow u \gtrdot v$, and
- $\sum_{v \in \Omega} T(u \rightarrow v) = 1$ for any u with non-minimal rank.

Throughout, we shall use U and D to denote an up rule and a down rule, respectively.

Example 1. Figure 1 shows an example of a $[0, 2]$ -graded poset with two sets of probabilities forming an up rule (left) and a down rule (right).

Up rules and down rules define classes of Markov chains on Ω . We will consider such Markov chains $(X_i)_{i \in J}$ on Ω for any (possibly infinite) subinterval $J \subseteq I$.

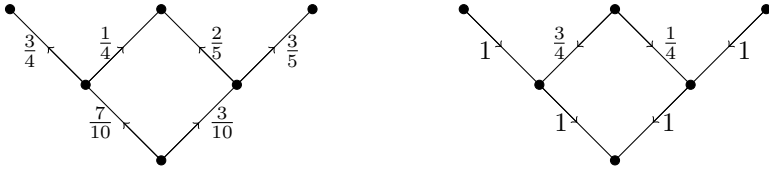


Figure 1: An up rule (left) and a down rule (right) on some Ω .

Definition 1. Let U be an up rule and let D be a down rule on an I -graded poset Ω . A Markov chain $(X_i \in \Omega_i)_{i \in J}$ is a U -chain (with time interval J) if

$$\text{Prob}(X_i = u \text{ and } X_{i+1} = v) = \text{Prob}(X_i = u)U(u \rightarrow v) \quad (1)$$

for any non-maximal $i \in J$ and any $u, v \in \Omega$. Analogously, it is a D -chain if

$$\text{Prob}(X_i = u \text{ and } X_{i-1} = v) = \text{Prob}(X_i = u)D(u \rightarrow v) \quad (2)$$

for any non-minimal $i \in J$ and any $u, v \in \Omega$.

A U - or D -chain is *maximal* if its time interval is I , and it is *positive* if $\text{Prob}(X_i = u) > 0$ for any $i \in J$ and any $u \in \Omega_i$.

In this paper we shall examine when an up rule and a down rule are compatible with each other. We shall distinguish between a weaker and a stronger notion of compatibility.

Definition 2. An up rule U and a down rule D on Ω are *compatible* if there is a maximal U -chain $(X_i \in \Omega_i)_{i \in I}$ and a maximal D -chain $(Y_i \in \Omega_i)_{i \in I}$ such that, for any $i \in I$, the random variables X_i and Y_i are equally distributed.

U and D are *strongly compatible* if there is a Markov chain that is both a maximal U -chain and a maximal D -chain.

Example 2. Let U and D be the up and down rules depicted in Figure 2. Clearly, U and D are compatible — just assign the probability $1/2$ to each element in the poset. But they are not strongly compatible since the probability of going diagonally is $3/4$ for the up rule and $1/4$ for the down rule.

In order to study probability distributions on the poset Ω we introduce some notation.

Let $\ell_1(\Omega)$ denote the Banach space of real-valued functions f on Ω such that the norm $\|f\| = \sum_{u \in \Omega} |f(u)|$ is finite. Let

$$\ell_1(\Omega_i) = \{f \in \ell_1(\Omega) : f^{-1}(\mathbb{R} \setminus \{0\}) \subseteq \Omega_i\}$$

denote the subspace consisting of functions with support on the level set Ω_i and let

$$M(\Omega_i) = \{\pi \in \ell_1(\Omega_i) : \pi(u) \geq 0 \text{ for any } u \in \Omega_i \text{ and } \sum_{u \in \Omega_i} \pi(u) = 1\}$$

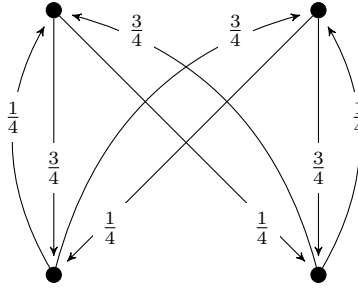


Figure 2: The up rule (bent arrows) and the down rule (straight arrows) are compatible but not strongly compatible.

denote the set of probability distributions on Ω_i .

An up or down rule T induces a linear operator (which, by abuse of notation, also is called T) on $\ell_1(\Omega)$ defined by

$$(T\pi)(v) = \sum_{u \in \Omega} T(u \rightarrow v)\pi(u).$$

Stepwise application of this operator defines sequences of probability distributions with support on one level set at a time, as follows.

Definition 3. Let U be an up rule and let D be a down rule on an I -graded poset Ω . A sequence $(\pi_i \in M(\Omega_i))_{i \in J}$ for some (possibly infinite) interval $J \subseteq I$ is a U -sequence if $U\pi_i = \pi_{i+1}$ for any non-maximal $i \in J$, and it is a D -sequence if $D\pi_i = \pi_{i-1}$ for any non-minimal $i \in J$.

A U - or D -sequence $(\pi_i)_{i \in J}$ is *positive* if $\pi_i(u) > 0$ for any $i \in J$ and any $u \in \Omega_i$.

Clearly, there is a one-to-one correspondence between U -sequences and U -chains and between D -sequences and D -chains.

Observation 1. U and D are compatible if and only if there exists a U -sequence that is also a D -sequence.

1.2. Up-and-down processes

Next we turn to alternatingly directed Markov chains on Ω .

Given an up rule U and a down rule D , we define a UD -chain as a Markov chain with state space Ω and with transitions induced by U and D alternately.

Definition 4. Let U be an up rule and let D be a down rule on an I -graded poset Ω . A UD -chain is a Markov chain $(X^{(0)}, X^{(1)}, \dots)$ on Ω such that

$$\text{Prob}(X^{(t)} = u \text{ and } X^{(t+1)} = v) = \begin{cases} \text{Prob}(X^{(t)} = u)U(u \rightarrow v) & \text{if } t \text{ is even} \\ \text{Prob}(X^{(t)} = u)D(u \rightarrow v) & \text{if } t \text{ is odd} \end{cases} \quad (3)$$

for any $u, v \in \Omega$.

Figure 3 depicts the level sets $\Omega_1 = \{u_1, u_2\}$ and $\Omega_2 = \{v_1, v_2, v_3\}$ of the poset Ω in Figure 1 with the same up rule U and down rule D . The up rule U and the down rule D define a UD -chain with state space $\Omega_1 \cup \Omega_2$.

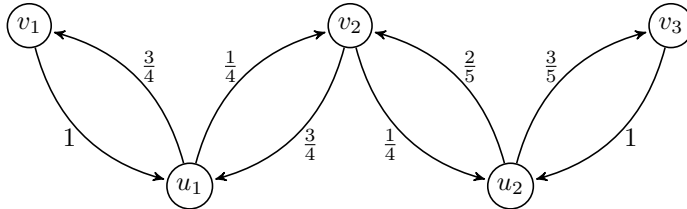


Figure 3: A UD -chain.

1.3. Results

Our first result relates the UD -process to the compatibility of U - and D -chains. We state this as two theorems, one about strong compatibility and one about compatibility. These theorems follow almost immediately from the definitions, but nevertheless they can be powerful tools. In Section 2 we use Theorem 2 to prove a conjecture of Eriksson and Sjöstrand [5] about the limit shape of a Markov process on Young's lattice.

Theorem 1. *Let Ω be an I -graded poset with an up rule U and a down rule D , and let $(X_i \in \Omega_i)_{i \in I}$ be a Markov chain. The following are equivalent.*

- (a) *$(X_i)_{i \in I}$ is both a U -chain and a D -chain (and hence U and D are strongly compatible).*
- (b) *For any adjacent levels $i, i+1 \in I$ it holds that $X_i, X_{i+1}, X_i, X_{i+1}, \dots$ is a UD -chain.*

Proof. In the sequence $(X_i, X_{i+1}, X_i, X_{i+1}, \dots)$ the even-indexed and odd-indexed variables are X_i and X_{i+1} , respectively. The two cases in (3) in Definition 4 with $t = i$ for even t and $t = i+1$ for odd t are equivalent to (1) together with (2) in Definition 1. $\square$

In other words, Theorem 1 says that if and only if random variables can be defined on each level set such that they correspond both to a Markov chain following the up rule and a Markov chain following the down rule, they also correspond to all Markov chains following the up and down rule alternatingly.

Theorem 2. *Let Ω be an I -graded poset with an up rule U and a down rule D , and let $(\pi_i \in M(\Omega_i))_{i \in I}$ be a sequence of probability distributions. The following are equivalent.*

- (a) *$(\pi_i)_{i \in I}$ is both a U -sequence and a D -sequence (and hence U and D are compatible).*
- (b) *For any adjacent levels $i, i+1 \in I$ it holds that $(\pi_i + \pi_{i+1})/2$ is a stationary distribution of the UD -process.*

Proof. Note that $(\pi_i + \pi_{i+1})/2$ is a stationary distribution of the UD -process if and only if $U\pi_i = \pi_{i+1}$ and $D\pi_{i+1} = \pi_i$. Now the theorem follows directly from Definition 3 (with $J = I$). $\square$

In words, Theorem 2 says that a given sequence of probability distributions is both a U -sequence and a D -sequence if and only if the average (in $\ell_1(\Omega)$) of two adjacent such distributions is a stationary distribution of the UD -process.

The second part of this paper will answer the following questions that arise naturally from the notion of compatibility.

- Q1. Given an up rule U and a U -chain (X_i) , is there a down-rule D such that (X_i) is a D -chain?
- Q2. Given an up or down rule T , is there a maximal T -chain?

The answer to Q1 is that every *positive* U -chain is a D -chain for some D . This is Theorem 3 in Section 3 and we give a constructive proof using Bayesian updating of probability distributions between adjacent levels.

The answer to Q2 is more complex and is presented as three theorems in Section 4. If all level sets Ω_i are finite, it turns out that there always exists a T -chain, but not necessarily a positive one. The proofs rely on topological arguments and require the axiom of choice.

1.4. Applications

We will apply the results to processes on two particular posets: (i) Young's lattice; and (ii) the d -dimensional nonnegative integer lattice.

In [5], Eriksson and Sjöstrand study stochastic processes on Young diagrams of integer partitions, in particular their limit shapes. In the framework of this paper, these processes are up processes and up-and-down processes, respectively, where the underlying poset is Young's lattice. A motivation for this work is a limit shape conjecture in [5] for a certain up-and-down process (described in Section 1.5 below). Using the results in the current paper, in particular Theorem 2, we will prove this conjecture. This is done in Corollary 2 in Section 2.

Applications of our theory to the d -dimensional nonnegative integer lattice are presented in Section 3.1.

1.5. An application of Theorem 2 to a process on Young's lattice

One of the processes studied in [5], called DEROW-ROW(μ), is a *UD*-chain on Young's lattice. We will first introduce some notation for integer partitions and Young diagrams and then describe the up rule ROW(μ) and the down rule DEROW used in this process.

1.5.1. Notation

An introduction to the theory of integer partitions can be found in [1].

With $\mathcal{P}_n$ we mean the set of all partitions of the positive integer n . For $\lambda \in \mathcal{P}_n$, we write $|\lambda| = n$. Denote the parts of the partition by $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_N)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$ and $N = N(\lambda)$ is the number of parts of λ . Let $r_i = r_i(\lambda) \geq 0$ denote the number of parts of size i . Thus, $N = \sum_{i=1}^n r_i$ and $n = \sum_{i=1}^N \lambda_i = \sum_{i=1}^n i r_i$.

An integer partition λ can be represented by a Young diagram drawn as left-aligned rows of squares in the first quadrant such that the i th row from the bottom has length λ_i . The diagrams in the first three levels of Young's lattice can be seen in Figure 4. With $\lambda \in \mathcal{P}_n$ we mean either the partition or its corresponding Young diagram.

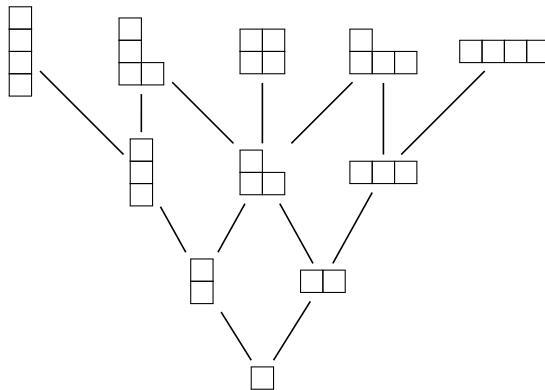


Figure 4: Young's lattice.

1.5.2. Description of the process

To describe the *UD*-chain DEROW-ROW(μ), we need to associate squares with row-lengths and row-lengths with corners in a Young diagram as follows.

Definition 5. See Figure 5. Consider some given Young diagram λ . For any of its squares s let $\kappa(s)$ denote the length of the row to which s belongs. If κ is a row length, let $\omega(\kappa)$ and $\iota(\kappa)$ denote the unique outer corner and inner corner, respectively, for which the row coordinate is κ :

$$\omega(\kappa) = (\kappa, \max\{i \mid \lambda_i = \kappa\}),$$

$$\iota(\kappa) = \begin{cases} (\kappa, \max\{i \mid \lambda_i > \kappa\}) & \text{if } \kappa < \lambda_1 \\ (\lambda_1, 0) & \text{if } \kappa = \lambda_1 \end{cases}.$$

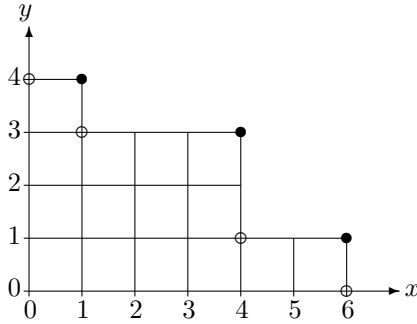


Figure 5: Inner and outer corners in the Young diagram of the partition $\lambda = (6, 4, 4, 1)$. The outer corners (filled circles) are from left to right $\omega(1)$, $\omega(4)$ and $\omega(6)$. The inner corners (unfilled circles) are from left to right $\iota(0)$, $\iota(1)$, $\iota(4)$ and $\iota(6)$.

Consider a current Young diagram λ . The action of the down rule DEROW is defined by choosing a non-empty row i uniformly at random and removing the corresponding outer corner $\omega(\lambda_i)$.

The action of the up rule ROW(μ) is defined as follows: With probability μ create a new row of length 1. Otherwise make a uniformly random choice of a row i among the $N(\lambda)$ non-empty rows and insert a new square at the corresponding inner corner $\iota(\lambda_i)$. See Definition 5.

The process DEROW-ROW(μ) is the up-and-down process on $\mathcal{P}_n$ (for some $n \geq 2$) using these up and down rules.

1.5.3. The stationary distributions

Lemma 1. For any $0 < \mu < 1$, the stationary distribution over the partitions in $\mathcal{P}_i \cup \mathcal{P}_{i+1}$ in the process ROW-DEROW(μ) is

$$\pi_i^{UD}(\lambda) = \frac{1}{2}(1 - \mu)^{|\lambda| - N(\lambda)} \mu^{N(\lambda) - 1} \frac{N(\lambda)!}{\prod_k r_k(\lambda)!}. \quad (4)$$

Proof. In [5] (equations (12) and (13)) it is proved that, up to a normalization constant, $\pi_i^{UD}(\lambda)$ is given by

$$\pi_i^{UD}(\lambda) = \frac{\left(\frac{\mu}{1-\mu}\right)^{N(\lambda)} N(\lambda)!}{\prod_k r_k(\lambda)!} \cdot \begin{cases} \frac{1}{2}c_i & \text{if } |\lambda| = i+1, \\ \frac{1}{2}c_i/(1-\mu) & \text{if } |\lambda| = i. \end{cases} \quad (5)$$

for some constant c_i (independent of λ). It follows that $c_{i+1}/(1-\mu) = c_i$ for each i , and since $c_1 = (1-\mu)^2/\mu$ we must have $c_i = (1-\mu)^{i+1}/\mu$. Inserting this expression for c_i in (5) yields (4). $\square$

Combining the lemma with Theorem 2 yields a formula for the distributions of the U -chain induced by the up rule $\text{ROW}(\mu)$.

Corollary 1. *The up rule $\text{ROW}(\mu)$ and the down rule DEROW are compatible and the distribution of the $\text{ROW}(\mu)$ process on $\mathcal{P}_i$ is given by the probability function*

$$p(\lambda) = (1-\mu)^{i-N(\lambda)} \mu^{N(\lambda)-1} \frac{N(\lambda)!}{\prod_k r_k(\lambda)!}.$$

In fact the derivation of equations (12) and (13) in [5] reveals that the stationary $\text{DEROW-ROW}(\mu)$ process is reversible, and hence, by Theorem 1, $\text{ROW}(\mu)$ and DEROW are strongly compatible.

2. Limiting objects

The original motivation for the current work has been the study of limiting objects, in particular limit shapes of Young diagrams under stochastic processes as initiated in [5]. As a corollary to Theorem 2, we can now prove a limit shape conjecture in [5] (Conjecture 2).

First, let us remind ourselves what is meant by a *scaling* of a Young diagram. As the number of squares n grows we need to rescale the diagram to achieve any limiting behaviour. Following [5] and [11], a diagram is rescaled using a *scaling factor* $a_n > 0$ such that all row lengths are multiplied by $1/a_n$ and all column heights are multiplied by a_n/n , yielding a constant diagram area of 1.

Corollary 2. *Let $\mu_n \log(\mu_n n) \rightarrow 0$ and $\mu_n n \rightarrow \infty$ as $n \rightarrow \infty$ and choose the scaling $a_n = 1/\mu_n$. Then the stationary distribution of the $\text{DEROW-ROW}(\mu_n)$ process has the limit shape*

$$y(x) = e^{-x}.$$

Proof. By Corollary 1, the stationary distribution for $\text{DEROW-ROW}(\mu)$ equals the distribution for $\text{ROW}(\mu)$. By Theorem 4 in [5], the limit shape for this distribution with $\mu = \mu_n$ is $y = e^{-x}$ as long as $\mu_n \log(\mu_n n) \rightarrow 0$ and $\mu_n n \rightarrow \infty$ as $n \rightarrow \infty$. Therefore, under these conditions the limit shape for $\text{DEROW-ROW}(\mu_n)$ must also be $y = e^{-x}$. $\square$

2.1. The asymptotics of μ_n in DEROW-ROW(μ_n)

Theorem 4 in [5] has the conditions $\mu_n n \rightarrow \infty$ and $\mu_n \log(\mu_n n) \rightarrow 0$ as $n \rightarrow \infty$. After scaling a Young diagram with scaling factor $a_n = 1/\mu_n$, each square will have width $1/a_n = \mu_n$ and height $a_n/n = \frac{1}{\mu_n n}$. In order for the boundary of the Young diagram (which is the object whose limiting behaviour one considers) to resolve properly as $n \rightarrow \infty$, we must at least have

$$\mu_n \rightarrow 0 \quad \text{and} \quad \mu_n n \rightarrow \infty \quad \text{as} \quad n \rightarrow \infty. \quad (6)$$

However, Theorem 4 in [5] uses the stronger assumption $\mu_n \log(\mu_n n) \rightarrow 0$ as $n \rightarrow \infty$. Whether this can be relaxed in Corollary 2 is an open problem.

2.2. A generalization of limiting objects

One may also generalize the concepts of limiting object and limit distribution. In order to be able to talk about a generic process having a limiting object and a limit distribution, let S be a separable metric space, called a *limit space*, with the metric d_S .

For $n = 0, 1, \dots$ let R_n be the countable set of states reachable after n steps, and let $f_n: R_n \rightarrow S$ be a “scaling” function. A stochastic process $(X_n \in R_n)_{n=0}^\infty$ having a limit distribution (with respect to the metric d_S) corresponds to the convergence of the random variables $\{f_n(X_n)\}$ *in distribution* to a random variable $X \in S$; the distribution of X is the limit distribution.

Further, if the limit distribution has all probability mass concentrated in a single point in S , in other words, if X is constant, then this point is the *limiting object* of the process (in which case we may equivalently talk about convergence *in probability* of $\{f_n(X_n)\}$ to the limiting object).

In this paper, we have made this concept tangible by studying processes on Young’s lattice, where the limit space is the set of decreasing functions in the first quadrant with integral 1. In the next section, we will study processes on the d -dimensional nonnegative integer lattice $\mathbb{N}^d$ where the limit space is the set of non-negative real d -tuples adding to 1. We see a further possible application to this generalization in the study of limits of permutation sequences (see for example [7]). In this case S is the set of Lebesgue measurable functions $[0, 1]^2 \rightarrow [0, 1]$ with certain properties.

3. Given an up rule, is there a compatible down rule?

In this section we will prove the existence of a down rule compatible with a given up rule. The proof is constructive and we will use the construction in some examples.

Theorem 3. *Let U be an up rule on an I -graded poset Ω , and let $(X_i)_{i \in I}$ be a U -chain. Then (X_i) is a D -chain for some down rule D on Ω if and only if $\text{Prob}(X_i = u) = 0$ implies $\text{Prob}(X_{i+1} = v) = 0$ for any $i, i+1 \in I$ and any $\Omega_i \ni u \lessdot v \in \Omega_{i+1}$. In particular this holds when (X_i) is positive.*

Proof. By Theorem 1, (X_i) is a D -chain if and only if $X_i, X_{i+1}, X_i, X_{i+1}, \dots$ is a UD -chain for any $i \in I$. Since $X_i, X_{i+1}, X_i, X_{i+1}, \dots$ is a reversible Markov chain (see any text book on Markov chains, for instance [8]), it is a UD -chain if and only if $\text{Prob}(X_i = u)U(u \rightarrow v) = \text{Prob}(X_{i+1} = v)D(v \rightarrow u)$ for any $u, v \in \Omega$. A D satisfying that equation can be chosen by letting

$$D(v \rightarrow u) = \frac{\text{Prob}(X_i = u)U(u \rightarrow v)}{\text{Prob}(X_{i+1} = v)} \quad (7)$$

but this is possible only unless $\text{Prob}(X_i = u) = 0$ and $\text{Prob}(X_{i+1} = v) > 0$ for some $u \lessdot v$ or $\text{Prob}(X_i = u) > 0$ and $\text{Prob}(X_{i+1} = v) = 0$ for some $u \lessdot v$. The latter is impossible since $\text{Prob}(X_{i+1} = v) \geq \text{Prob}(X_i = u)U(u \rightarrow v)$. $\square$

3.1. Processes on the d -dimensional nonnegative integer lattice

In this section we will demonstrate two applications of Theorem 3 on the d -dimensional nonnegative integer lattice $\mathbb{N}^d$. We will construct down rules compatible with given up rules. We shall also see examples of processes both with and without a limiting object.

3.1.1. The lattice $\mathbb{N}^d$

For a positive integer d , let $(\mathbb{N}^d, \leq)$ be the poset of nonnegative integer d -tuples $(x_1, \dots, x_d)$ ordered component-wise, i.e. for $x = (x_1, \dots, x_d), y = (y_1, \dots, y_d) \in \mathbb{N}^d$, we have $x \leq y$ if $x_i \leq y_i$ for all $i = 1, \dots, d$. For $x, y \in \mathbb{N}^d$, we have

$$\begin{aligned} x \vee y &= (\max(x_1, y_1), \dots, \max(x_d, y_d)) \text{ and} \\ x \wedge y &= (\min(x_1, y_1), \dots, \min(x_d, y_d)), \end{aligned}$$

so $\mathbb{N}^d$ is a lattice, which we will refer to as *the d -dimensional nonnegative integer lattice*.

Obviously, $\mathbb{N}^d$ is graded with the rank function

$$|x| = \sum_{j=1}^d x_j$$

and for $n \geq 0$, the level set $\mathbb{N}_n^d = \{x \in \mathbb{N}^d : |x| = n\}$ is the set of weak compositions of n .

3.1.2. A limiting object on $\mathbb{N}^d$

As an analogy to the concepts of limit shapes for birth- and birth-and-death processes on Young diagrams we will here define a limiting object for processes on $\mathbb{N}^d$.

If we divide the coordinates of a lattice point reached in an up process on $\mathbb{N}^d$ by the number of steps taken in the process, the result is a point in the $(d-1)$ -dimensional simplex $\Delta^{d-1} = \{(x_1, \dots, x_d) \in [0, 1]^d \mid x_1 + \dots + x_d = 1\}$. Under this scaling we define a limiting object as follows.

Definition 6. For an up process on $\mathbb{N}^d$, let $X_n = (X_n^1, \dots, X_n^d)$ be the lattice point after n steps. A point $x \in \Delta^{d-1}$ is a limit point of the up process if for any $\varepsilon > 0$ we have

$$\lim_{n \rightarrow \infty} \text{Prob} \left(\left| \frac{1}{n} X_n - x \right| < \varepsilon \right) = 1.$$

3.1.3. Notation

For $x = (x_1, \dots, x_d) \in \mathbb{N}^d$, let

$$\begin{aligned} x_{(i)} &= (x_1, \dots, x_i - 1, \dots, x_d) \text{ and} \\ x^{(i)} &= (x_1, \dots, x_i + 1, \dots, x_d) \end{aligned}$$

for $i = 1, \dots, d$. An up process on $\mathbb{N}^d$ starts at $(0, \dots, 0)$. In each step, the up rule U increases the rank of the current state by 1, i.e. increments exactly one component. Thus, U is determined by the probabilities $U(x \rightarrow x^{(i)})$ for $i = 1, \dots, d$.

For $n \in \mathbb{N}$, let $p_n: \mathbb{N}_n^d \rightarrow [0, 1]$ be the probability function on $\mathbb{N}_n^d$ induced by the up rule U .

3.1.4. An up process on $\mathbb{N}^d$ with a limiting object

Consider the up rule U on $\mathbb{N}^d$ governed by the constant transition probabilities $U(x \rightarrow x^{(i)}) = \nu_i$. We may use Theorem 3 to conclude that there is a down rule D compatible with U . Let us construct it!

First of all, the probability to reach $x = (x_1, \dots, x_d)$ (after $n = |x|$ steps in the up process induced by U) is clearly

$$p_n(x_1, \dots, x_d) = \frac{n!}{x_1! \dots x_d!} \nu_1^{x_1} \dots \nu_d^{x_d}.$$

As a consequence,

$$p_{n-1}(x_{(i)}) = \frac{(n-1)!}{x_1! \dots (x_i - 1)! \dots x_d!} \nu_1^{x_1} \dots \nu_i^{x_i - 1} \dots \nu_d^{x_d}.$$

We use (7) to compute the transition probability $D(x \rightarrow x_{(i)})$ from $x \in \mathbb{N}_n^d$ to $x_{(i)} \in \mathbb{N}_{n-1}^d$:

$$D(x \rightarrow x_{(i)}) = \frac{p_{n-1}(x_{(i)})U(x_{(i)} \rightarrow x)}{p_n(x)} = \frac{p_{n-1}(x_{(i)})\nu_i}{p_n(x)} = \frac{x_i}{n} = \frac{x_i}{|x|}.$$

Here we see that an up rule employing the degree of freedom parameters $\nu_1, \dots, \nu_d$ has a compatible down rule with no such degree of freedom present. As we saw in Corollary 1, this is also the case with the up rule in the process $\text{ROW}(\mu)$ having a compatible down rule DEROW , void of the degree of freedom parameter μ .

Proposition 1. *The up process on $\mathbb{N}^d$ using the up rule $U(x \rightarrow x^{(i)}) = \nu_i$ has the limit point $(\nu_1, \dots, \nu_d)$.*

Proof. For this process we have $\text{Prob}(X_n^i = k) = \binom{n}{k} \nu^k (1 - \nu)^{n-k}$ for $i = 1, \dots, d$, i.e. $X_n^i \sim \text{Bin}(n, \nu_i)$, which means $E(X_n^i)/n = \nu_i$. Thus, the limit point is the d -tuple $(\nu_1, \dots, \nu_d)$. $\square$

3.1.5. An up process on $\mathbb{N}^d$ without a limiting object

We will now consider an up process on $\mathbb{N}^d$ starting at $(0, \dots, 0)$ induced by an up rule that depends on the current level.

Proposition 2. *The process on $\mathbb{N}^d$ induced by the up rule U governed by the transition probabilities*

$$U(x \rightarrow x^{(i)}) = \frac{x_i + 1}{x_1 + \dots + x_d + d} = \frac{x_i + 1}{|x| + d} \quad (8)$$

has a uniform distribution on $\mathbb{N}_n^d$, for all $n \geq 0$.

Proof. We prove this by induction over the number of steps n in the process. First of all we observe that $|\mathbb{N}_n^d| = \binom{d+n-1}{d-1}$.

For $n = 0$, we have a trivial uniform distribution.

For $n \geq 0$, assume that the distribution on $\mathbb{N}_n^d$ under this process is uniform, i.e., $p_n(x) = |\mathbb{N}_n^d|^{-1} = \binom{d+n-1}{d-1}^{-1}$ for all $x \in \mathbb{N}_n^d$. We want to prove that the distribution over $\mathbb{N}_{n+1}^d$ is also uniform, i.e., $p_{n+1}(x) = |\mathbb{N}_{n+1}^d|^{-1} = \binom{d+n}{d-1}^{-1}$.

Let $x = (x_1, \dots, x_d) \in \mathbb{N}_{n+1}^d$ be a fixed element. Of course $|x| = n + 1$. Now, x can be reached from any of the elements $x_{(1)}, \dots, x_{(d)} \in \mathbb{N}_n^d$. By (8),

$$U(x_{(j)} \rightarrow x) = \frac{(x_j - 1) + 1}{|x_{(j)}| + d} = \frac{x_j}{n + d},$$

and by the induction hypothesis, $p_n(x_{(j)}) = \binom{d+n-1}{d-1}^{-1}$, so the probability $p_{n+1}(x)$ for $x \in \mathbb{N}_{n+1}^d$ is

$$\begin{aligned} p_{n+1}(x) &= \sum_{j=1}^d p_n(x_{(j)}) P(x_{(j)} \rightarrow x) \\ &= \binom{d+n-1}{d-1}^{-1} \sum_{j=1}^d \frac{x_j}{n+d} \\ &= \frac{(d-1)!n!(n+1)}{(d+n-1)!(n+d)} = \frac{(d-1)!(n+1)!}{(n+d)!} = \binom{d+n}{d-1}^{-1}, \end{aligned}$$

so the distribution is uniform also on $\mathbb{N}_{n+1}^d$, and the result follows by induction. $\square$

Since the distribution is uniform, this process cannot have a limiting object.

As in Section 3.1.4, let us use (7) to construct the down rule compatible with the up rule in Theorem 2. We get the probability $D(x \rightarrow x_{(i)})$ for moving from $x \in \mathbb{N}_n^d$ to $x_{(i)} \in \mathbb{N}_{n-1}^d$ by

$$\begin{aligned} D(x \rightarrow x_{(i)}) &= \frac{p_{n-1}(x_{(i)}) P(x_{(i)} \rightarrow x)}{p_n(x)} \\ &= \frac{\binom{d+n-2}{d-1}^{-1} \frac{x_i}{n-1+d}}{\binom{d+n-1}{d-1}^{-1}} = \frac{\binom{d+n-1}{d-1}}{\binom{d+n-2}{d-1}} \frac{x_i}{n-1+d} = \frac{x_i}{n} = \frac{x_i}{|x|}. \end{aligned}$$

4. Given an up or down rule T , is there a maximal T -chain?

Let T be an up or down rule on an I -graded poset Ω . Recall that there is a one-to-one correspondence between T -chains and T -sequences, so questions about the existence of maximal T -chains are equivalent to questions about the existence of maximal T -sequences.

If I has a minimal element m there obviously exists a maximal U -sequence for a given up rule U : Just choose any probability distribution $\pi_m \in M(\Omega_m)$ on level m , and the up rule U will induce a U -sequence $(U^{i-m}\pi_m)_{i \in I}$. However, if I has no lower bound it is not obvious whether there exists a maximal U -sequence, and, by symmetry, if I has no upper bound it is not obvious whether there exists a maximal D -sequence for a given down rule D . Here is an example where no such D -sequence exists.

Example 3. Let Ω be the two-dimensional integer lattice $\mathbb{Z}^2$ with the partial order $(x, y) \leq (x', y')$ if $x \leq x'$ and $y \leq y'$. It is $\mathbb{Z}$ -graded by $\rho(x, y) = x + y$. Let D be the down rule with probability $1/2$ for each edge in the Hasse diagram. If we start

at any element of high rank n and follow down edges randomly according to the down rule, the chance of hitting any particular element of rank zero is very small; it tends to zero as n grows. Thus, in a maximal D -sequence any rank-zero element must be given the probability zero, which is impossible.

The phenomenon in the above example cannot happen if the level sets Ω_i are finite, and, as we will see in Theorem 6, in this case there is always a maximal T -sequence. As the following example reveals, however, the existence of a *positive* maximal T -sequence is not guaranteed.

Example 4. Look at the two-dimensional nonnegative integer lattice $\mathbb{N}^2$ with the down rule given by the probability $1/2$ at every edge in the Hasse diagram, except for the leftmost and rightmost edges which must have probability one; see Figure 6. If we start at any element of high rank n and follow down edges randomly according to the down rule, with very high probability we will walk into the left or right border before we reach level 2. Thus, the probability of reaching the middle element $(1, 1)$ at level 2 tends to zero as n grows. This means that every D -chain (X_i) must have $\text{Prob}(X_2 = (1, 1)) = 0$.

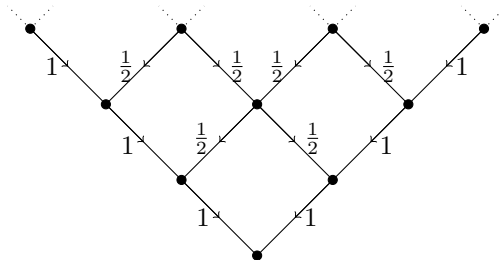


Figure 6: The down rule on $\mathbb{N}^2$ in Example 4.

We will present three theorems about the existence of T -sequences and positive T -sequences. Their proofs all depend on the following lemma.

Lemma 2. *Let T be an up or down rule on an I -graded poset Ω . Let $(C_i \subseteq M(\Omega_i))_{i \in I}$ be a sequence of compact sets and suppose for any $m \leq n$ in I there exists a T -sequence $(\pi'_i \in C_i)_{i \in [m, n]}$. Then there exists a maximal T -sequence $(\pi_i \in C_i)_{i \in I}$.*

Proof. Without loss of generality, we may assume that $T = U$ is an up rule.

Let Π be the product space $\Pi = \prod_{i \in I} C_i$. For any non-maximal $k \in I$, let

$$\Pi_k := \{(\pi_i) \in \Pi \mid U\pi_k = \pi_{k+1}\}.$$

Let us first show that the set Π_k is closed.

Clearly, Π_k is homeomorphic to the product of $\prod_{i \in I \setminus \{k, k+1\}} C_i$ and the graph $G \subseteq C_k \times C_{k+1}$ of the restriction of U to $C_k \cap U^{-1}C_{k+1}$. The map U has operator norm 1, so it is continuous. Hence, the preimage $U^{-1}C_{k+1}$ is closed and so is $C_k \cap U^{-1}C_{k+1}$. It follows that the graph G is closed and hence Π_k is closed, being homeomorphic to a product of closed sets.

By Tychonoff's theorem Π is compact, and therefore the closed subsets Π_k are compact, and so is the intersection Λ of all Π_k . Clearly, Λ is precisely the set of maximal U -sequences, so our task is to show that Λ is nonempty. To that end, suppose it is empty and define $U_k = \Pi \setminus \Pi_k$. Then $\bigcup U_k = \Pi \setminus \bigcap \Pi_k = \Pi$ so the family of all sets U_k is an open cover of Π . Since Π is compact there is an open subcover $\{U_k\}_{k \in F}$, where F is a finite set of non-maximal elements in I . Choose m and n in I so that $m \leq k < n$ for any $k \in F$. Now, by the assumption in the theorem there is a U -sequence $(\pi'_i \in C_i)_{i \in [m, n]}$. For $i \in I \setminus [m, n]$, let π'_i be an arbitrary element in C_i . The sequence $(\pi'_i)_{i \in I}$ so obtained is a point outside the union $\bigcup_{k \in F} U_k$, which contradicts the fact that $\{U_k\}_{k \in F}$ covers Π . Hence, our supposition that Λ is empty is false. $\square$

Our first existence theorem states that if there exists a T -sequence for any finite subinterval of I , and these sequences are uniformly bounded in a certain sense, then there exists a maximal T -sequence.

For $f, g \in \ell_1(\Omega)$ we will write $f \leq g$ if $f(u) \leq g(u)$ for all $u \in \Omega$.

Theorem 4. *Let T be an up or down rule on an I -graded poset Ω .*

Suppose there is a sequence $(\hat{b}_i \in \ell_1(\Omega_i))_{i \in I}$ such that for any $m \leq n$ there exists a T -sequence $(\pi_i)_{i \in [m, n]}$ with $\pi_i \leq \hat{b}_i$ for any $i \in [m, n]$. Then there exists a maximal T -sequence.

Proof. For any $i \in I$, let $C_i = \{\pi \in M(\Omega_i) : \pi \leq \hat{b}_i\}$. If we can show that C_i is compact, the theorem will follow from Lemma 2. We can write $C_i = M(\Omega_i) \cap L_i$ where

$$L_i = \{\pi : \Omega_i \rightarrow \mathbb{R} \mid 0 \leq \pi \leq \hat{b}_i\} \subset \ell_1(\Omega_i).$$

Since $M(\Omega_i)$ is closed it suffices to show that L_i is compact. By the dominated convergence theorem L_i is homeomorphic to the product space $\prod_{u \in \Omega_i} [0, \hat{b}_i(u)]$ which is compact by Tychonoff's theorem. $\square$

For the existence of a *positive* maximal T -sequence, it is not enough to assume the existence of finite T -sequences that are uniformly bounded from above; they must be uniformly bounded from above and from below simultaneously!

Let $\ell_1^+(\Omega_i) = \{f \in \ell_1(\Omega_i) : f(u) > 0 \text{ for any } u \in \Omega_i\}$ denote the set of strictly positive ℓ_1 -functions on Ω_i .

Theorem 5. *Let T be an up or down rule on an I -graded poset Ω .*

Suppose there are sequences $(\check{b}_i \in \ell_1^+(\Omega_i))_{i \in I}$ and $(\hat{b}_i \in \ell_1^+(\Omega_i))_{i \in I}$ such that for any $m \leq n$ in I there exists a T -sequence $(\pi_i)_{i \in [m, n]}$ with $\check{b}_i \leq \pi_i \leq \hat{b}_i$ for any $i \in [m, n]$. Then there exists a positive maximal T -sequence.

Proof. The proof is completely analogous to that of Theorem 4, but with $L_i = \{\pi : \Omega_i \rightarrow \mathbb{R} : \check{b}_i \leq \pi \leq \hat{b}_i\} \subset \ell_1(\Omega_i)$. $\square$

In most combinatorial applications, the level sets Ω_i are finite. In that case, the uniform upper bound $(\hat{b}_i)$ in the assumption in Theorem 4 automatically exists. Furthermore, the requirement of uniformity of the lower bound $(\check{b}_i)$ in Theorem 5 can be relaxed.

The following theorem is stated for an up rule, but the dual statement for a down rule is of course equivalent.

Theorem 6. *Let U be an up rule on an I -graded poset Ω with finite level sets Ω_i . Then there exists a maximal U -sequence, and there exists a positive maximal U -sequence if and only if there is a sequence $(b_i : \Omega_i \rightarrow (0, \infty))_{i \in I}$ with the property that for any $m \leq n$ in I there is a distribution $\pi_{m, n} \in M(\Omega_m)$ such that $U^{n-m}\pi_{m, n} \geq b_n$.*

Proof. Define a sequence $(\hat{b}_i \in \ell_1(\Omega_i))_{i \in I}$ by letting $\hat{b}_i(u) = 1$ for any $u \in \Omega_i$. Now Theorem 4 guarantees the existence of a maximal U -sequence.

Next, suppose there is a sequence $(b_i : \Omega_i \rightarrow (0, \infty))_{i \in I}$ and distributions $(\pi_{m, n} \in M(\Omega_m))_{m \leq n}$ with the property given in the theorem. Let $(\gamma_i)_{i \in I}$ be positive numbers adding to one, and for each $i \in I$ put $\check{b}_i = \gamma_i b_i$.

Now, fix $m \leq n$ in I . Let $\Gamma = \sum_{i \in [m, n]} \gamma_i$ and define

$$\pi = \frac{1}{\Gamma} \sum_{i \in [m, n]} \gamma_i \pi_{m, i}.$$

Since U is a linear operator, for any $j \in [m, n]$ we have

$$U^{j-m}\pi = \frac{1}{\Gamma} \sum_{i \in [m, n]} \gamma_i U^{j-m}\pi_{m, i} \geq \frac{\gamma_j}{\Gamma} U^{j-m}\pi_{m, j} \geq \frac{\gamma_j}{\Gamma} b_j = \frac{\check{b}_j}{\Gamma} \geq \check{b}_j$$

and hence $(U^{j-m}\pi)_{j \in [m, n]}$ is a T -sequence such that $U^{j-m}\pi \geq \check{b}_j$ for any $j \in [m, n]$. Theorem 5 now yields the existence of a positive maximal U -sequence. $\square$

Theorem 6 guarantees the existence of a maximal T -sequence if all level sets are finite, but in general it is not possible to extend a T -sequence for a subinterval $J \subset I$ to a maximal T -sequence. For example, consider the poset in Figure 7 with a defined down rule D . The distributions $\pi_0 \in M(\Omega_0)$ and $\pi_1 \in M(\Omega_1)$ given by $\pi_0(\hat{0}) = 1$, $\pi_1(s_1) = 1/4$ and $\pi_2(s_2) = 3/4$ constitute a D -chain, but this D -chain can obviously not be extended to the top level.

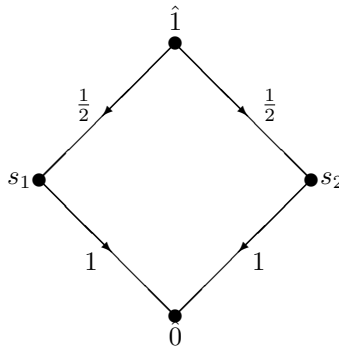


Figure 7: A graded poset with a given down rule D .

5. General homogeneous Markov chains

Finally, we shall use Theorem 4 to prove a more general result about the existence of Markov chains with time interval $\mathbb{Z}$. As far as we know, this has not been treated before.

Let S be a countable (or finite) state space and let P be a transition function on S , that is, an assignment of a probability $P(s \rightarrow s')$ to each pair $(s, s') \in S \times S$ such that $\sum_{s' \in S} P(s \rightarrow s') = 1$ for any $s \in S$. For any initial (random) state X_0 , there is a unique Markov chain $X_0, X_1, \dots$ with transition function P , but is there a Markov chain $\dots, X_{-1}, X_0, X_1, \dots$ with time interval $\mathbb{Z}$? The following “homogeneous” version of Theorem 4 answers that question.

Theorem 7. *Let S be a state space and let P be a transition function on S . Suppose there exists a sequence of functions $(\hat{b}_i \in \ell_1(S))_{i \in \mathbb{Z}}$ such that for any integers $m \leq n$ there is a Markov chain $X_m, \dots, X_n$ on S with transition function P such that $\text{Prob}(X_i = s) \leq \hat{b}_i(s)$ for any $i \in [m, n]$ and any $s \in S$. Then there exists a Markov chain $\dots, X_{-1}, X_0, X_1, \dots$ on S with transition function P .*

Proof. Define a $\mathbb{Z}$ -graded poset $\Omega = S \times \mathbb{Z}$ with $(s, i) \prec (s', i+1)$ if $P(s, s') > 0$ and define an up rule U on Ω by letting

$$U((s, i) \rightarrow (s', i+1)) = P(s \rightarrow s')$$

for any $i \in \mathbb{Z}$ and any $s, s' \in S$. Now the theorem follows from Theorem 4. $\square$

6. Discussion

We have applied the results in this paper to Young’s lattice (in Section 2) and to the d -dimensional nonnegative integer lattice (in Section 3.1).

As mentioned in Section 2.2, a possible application to our generalization of limiting object is in the study of sequences of permutations [7]. Similarly to the well-studied *Tsetlin library Markov chain* (see for example [6]), such a sequence can be generated by a stochastic process in which the number $n + 1$ is inserted into a permutation of $\{1, 2, \dots, n\}$ to create a permutation of $\{1, 2, \dots, n + 1\}$, wherein each of the possible $n + 1$ “slots” is assigned a probability to create a probability distribution. In terms of this paper, such probabilities constitute an up-rule U on the $\mathbb{N}$ -graded poset Ω , where each level set $\Omega_i = \mathfrak{S}_i$ is the set of permutations of $\{1, 2, \dots, i\}$. One direction for future research is to investigate the relation between U and possible limiting objects of resulting permutation sequences in terms of [7].

Other Markov chains whose state-space has a partial order occur in the study of certain semigroups called *left-regular bands* (for example [2]) as well as in the study of poset block structures ([3, 4]). Another direction for future research may thus include investigating whether the results of the present paper may be applied also to these families of Markov chains.

References

- [1] Andrews, G.E., Eriksson, K.: Integer partitions. Cambridge University Press (2004)
- [2] Brown, K.S.: Semigroups, rings, and Markov chains. *Journal of Theoretical Probability* **13**(3), 871–938 (2000)
- [3] D’Angeli, D., Donno, A.: Markov chains on orthogonal block structures. *European Journal of Combinatorics* **31**(1), 34–46 (2010)
- [4] D’Angeli, D., Donno, A.: The lumpability property for a family of Markov chains on poset block structures. *Advances in Applied Mathematics* **51**(3), 367–391 (2013)
- [5] Eriksson, K., Sjöstrand, J.: Limiting shapes of birth-and-death processes on Young diagrams. *Adv. in Appl. Math.* **48**(4), 575 – 602 (2012). DOI 10.1016/j.aam.2011.12.001
- [6] Fill, J.A.: An exact formula for the move-to-front rule for self-organizing lists. *Journal of Theoretical Probability* **9**(1), 113–160 (1996)
- [7] Hoppen, C., Kohayakawa, Y., Moreira, C.G., Ráth, B., Sampaio, R.M.: Limits of permutation sequences. *Journal of Combinatorial Theory, Series B* **103**(1), 93–113 (2013)
- [8] Ross, S.M.: Introduction to probability models. Academic press (2014)

- [9] Stanley, R.P.: Differential posets. J. Amer. Math. Soc. **1**(4), 919–961 (1988)
- [10] Stanley, R.P.: Enumerative combinatorics, vol. 1, Wadsworth and Brooks/Cole, Pacific Grove, CA, 1986; second printing (1996)
- [11] Vershik, A.: Statistical mechanics of combinatorial partitions, and their limit shapes. Funct. Anal. Appl. **30**(2), 90–105 (1996)



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